

Cardy's formula for critical Poisson–Voronoi percolation

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Abstract

We prove Cardy's formula for annealed crossing probabilities in critical planar Poisson–Voronoi percolation in every bounded Jordan quadrilateral. This proves the annealed crossing-probability form of the conformal-invariance conjecture for this model.

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1 Introduction

Cardy's formula predicts that a critical two-dimensional percolation crossing probability is determined by a conformal parameter of the marked domain [6]. Smirnov proved the formula for site percolation on the triangular lattice [18]. Schramm's Problem 2.12 asks for the corresponding conformal-invariance theorem for Voronoi percolation [16]. We prove the annealed Cardy-crossing aspect of this conjecture for the Poisson model, with no regularity assumption on the boundary beyond the Jordan property.

1.1 The crossing law

Identify the oriented plane with $\mathbb{C}$. For $\varepsilon > 0$, let $\eta = X_\varepsilon$ be a homogeneous Poisson point process of intensity ε^{-2} . For $v \in \eta$ its Voronoi cell is the closed set

$$\mathcal{V}_v = \{z \in \mathbb{C} : |z - v| \leq |z - u| \text{ for every } u \in \eta\}.$$

Independently of η , color its sites black or white with probability $1/2$ each, and give each cell the color of its site. Let $\mathcal{B}_\varepsilon$ be the union of the closed black cells. Probabilities and expectations under the joint law of the points and colors are denoted by $\mathbb{P}$ and $\mathbb{E}$. Conditional probabilities given the points are denoted by $\mathbb{P}^\eta$; $\mathbb{E}_{\text{sp}}$ denotes averaging over independent fair colors on a fixed graph. Thus all probabilities in the main theorem are annealed.

Let D be a bounded Jordan domain and let a, b, c, d be distinct points of ∂D in counterclockwise order. Boundary arcs always include their endpoints. Define $\mathcal{C}_\varepsilon(D; a, b, c, d)$ to be the event that $\mathcal{B}_\varepsilon \cap \overline{D}$ contains a connected subset meeting both the arc ab and the arc cd . The conformal map from D to $\mathbb{H} = \{z : \text{Im } z > 0\}$ which sends a, c, d to $0, 1, \infty$ extends to the boundary in the sense of Carathéodory [13, Section 2.1, p. 18]. Its value at b is a uniquely determined number $x \in (0, 1)$.

Theorem 1.1 (Annealed Voronoi crossing law). *For every bounded Jordan domain D and every four distinct counterclockwise boundary marks a, b, c, d , one has*

$$\lim_{\varepsilon \downarrow 0} \mathbb{P}(\mathcal{C}_\varepsilon(D; a, b, c, d)) = F(x) := \frac{\int_0^x [u(1-u)]^{-2/3} du}{\int_0^1 [u(1-u)]^{-2/3} du} = \frac{\Gamma(2/3)x^{1/3}}{\Gamma(1/3)\Gamma(4/3)} {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; \frac{4}{3}; x\right). \quad (1.1)$$

The theorem resolves positively the annealed Cardy-crossing conjecture for critical Poisson–Voronoi percolation, the crossing-probability aspect of Schramm’s Problem 2.12 above. Its scope is a limit of crossing probabilities averaged over geometry and colors. The statement does not assert convergence of the full interface law or a quenched crossing limit for arbitrary Jordan domains. In particular, the result concerns the specified closed-cell crossing event in every fixed Jordan domain, including domains for which a smooth boundary approximation alone would not immediately identify that event.

For a fixed rectangle, the established concentration of conditional crossing probabilities also gives a quenched L^2 limit; see Corollary 1.2 below.

1.2 Prior work

The conformal-invariance conjecture for critical crossings developed from numerical and universality investigations. Langlands, Pouliot and Saint-Aubin explicitly credit Aizenman with the conformal formulation [9, Sections 2.1 and 2.4]. Cardy’s conformal-field-theory calculation identifies the predicted crossing function [6]. Smirnov’s proof on the triangular lattice uses color-switching identities to construct conformal observables [18, 17]. Carleson’s representation of Cardy’s formula as a linear boundary value in an equilateral triangle, explained in [17, Corollary 3], also describes the boundary problem at the end of our argument.

The same Cardy function is obtained for critical square-lattice bond percolation under uniform marked-Jordan approximation [12, Theorem 1.2]. There the event is an actual open-edge path with all boundary edges sampled independently and no external connections; here it is an

annealed closed-cell crossing in a fixed Jordan domain. The square-bond theorem is not used in our proof.

Voronoi percolation provides a random planar geometry whose law has no preferred lattice direction. Benjamini and Schramm proved a distinct high-intensity invariance under smooth conformal changes of metric, with the sampling measure held fixed [4]. Bollobás and Riordan established that the critical probability of the planar Poisson model is $1/2$ [5]. Uniform box-crossing estimates were established by Tassion [19]; quenched crossing estimates were developed by Ahlberg, Griffiths, Morris and Tassion [1]. Vanneuville’s quantitative work supplies comparisons between quenched and annealed arm probabilities, as well as the crossing estimates used here [22]. The conformal-invariance conjecture and its relation to these partial results are discussed in [21, Section 1.2]. These estimates control crossings and exceptional geometry; identifying the limiting function requires an additional argument. Proposition 3.1 states the established estimates at the precise hypotheses used below.

The challenge is to obtain an analytic relation that survives the irregular embedding. In a related setting, Beffara separates a color-switching identity from the geometric defect of a non-equilateral embedding [3, Sections 3.2–3.3, equations (13)–(14)]. His crossing theorem concerns periodic triangulations subdivided into triangular-lattice pieces, under uniform box-crossing and a specified relation between the two scales; its limit agrees with triangular-lattice percolation up to a real-linear transformation determined by the initial embedding. Here the geometry is Poisson–Delaunay; we recover the needed relation by transporting exact closed-circuit identities to open portions of random boundaries.

1.3 From circuit identities to the crossing law

The proof begins with the triangular graph joining sites of neighboring Voronoi cells and its dual graph. Dual edges separating opposite colors form ordinary cluster-boundary circuits. For each such circuit L , we delete all points of L , including its vertices, and retain each connected component C of the remaining dual graph. An unused edge whose two endpoints lie on L then leaves an isolated open-edge component. The label is the pair (L, C) , not just the geometric side of L . Except for isolated open-edge components, these components have simple monochromatic primal boundaries P , called *rims*, oriented with their component on the left. Keeping the full component label, including isolated edges, is what makes the exact orthogonality identity of Section 2 hold.

Each triangle is the image of a reference equilateral triangle under an orientation-preserving real-affine map. The complex-linear and antilinear parts of this map split each clockwise side vector into two terms. For a smooth test function f , take the contributions for the side belonging to P in each triangle immediately to its right, and weight them by f at the triangle’s barycenter. Summing the complex-linear terms gives $U_P(f)$; summing the antilinear terms gives $Q_P(f)$. Thus Q records the departure from equilateral geometry, while $V_P(f) = U_P(f) + Q_P(f)$ is the weighted tangent displacement. Orthogonality gives a separate exact second-moment identity for each Fourier field when summed over all component labels, including isolated open edges. The complete tangent sum is $V_P(1) = 0$ on every closed rim. Stationarity makes the corresponding macroscopic closed-rim sums of Q small. The main task is to open that identity: along a common microscopic subsequence, we prove

$$Q_P(f) - aV_P(f) \longrightarrow 0$$

in probability, uniformly over rims extending beyond a fixed collar of $\text{supp } f$. The scalar a is

independent of the later test and bulk patch; Theorem 4.1 states the precise local formulation.

For this passage to open arcs, we retain local colored configurations near endpoints in deterministic regions called *plugs*. Separated two-color corridors join them into closed experiments. Surrounding circuits restrict the path to the same two contacts, and the absence of an alternative monochromatic path on its right forces agreement when the exterior configuration is changed. For each fixed connector, conditional resampling produces a function of the two endpoint states, common to its completions, whose sum around each fixed cycle is small. Comparing closures and inserting intermediate plugs decomposes this function into an affine displacement, two local endpoint potentials, and a controlled remainder. Sections 4 and 5 construct this endpoint transport before it is used on a random arc.

Two issues remain in returning to actual rims. The local geometry and colors have already selected the arc, so the comparison must retain their conditioning. At each fixed comparison scale, arm estimates provide a finite menu of plug models densely along every relevant rim and a finite deterministic family of connecting patterns. The comparison is made with unnormalized subprobabilities dominated by the original law; there is no division by a configuration-dependent fitting probability. A second issue is cancellation: the fields are signed, so a small change in their coefficients is not controlled by a path-length estimate. Deterministic spatial refinement, a near-return bound, and the orthogonality identity supply that control. Sections 6 and 7 complete these two steps in the order needed for bulk transport.

The final steps prove $1 - 2\operatorname{Re} a > 0$ for the scalar a appearing in the transport relation. In particular, the coefficient $1 - a$ needed by the observable is nonzero. Complete a finite discrete disk by three exterior marked vertices; a component observable at this closing triangle records which boundary connection occurs. One of its boundary coefficients is exactly the crossing probability. Orthogonality and the transport relation give the contour cancellation needed for a holomorphic limit. The scalar a need not be identified: once $1 - a \neq 0$, it disappears from the limiting Cauchy–Riemann equation. The resulting boundary-value problem has the conformal map to an equilateral triangle as its unique solution (Propositions 9.2 and 9.4, and Lemma 9.5). This common boundary problem identifies every subsequential crossing limit. A deterministic comparison at fixed buffers in a Schoenflies chart then transfers this crossing law to the exact closed-cell event in arbitrary Jordan domains.

1.4 A quenched consequence for rectangles

Corollary 1.2 (Quenched Cardy limit for fixed rectangles). *Let D be a fixed bounded nondegenerate open rectangle and let a, b, c, d be its corners in counterclockwise order, with ab and cd the chosen opposite sides. Write x_D for its conformal parameter above and set*

$$Q_\varepsilon(\eta) = \mathbb{P}^\eta(\mathcal{C}_\varepsilon(D; a, b, c, d)), \quad \eta = X_\varepsilon.$$

If $\mathbb{P}_{\text{geom},\varepsilon}$ and $\mathbb{E}_{\text{geom},\varepsilon}$ denote probability and expectation for the Poisson geometry at mesh ε , then

$$\lim_{\varepsilon \downarrow 0} \mathbb{E}_{\text{geom},\varepsilon}[|Q_\varepsilon - F(x_D)|^2] = 0.$$

In particular, for every $t > 0$, $\mathbb{P}_{\text{geom},\varepsilon}(|Q_\varepsilon - F(x_D)| > t) \rightarrow 0$. These limits use the geometry law at each mesh, without a coupling across meshes.

Proof. Almost surely only finitely many Voronoi cells meet $\overline{D}$, and each cell clipped to $\overline{D}$ is closed and convex. Every connected component of a finite union of such clipped cells is path connected,

by concatenating segments through a chain of intersecting cells. Thus, almost surely in the geometry and for every coloring, our connected-set crossing event agrees with the continuous closed-cell crossing event in [22, Definition 1.1].

After an orientation-preserving isometry, write $\overline{D} = [-w, w] \times [-h, h]$, with the chosen sides vertical. Scaling by ε^{-1} sends the full-plane intensity- ε^{-2} Poisson process to unit intensity and this rectangle to $[-\lambda R, \lambda R] \times [-R, R]$, where $\lambda = w/h$ and $R = h/\varepsilon$. The Voronoi cells and their conditional fair coloring commute with this similarity. Therefore [22, Theorem 1.3] gives an absolute $\alpha > 0$ and a constant $C_\lambda < \infty$ such that, for $0 < \varepsilon \leq h$,

$$\text{Var}_{\text{geom}, \varepsilon}(Q_\varepsilon) \leq C_\lambda(\varepsilon/h)^\alpha.$$

Let $m_\varepsilon = \mathbb{P}(\mathcal{C}_\varepsilon(D; a, b, c, d))$. Conditioning on the geometry gives $\mathbb{E}_{\text{geom}, \varepsilon} Q_\varepsilon = m_\varepsilon$, while Theorem 1.1 gives $m_\varepsilon \rightarrow F(x_D)$. Hence

$$\mathbb{E}_{\text{geom}, \varepsilon} |Q_\varepsilon - F(x_D)|^2 = \text{Var}_{\text{geom}, \varepsilon}(Q_\varepsilon) + |m_\varepsilon - F(x_D)|^2 \leq C_\lambda(\varepsilon/h)^\alpha + |m_\varepsilon - F(x_D)|^2 \rightarrow 0.$$

Markov's inequality gives the assertion in probability. $\square$

1.5 Geometry, bulk patches and notation

Write $\omega = \exp(2\pi i/3)$. Unless specified otherwise, sites, edges and triangles belong to the primal Delaunay triangulation. Either color can be the path color; we describe it as black when making a construction. Distances, diameters and balls are Euclidean, and complex vectors use the positive orientation of the plane.

We recall the geometric facts behind this convention. Almost surely, the Poisson process is locally finite and has neither collinear triples nor cocircular quadruples, by diffuseness and a countable exhaustion. For a finite set in general position, empty-circumdisk triangles form the triangulation of its convex hull: lift the sites to the paraboloid and project the lower faces. Their circumcenters are Voronoi vertices, and shared cell sides correspond to primal edges. The local determination estimate in Lemma 3.3 makes these finite constructions stabilize on every compact set along an exhaustion. It also bounds the cells of sites in any fixed compact set. Thus the full-plane cells form a locally finite polygonal tiling with trivalent meeting points. Sites of touching cells are primal neighbors, including when the cells meet at a vertex. The primal graph can equivalently be drawn through the tiling by joining sites to interior points of shared cell sides. This drawing has the same planar orders as the straight Delaunay drawing.

Definition 1.3 (Bulk setting). A bulk setting is either the full-plane sample or a finite simple triangulation of the sphere carrying independent fair site colors conditional on its graph and geometry. In the latter case an oriented open planar patch is coupled to a full-plane sample so that, on every fixed compact subset of the patch, the graphs and their colors agree with probability tending to one as $\varepsilon \downarrow 0$. The graph and chart agreement event is independent of color sampling conditional on the geometry. Spatial weights are used only on triangles in the agreement patch. Compactly supported tests may be set to zero on the exceptional geometry event.

The finite completions will be useful when the observable is formed near the boundary. No geometric control of their triangles outside the patch is required for the local field identities.

We use the Jordan–Schoenflies theorem in its prescribed-boundary form [10, Theorem 10.4], its extension to a simple planar arc [2, Première Partie, Chapitre I, Section 17], and Carathéodory's

boundary extension theorem. We also use the following form of the Radó convergence theorem [14, Section 1, pp. 182–183]. If Jordan boundaries have parametrizations converging uniformly and in order to a Jordan boundary, their Riemann maps, normalized at a common interior point and with positive derivative, converge uniformly in the closed-disk parametrization to the limiting chart. When a prescribed ordered cut is to be realized by an ambient change, we give the extension argument explicitly in Lemma 6.3. Paths, band sides and gates are always considered together with their planar embeddings.

1.6 Reading the proof and the order of limits

Section 2 establishes the exact fields and the stationary closed-sum estimate. Section 3 states the percolation inputs and derives the local arm and mesh estimates. Sections 4 and 5 construct the state transport and its affine decomposition. Sections 6 and 7 place plugs on actual rims and prove the bulk transport theorem. Section 8 excludes its degenerate scalar. Section 9 constructs the holomorphic limit and proves Theorem 1.1.

Throughout the transport construction, the microscopic limit $\varepsilon \downarrow 0$ is taken first, at a fixed comparison scale h and for finitely many fixed patterns. Pattern-specific positive conditioning probabilities may be small. Bounds asserted to be uniform of order h use only a fixed finite family of short normalized templates. Spatial refinement depth is also fixed during the microscopic limit and is subsequently increased before h decreases. The final Jordan-domain buffers are fixed until the microscopic crossing limit has been taken. Each argument below specifies which of these limits is in use.

2 Circuit components and orthogonal fields

The finite identity in this section is independent of the geometry of a triangulation. Geometry enters only when we assign displacement vectors to its sides. A component beside a dual circuit, rather than the whole side of that circuit, is the coordinate of the identity. We first obtain the separate second-moment identities for the two displacement fields and then use stationarity to control their complete macroscopic rim sums.

2.1 An exact identity on a triangulated sphere

Fix a finite simple triangulation T of the oriented sphere, and give its vertices independent fair colors. Draw its trivalent dual Γ through the interiors of the triangles. A dual edge is *occupied* when the endpoints of the primal edge it crosses have different colors. At a dual vertex there are either zero or two occupied edges. The occupied subgraph is therefore a union of vertex-disjoint simple circuits.

In fact its law is uniform on the even subgraphs of Γ . To see this, prescribe a set of primal edges across which the spin changes. Spins can be recovered, starting from one vertex, precisely when the number of prescribed changes around every primal cycle is even. The face boundaries generate the cycle space of a spherical triangulation, so it is enough to check the triangular faces. These are exactly the even-degree conditions in the dual. Every allowable prescription has the two spin assignments obtained from one another by interchanging the colors.

For each occupied circuit L , take the connected components C of the topological graph $\Gamma \setminus L$: all points of L , including its vertices, are removed. In particular, an open edge whose two endpoints have been removed can be a component by itself. Figure 1 illustrates the deletion

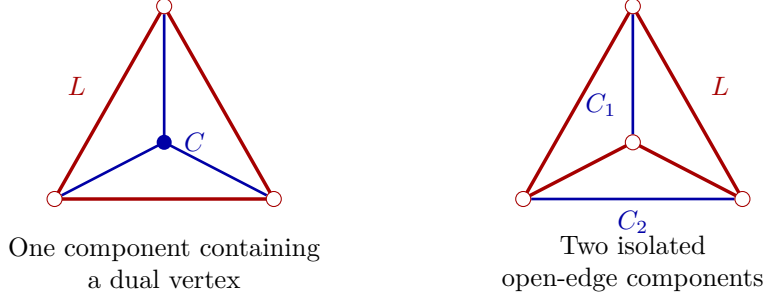


Figure 1: Components in the tetrahedral dual $\Gamma = K_4$. The heavy red circuit L and its hollow vertices are shown for reference but are deleted. The remaining blue graph has one component in the left panel and two in the right panel; its endpoints at deleted vertices are excluded. Components are determined by connectivity in $\Gamma \setminus L$.

convention in the tetrahedral dual. Put $\lambda = (L, C)$ and, for a point m in the interior of a dual edge, define

$$H_\lambda(m) = \mathbf{1}_{\{m \in C\}}.$$

Vectors indexed by these pairs, with the index set taken separately in each configuration, have the scalar product

$$\langle F, G \rangle = \mathbb{E}_{\text{sp}} \sum_{\lambda} F_{\lambda} \overline{G_{\lambda}}.$$

Here and throughout the finite discussion, $\mathbb{E}_{\text{sp}}$ averages colors with the triangulation fixed. Write $H(m) = (H_{\lambda}(m))_{\lambda}$ and $K(m, n) = \langle H(m), H(n) \rangle$.

Lemma 2.1 (Two defects). *For distinct edge-interior points m, n , subdivide the dual edges at these points and take a uniform subgraph with odd degree at m, n and even degree elsewhere. Then $K(m, n)$ is the expected number of circuits in this subgraph. The assertion also applies to distinct points on the same edge.*

Proof. For any even set D of vertices of a connected finite graph, the subgraphs with boundary D , when nonempty, form an affine translate of the cycle space. Toggling a path between two prescribed defects gives such a translation. In particular, the even and two-defect ensembles have the same cardinality; subdividing an edge does not change that cardinality.

In the present two-defect ensemble, ordinary vertices have occupied degree zero or two, while the degree-two marked vertices have occupied degree one. Thus there is one simple defect path and a collection of disjoint circuits. Fix a possible circuit L . When it is occupied, every third edge at a vertex of L is unoccupied. A defect path must consequently lie in $\Gamma \setminus L$. The circuit can occur in the defect ensemble only if m, n lie in the same component of this complement. If they do, fix a path between them in that component. Toggling it bijects even subgraphs containing L with two-defect subgraphs containing L . Hence

$$\mathbb{P}_{m,n}(L \text{ occupied}) = \mathbb{P}_{\emptyset}(L \text{ occupied}) \mathbf{1}_{\{m,n \text{ in one component of } \Gamma \setminus L\}}.$$

Summing over L proves the assertion. If the two marks lie on one edge, its open interior connects them whenever that edge is not in L , even when both endpoints of the edge belong to L . $\square$

At a dual vertex corresponding to a triangle t , choose points $m_j(t)$ on its three spokes, indexed by $j = 0, 1, 2$ in positive cyclic order. Choose the star neighborhoods disjoint. With $\omega = \exp(2\pi i/3)$, set

$$Z_t = \sum_{j=0}^2 \omega^j H(m_j(t)).$$

Proposition 2.2 (Finite orthogonality). *For distinct triangles t, u ,*

$$\langle Z_t, Z_u \rangle = 0, \quad \|Z_t\|^2 = \frac{3}{4}. \quad (2.1)$$

Proof. For the off-diagonal identity, start with uniform oddness at the two dual vertices t, u . Toggle the short segment from t to $m_j(t)$ and that from u to $m_k(u)$. This moves the defects to the two marked points and preserves uniformity. By [Theorem 2.1](#), it suffices to take the $\sum_{j,k} \omega^{j-k}$ Fourier sum of the resulting circuit counts.

Delete the interiors of the two small stars. For a fixed initial configuration, what remains consists of circuits and paths pairing its occupied star ports. Each star has either one or three such ports. The outside circuits contribute a constant. If a star has one occupied port, the modified defect path uses that port regardless of which short segment was toggled, so the circuit count is independent of that star's index. These terms vanish in the Fourier sum.

It remains to consider three occupied ports at each star. There are two possibilities. With one through path, the two other ports at each star are paired by an outside cap. If r_t, r_u are the through ports, the additional circuit count after moving the defects is

$$\mathbf{1}_{\{j=r_t\}} + \mathbf{1}_{\{k=r_u\}}.$$

It is a sum of one-index terms and again has zero double Fourier sum. With three through paths, let π be their matching of ports. An additional circuit is formed exactly when $k = \pi(j)$: the other two through paths then join into one circuit. Disjoint paths between the two boundary circles of the punctured sphere reverse their positive cyclic orders, so $\pi(j) = \ell - j \pmod{3}$ for some ℓ . Their contribution is

$$\sum_{j=0}^2 \omega^{j-\pi(j)} = \omega^{-\ell} \sum_{j=0}^2 \omega^{2j} = 0.$$

This establishes the cancellation for each initial configuration, including when t, u are adjacent.

For the norm, a circuit avoiding t places the three nearby spoke points in one component, whose Fourier sum is zero. If t is occupied, exactly one circuit passes through it. The two used spokes belong to that circuit, while the missing spoke belongs to exactly one complementary component. Thus Z_t has precisely one nonzero coordinate, of modulus one, in this case. The three distinct primal vertices of t are not all the same color with probability $3/4$, proving the second identity. Equivalently, any diagonal kernel in this calculation can be evaluated using two distinct points of the same spoke, which have identical H -vectors; coincident defects are never needed. $\square$

2.2 The two displacement fields

Suppose t is drawn as a straight triangle in an oriented plane chart. Let z_t be its barycenter, and let d_{tj} be the complex side vector dual to spoke j , traversed clockwise around t . Since the

three side vectors sum to zero, there are unique $p_t, q_t \in \mathbb{C}$ such that

$$d_{tj} = p_t \omega^j + q_t \bar{\omega}^j, \quad p_t = \frac{1}{3} \sum_j d_{tj} \omega^{-j}, \quad q_t = \frac{1}{3} \sum_j d_{tj} \omega^j. \quad (2.2)$$

Applying the real affine map from an equilateral triangle to t gives

$$|p_t|^2 - |q_t|^2 = \frac{4}{\sqrt{3}} \text{Area}(t), \quad |p_t| + |q_t| \leq C \text{diam}(t). \quad (2.3)$$

Indeed the two Fourier coefficients are the complex-linear and antilinear parts of this map, whose Jacobian is their squared-modulus difference; the equilateral case fixes the constant and sign.

A triangle t is *tagged* for $\lambda = (L, C)$ by j if L uses t and its missing spoke j belongs to C . For a coefficient function g supported where the side vectors are defined, put

$$\begin{aligned} U_\lambda(g) &= \sum_{t \text{ tagged by } j} g(z_t) p_t \omega^j, \\ Q_\lambda(g) &= \sum_{t \text{ tagged by } j} g(z_t) q_t \bar{\omega}^j, \\ V_\lambda(g) &= U_\lambda(g) + Q_\lambda(g). \end{aligned}$$

Changing the first spoke index changes the Fourier coefficients by the inverse phases, so the summands are well defined. A rotation of the chart multiplies each of the two summands by the corresponding complex rotation.

The λ -coordinate of Z_t is exactly ω^j when t has that tag, and is zero otherwise. Its conjugate gives the other phase. Consequently Equation (2.1) gives the exact identities

$$\mathbb{E}_{\text{sp}} \sum_\lambda |Q_\lambda(g)|^2 = \frac{3}{4} \sum_t |g(z_t) q_t|^2, \quad \mathbb{E}_{\text{sp}} \sum_\lambda |U_\lambda(g)|^2 = \frac{3}{4} \sum_t |g(z_t) p_t|^2. \quad (2.4)$$

In particular,

$$\mathbb{E}_{\text{sp}} \sum_\lambda |V_\lambda(g)|^2 \leq \frac{3}{2} \sum_t |g(z_t)|^2 (|p_t|^2 + |q_t|^2). \quad (2.5)$$

We call these the *Bessel bounds*. The coefficients may be arbitrary complex numbers depending on the graph and its geometry, but must be independent of its colors. Continuity is not required. In particular, these bounds apply to deterministic spatial masks. The sums include every component, including isolated open edges.

2.3 Rims and their extremality

Lemma 2.3 (Component rims). *If C contains a dual vertex, the union of the closed primal triangles whose centers belong to C has one simple monochromatic boundary circuit P . Orient P with C on its left. Its right-hand triangles are exactly the tags of (L, C) , one per edge, and their facing third vertices belong to one cluster of the opposite color. The oriented circuit P determines (L, C) .*

If C contains no dual vertex, it is an isolated open edge. It has exactly two tags. When they are in one plane chart, their contributions satisfy

$$V_{(L,C)}(g) = (g(z_t) - g(z_u)) d_{te}, \quad V_{(L,C)}(1) = 0, \quad (2.6)$$

where t, u are its endpoint triangles and e their common missing side.

Proof. The triangles with centers in C are connected through shared sides. A boundary side separates such a triangle from a triangle visited by L ; the latter's missing spoke leads into C . This identifies the boundary sides with the tags.

At a primal vertex, the sectors belonging to C form one cyclic interval, unless they fill its entire neighborhood. Otherwise choose two separated runs of C -sectors. Each intervening interval contains an L -sector: an immediately adjacent untouched triangle would still connect to C . Extend two selected C -sectors and two intervening L -sectors to alternating points of a sufficiently small circle about the primal vertex, using short segments inside the sectors. Outside that circle they are joined in the two disjoint connected sets C and L . Two disjoint connected sets in the complementary disk cannot join alternating boundary contacts, a contradiction.

It follows that the union of the closed C -triangles is a connected subsurface of the sphere with disjoint simple boundary circuits. Each boundary circuit faces a different complementary disk. The connected circuit L avoids this subsurface and faces it across every boundary side, so there can be only one such disk and one boundary circuit. This is P . The missing side at an occupied triangle has equal-colored endpoints; hence all vertices of P have the same color. Following either side of a disagreement circuit gives a connected walk of constant color: within a visited triangle the adjacent site either stays the same or changes across an edge joining two sites of that color. The vertices facing P therefore lie in one opposing cluster.

Any directed edge of P identifies its occupied right triangle and thus L ; its left triangle identifies C . Finally, a component with no dual vertex must be one open edge with both endpoints on L . Those endpoints give its two tags, whose side vectors are opposite. This proves (2.6). Their separate U - and Q -contributions need not vanish. $\square$

We call P an oriented *rim* and write U_P, Q_P, V_P for the corresponding fields. If its entire drawing is in plane coordinates, $V_P(1)$ is the sum of its oriented edge displacements, and hence is zero. Restricting (2.4) to genuine rims gives inequalities; the equalities themselves retain the isolated components.

Lemma 2.4 (No right bypass and its converse). *A rim has no path of its own color joining two distinct rim vertices with all its other points in the open right-hand disk. Conversely, a simple monochromatic circuit with this property, and with an opposite-colored site in its right-hand disk, is a rim.*

Proof. An alleged bypass may be made simple. It is a crosscut of the right-hand disk and separates its two nonempty rim arcs. Choose an edge on each arc. Its facing triangle lies on that arc's side of the crosscut, since a primal graph path cannot cross the interior of a face. Its opposing vertex does not lie on the bypass. Thus the two facing vertices are in different parts of the cut disk. Their opposing cluster can cross neither the monochromatic rim nor the monochromatic bypass, contradicting Theorem 2.3.

For the converse, let uv be an edge of the given circuit and w the third vertex of its right triangle. If w had the circuit color and lay strictly on the right, u, w, v would be a bypass. If w belonged to the circuit, one of uw, vw would be a right-side chord, unless the right disk were that single triangular face. The latter possibility has no opposite-colored site and is excluded by hypothesis. Every facing vertex is therefore of the opposite color.

Choose the occupied dual circuit L through one facing triangle. It cannot cross the monochromatic circuit, so it stays on its right. The triangles on the left are untouched and form part of one component C off L . Its rim P^* includes the chosen edge and lies in the closed right side of the original circuit, since its left subsurface contains the original left disk. If the two circuits differed,

a segment of P^* between successive contacts with the original circuit would be a right bypass. Such a segment has distinct endpoints by simplicity: follow P^* from the common edge to its first departure and then to its next contact. This contradicts the hypothesis, proving equality. $\square$

Lemma 2.5 (Sandwich). *In an annular band, suppose that there are disjoint essential simple black and white circuits, with the white circuit on the right of the black one for the direction of the band. There is a black rim, with this direction, in the closed region between them.*

Proof. Take a maximal biconnected block of the black graph containing the given black circuit. The face of this block toward the white circuit has a simple boundary. That boundary lies between the two given circuits: the block contains the black one and cannot cross the white one. A black bypass in this face would add an ear between two distinct vertices of the block and enlarge the block, contrary to maximality. The white circuit supplies an opposite-colored site in the face. Apply [Theorem 2.4](#). In the full plane the same argument uses the finite black cluster containing the given circuit. $\square$

2.4 Infinite volume and closed sums

The local estimates of [Section 3](#) imply the following facts for Poisson–Voronoi percolation. There are almost surely no infinite monochromatic clusters: the one-arm bound tends to zero as the outer radius tends to infinity, and a countable collection of inner boxes covers the plane. An infinite disagreement wall would have infinite connected site chains on both its sides, so every occupied wall is finite as well.

The dual graph has one end. Indeed, enclose any finite removed part and all its incident triangles in a compact set. Plane paths outside a larger compact set, avoiding primal vertices, connect sufficiently remote triangles by face adjacency without meeting the removed part. All such triangles are therefore in one component. This observation also shows that, for a fixed finite circuit L , component relations among finitely many points off L stabilize in sufficiently large spherical completions.

By [Theorem 3.3](#), increasing simple primal cycles can be chosen around any fixed compact set. Keep their interiors and triangulate the outside to obtain finite simple spherical triangulations, with independent fair colors on the additional vertices and the original colors inside. For any fixed finite set of triangles, their occupied walls eventually agree with the full-plane walls, and the preceding component relations agree. The bounded quantities

$$\sum_{\lambda} Z_{t,\lambda} \overline{Z_{u,\lambda}}$$

therefore converge. Conditional on almost every point configuration this holds with color probability one, by the absence of infinite clusters and Fubini’s theorem. Passing to the limit proves (2.1)–(2.5) for finite local tests in the full plane, and then for their square-summable limits. This uses only local tag products, not finiteness of the norm of an individual infinite-volume vector $H(m)$. The same identities hold in any bulk patch of a spherical completion, for weights supported where its geometry is defined. The rim statements follow either by this exhaustion or by compactifying the plane around a finite wall.

Proposition 2.6 (Closed-sum tail). *For every fixed bounded box $B \subset \mathbb{C}$ and every $a > 0$, the full-plane model of spacing ε satisfies*

$$\lim_{\varepsilon \downarrow 0} \mathbb{E} \sum_{\substack{P \text{ a rim, } P \subset B \\ \text{diam}(P) \geq a}} |Q_P(1)|^2 = 0. \quad (2.7)$$

The corresponding full sums for V_P vanish identically.

Proof. Work first at spacing one. Anchor each rim at the average of its vertex positions, denoted $b(P)$. This choice is translation equivariant. For $s > 0$, retain only those rims all of whose tagged barycenters are within distance s of their anchor, and define the stationary random measure

$$\mu_s = \sum_{P \text{ retained}} |Q_P(1)|^2 \delta_{b(P)}.$$

For anchors in a square W_R of side R , the full sum of every retained rim equals its sum against $\mathbf{1}_{W_R^{+s}}$, where W_R^{+s} is the s -neighborhood of the square. Thus (2.4) and stationarity give

$$\mathbb{E} \mu_s(W_R) \leq \frac{3}{4} \mathbb{E} \sum_{z_t \in W_R^{+s}} |q_t|^2 = c_q \text{Area}(W_R^{+s}).$$

The finite constant c_q exists by Theorem 3.3. Dividing by R^2 and letting R grow shows that the intensity of μ_s is at most c_q , uniformly in s . Every rim has finitely many tags, so monotone convergence gives a finite intensity for

$$\mu = \sum_P |Q_P(1)|^2 \delta_{b(P)}.$$

Let $\theta(M)$ be the intensity restricted to rims of diameter at least M . Since each rim is finite and μ has finite mean on a unit square, $\theta(M) \rightarrow 0$ as $M \rightarrow \infty$.

Rescaling a spacing-one configuration by ε multiplies each Q sum by ε and anchor area by ε^2 . Consequently, the expected squared mass of spacing- ε rims of diameter at least a , with anchors in B , is $\text{Area}(B)\theta(a/\varepsilon)$. A rim contained in the box has its anchor in the box, so this bounds the expression in (2.7). Finally, $V_P(1)$ is the tangent sum around a closed polygon and is exactly zero. $\square$

The restriction to complete rims in (2.7) is essential to its proof. The following sections will turn this closed-sum estimate into a statement on open pieces by comparing configurations with common endpoint data.

3 Probability input, local geometry, and two arm consequences

We separate the established percolation input from the consequences that will be needed for rims. All the input concerns the full-plane, unit-intensity Poisson process with independent fair colors. Write $\mathbb{P}^\eta$ for the color law conditional on the point set η .

3.1 The percolation input and its scope

For $1 \leq r \leq R$, let $A_j(r, R)$ be the arm event across the square annulus $[-R, R]^2 \setminus (-r, r)^2$. For even j the j arms have alternating colors in cyclic order. For odd $j > 1$ there are $j - 1$ alternating arms and an additional black arm such that no Voronoi cell intersects both that arm and any other arm. The event A_1 is the existence of a black arm. In these definitions paths can be chosen so that their used site sets are disjoint. Set

$$\alpha_j(r, R) = \mathbb{P}(A_j(r, R)), \quad p_j^\eta(r, R) = \mathbb{P}^\eta(A_j(r, R)).$$

Proposition 3.1 (Established input). *The following estimates hold at criticality.*

(i) *For each fixed rectangle aspect ratio, both colors have crossing probabilities bounded below by a positive constant, uniformly in its size. At sizes tending to infinity, the corresponding quenched crossing probabilities are bounded below with probability tending to one. More precisely, for each fixed finite family of buffered rectangle chains on fixed square grids, and each $\gamma > 0$, the required quenched crossing lower bounds hold outside a geometry event of probability $O_\gamma(R^{-\gamma})$ when the family is dilated by R . The positive lower bound may depend on the family and on γ .*

(ii) *For every fixed $j \geq 1$, there is $C_j < \infty$ such that, for all $1 \leq r \leq R$,*

$$\mathbb{E}[p_j^\eta(r, R)^2] \leq C_j \alpha_j(r, R)^2. \quad (3.1)$$

(iii) *There are constants $c_1, c_4 > 0$ and $C < \infty$ such that, for all $1 \leq r \leq R$,*

$$\alpha_1(r, R) \leq C(r/R)^{c_1}, \quad \alpha_5(r, R) \leq C(r/R)^2, \quad \alpha_4(r, R) \leq C(r/R)^{1+c_4}. \quad (3.2)$$

These estimates are due to Tassion [19], Ahlberg–Griffiths–Morris–Tassion [1], and Vanneuville. The annealed scaling relations, quasi-multiplicativity, and the five-arm estimate originate in [20]. For the full-plane Poisson estimates above we use [22, Theorems 1.2–1.4]; the arbitrarily polynomial exceptional-geometry estimate for the indicated grid families is [22, Proposition 3.14], followed by finite gluing. The powers in (3.2) are, respectively, [22, (3.1), Proposition 3.4(iii), and Proposition 6.2]. Only fixed relative grid meshes and a fixed finite list of buffered shapes are used in applying Proposition 3.14. Its minimum-size condition is then met after dilation. These assertions do not give crossing lower bounds for arbitrary conditioned point processes.

We will use positive association and BK for increasing events conditional on η . We also need disjoint occurrence for events with prescribed colors of both kinds. For completeness, the required consequence of Reimer’s theorem has the following elementary reduction.

Lemma 3.2 (Disjoint witnesses at fixed geometry). *For events A, B in a finite product of fair bits, let $A \sqcap B$ mean that there are disjoint sets of coordinates certifying A and B . Then*

$$\mathbb{P}(A \sqcap B) \leq \mathbb{P}(A)\mathbb{P}(B). \quad (3.3)$$

In particular this inequality applies to bounded percolation tests conditional on the Voronoi geometry, with disjoint used site sets.

Proof. Reimer’s fair-bit inequality [15] states that $|A \sqcap B| \leq |A \cap \overline{B}|$, where $\overline{B} = \{\mathbf{1} - b : b \in B\}$ denotes bit complementation, not set complementation. Apply it on slices of pairs of configurations. For each pair $(x, y) \in \{0, 1\}^n \times \{0, 1\}^n$, fix the set K on which they agree and their common values there. On the other coordinates the two strings are complementary. If $x \in A \sqcap B$, its disjoint certificates, restricted to the coordinates outside K , certify the two restricted events on that slice. Reimer’s inequality bounds the number of such pairs in the slice by the number with $x \in A$ and $y \in B$. Summing over the slices gives $2^n |A \sqcap B| \leq |A||B|$, which is (3.3). A fixed bounded region meets only finitely many Voronoi cells almost surely, so the conditional percolation tests indeed use a finite product. $\square$

No annealed disjoint-occurrence product bound is asserted: averaging the right side of (3.3) gives the expectation of a product of quenched probabilities. We will control that product using (3.1).

3.2 Localization, moments, and buffered paths

Lemma 3.3 (Poisson mesh estimates). *The following facts hold for the full-plane model of spacing ε .*

- (i) *On a fixed compact set, the Delaunay triangulation and the Voronoi cells meeting it are determined by the points in any fixed positive collar, outside an event of probability tending to zero. Their local diameters tend to zero in probability. The same determination statement holds when the outside point configuration is changed while the collar is retained.*
- (ii) *For a unit square at spacing one, the probability that an incident triangle has circumradius greater than $R > 2$ is at most Ce^{-cR^2} . The same form of bound holds for a Voronoi cell meeting the square whose site has distance greater than R from it. Counts of incident triangles and sums of any fixed polynomial diameter weights have moments of all orders.*
- (iii) *If S is a square of side s , $k \geq 0$, and $1 \leq p < \infty$, then*

$$\left\| \sum_{t \cap S \neq \emptyset} \text{diam}(t)^k \right\|_{L^p} \leq C_{p,k} \varepsilon^{k-2} (s + \varepsilon)^2. \quad (3.4)$$

For a deterministic Borel set E of finite area,

$$\mathbb{E} \sum_{z_t \in E} \text{diam}(t)^k = m_k \varepsilon^{k-2} \text{Area}(E), \quad m_k < \infty. \quad (3.5)$$

At every fixed $s > 0$, the sum of $\text{diam}(t)^2$ over triangles with barycenters in S converges in every finite moment norm to $m_2 s^2$ as $\varepsilon \downarrow 0$. The sum over triangles meeting S has the same limit. The corresponding statements hold for translation-covariant weights of degree two bounded by a constant times $\text{diam}(t)^2$, such as $|p_t|^2$ and $|q_t|^2$.

- (iv) *Within fixed positive margins, locally determined simple primal cycles can approximate a fixed circle arbitrarily closely and in its cyclic order. Finitely many separated circles admit disjoint such cycles.*

Proof. If an empty disk of radius at least ρ contains a point of a region K , it contains an empty subdisk of radius comparable to ρ with center in an $O(\rho)$ -collar of K . A grid of mesh a small multiple of ρ reduces this to finitely many empty-disk tests. The Poisson void probability of each test is at most $\exp(-c\rho^2/\varepsilon^2)$, and the number of tests for a bounded region is at most a constant times $(1 + \text{diam}(K)/\rho)^2$.

A triangle meeting K has an empty circumdisk containing its meeting point. The same observation applies to a cell meeting K with a remote site. Excluding the above subdisks makes all relevant circumdisks small and places them inside the retained collar. Both the positive empty-circle tests defining those triangles and the exclusion of larger ones are then local; they remain valid after any change outside that collar. This proves (i), including shrinking collar radii much larger than $\varepsilon\sqrt{\log(1/\varepsilon)}$, and yields the exponential bounds in (ii).

On the event that all incident circumradii are at most R , the relevant vertices lie in a box of side $O(R)$. The incident triangle count is bounded by a polynomial in the number of Poisson sites there, and every diameter weight is bounded by a polynomial in R . Splitting into successive values of R , the exponential tail and the Poisson count moments give all the asserted moments. Tile S by $O((s/\varepsilon + 1)^2)$ squares of side ε . Scaling the unit-square estimate and using the triangle

inequality in L^p proves (3.4); no independence between adjacent squares is needed. Stationarity and scaling give (3.5).

For a barycenter mask, the spatial ergodic theorem for the Poisson process, applied at spacing one to expanding squares, proves the degree-two limit after rescaling. The higher moment bounds imply convergence in every finite moment norm. Triangles meeting the square but with barycenter outside it are contained in a thin boundary neighborhood on a good mesh event. First fix the width of that neighborhood, use the same moment bound there, and then let the width tend to zero; the bad mesh event is removed by higher moments and its vanishing probability. This proves the statement for intersecting triangles and for the other indicated weights.

Finally, in a thin collar of a circle, follow successive triangles around the circle and use their primal edges to form a walk with nonzero winding. On a sufficiently fine mesh the walk stays in the collar and follows the circle in order. Decomposing this walk into simple cycles gives a simple cycle with nonzero winding. Choosing the collar within a prescribed margin gives (iv), with simultaneous locality for any fixed finite list. $\square$

At spacing ε , write $A_j^\varepsilon(r, R)$ for the corresponding physical arm event, $p_{j,\varepsilon}^\eta(r, R)$ for its quenched probability, and $\alpha_j^\varepsilon(r, R) = \alpha_j(r/\varepsilon, R/\varepsilon)$ for its annealed probability. The arm estimates apply whenever $r \geq \varepsilon$. Thus every fixed positive inner radius eventually satisfies their unit-scale lower cutoff. Translation and rotation are allowed by invariance of the Poisson law. Tests in regions with disjoint positive collars use independent point and color data after the local determination of Theorem 3.3, up to a vanishing mesh exception.

Primal vertex paths and paths through the corresponding Voronoi cells give the same buffered crossing tests. For the direction from vertices to cells, join sites to interior points of their shared cell sides, inside the convex cells. This gives an equivalent embedded drawing with the same planar orders and vanishing displacement on a good mesh event. Vertex-disjoint paths can be routed through just their own cells and the open sides shared by successive cells; no other cells need be used or intersected. Conversely a cell crossing gives a connected chain of its sites. Fixed margins absorb the displacement in both directions. Lower bounds in an open corridor are obtained by a finite chain of overlapping fixed-shape rectangle crossings, with enough overlap and margin to concatenate them. Positive association conditional on η and Theorem 3.1 give a quenched positive lower bound with high probability for each such fixed chain.

3.3 A planar color-order fact

Color-switching comparisons among prescribed arm colors are classical; for triangular-lattice half-plane and annular versions see [11, Propositions 19–20]. The rectangular graph statement needed here is proved directly below, so it remains valid after deleting vertices and does not invoke a lattice theorem on Voronoi geometry. We also use an elementary separation fact for triangulated disks. Boundary sides below include their endpoints. A path between opposite sides can always be trimmed to be *proper*, meeting those sides only at its first and last vertices.

Lemma 3.4 (Hex and indicator colorings). *In a triangulated disk with four successive vertex sides, either there is a black connection between the first and third sides or there is a white connection between the second and fourth sides. In particular, if a set of vertices X blocks every path between one pair of opposite sides, then X contains a path between the other pair.*

Proof. Attach four exterior vertices, successively colored black, white, black, white, and fan each onto its side; close the exterior to a sphere. The four color switches give four boundary ends of

disagreement strands. Their noncrossing pairing forces a connection between the two black fans or between the two white fans. Following the adjacent sites of a strand gives the corresponding primal connection inside the disk. For the last assertion, color *exactly* the vertices of X black and every other vertex white. A white crossing of the blocked pair is impossible, so the other crossing uses vertices of X only. $\square$

Lemma 3.5 (Ordered colors in a rectangle). *Let a finite planar graph be drawn in a rectangle, with independent fair colors on its vertices. For any $k \geq 1$, the probability of having k vertex-disjoint crossings between its two end sides, with a prescribed linear sequence of colors, depends only on k and not on that sequence. The assertion also holds for a planar subgraph with deleted vertices.*

Proof. Fan the two end sides to two uncolored exterior terminals. Crossings become simple terminal-to-terminal paths, internally disjoint and ordered from left to right. Conversely, first-hit and last-hit trimming recovers proper crossings in the same order.

Fix a first color. If there is a path of that color, add an artificial terminal-to-terminal edge on the exterior right. The union of this edge with all simple paths of the chosen color is biconnected: it is a union of simple cycles sharing that edge. Its left outer boundary is therefore a simple path of the chosen color. Call it the leftmost path. It has no same-colored bypass strictly on its left between distinct vertices, including the terminals. Conversely this property characterizes it. Indeed a portion of any other such path lying strictly to its left, between successive contacts, would be a bypass. Uniqueness also follows at the first divergence of two candidates, using the order at the start terminal or the order from their common incoming edge.

The event that a specified path is leftmost is determined by its own spins and the spins on its left. All vertices strictly to its right retain independent fair colors. The leftmost path can replace the first path of any ordered disjoint family: it lies on or to the left of that path, while all the others lie strictly to its right. Delete its internal vertices and apply induction in the graph on its right. For a fixed first color this shows that the remaining color sequence does not affect the probability. A global color interchange changes the first color as well. The proof never uses the presence of every vertex of the original rectangle, so also applies to a subgraph. $\square$

3.4 Four arms with both colors

The next two estimates are local statements. Let x be deterministic and $0 < s < S$ fixed before ε tends to zero. In a bulk patch assume a neighborhood of $\bar{B}(x, S)$ is within the region of agreement with the full-plane model. All local comparisons below allow fixed changes of the radii by positive constant factors. The constants do not depend on x, s, S , and bounded ratios S/s are covered by increasing them. The statements make no assertion for a center chosen from the colors or for an arbitrary microscopic choice $s = s_\varepsilon$.

Lemma 3.6 (Mixed-color four arms). *For some $c > 0$ and $C < \infty$,*

$$\limsup_{\varepsilon \downarrow 0} \mathbb{P} \left(\begin{array}{c} \text{four vertex-disjoint primal arms across } B(x, S) \setminus B(x, s), \\ \text{with both colors represented} \end{array} \right) \leq C(s/S)^{1+c}. \quad (3.6)$$

The four colors need not alternate.

Proof. We may take S/s large. Choose locally determined primal boundary cycles strictly inside the annulus and trim the four witnesses to disjoint proper paths between them. Every step

is made on a good mesh event; its complement has vanishing probability at these fixed scales. Select an adjacent pair of the four paths of opposite colors. In their gap there is a spanning disagreement strand γ : no strand can leave through a monochromatic side path, and the odd number of switches at each annular end forces a pairing from one end to the other. Stop γ at its boundary edges. The sites of its visited triangles are disjoint from the two remaining paths, also at the ends because all four paths are proper.

Sum over each fixed geometric candidate for γ and each consistent spin pattern on its visited sites. Its occurrence is determined by those spins. Conditional on them, delete those sites and slit the annulus along the strand. The remaining sites are free fair bits in a planar rectangle, and the two avoiding paths have a linear order. By [Theorem 3.5](#), the probability of the union of their four possible color sequences is at most four times the probability of any specified sequence. Choose the sequence compatible with alternating four arms. The two missing arms are supplied by the opposite-colored banks of γ . The sites beside each bank form a connected arm, are disjoint from the two avoiding paths, and lie on their respective extreme sides in the slit: they connect to the strand through its visited triangles without crossing either avoiding path. The resulting four arms thus have the required cyclic order.

Let N count proper spanning wall strands in the working annulus, each stopped at its first boundary edges and counted once. These are the spanning components after the walls are cut at all contacts with the two boundary cycles. Let G be the geometry event on which localization holds and a fixed buffered subannulus has quenched white-circuit probability at least $\delta > 0$. A fixed finite rectangle chain supplies this circuit bound by [Theorem 3.1](#). In particular, $\mathbb{P}(G^c) = o(1)$ for each fixed s, S , and δ is independent of S/s . The preceding sum over strands gives

$$\mathbb{P}(\text{mixed four arms}) \leq o(1) + C\mathbb{E}[\mathbf{1}_G N \mathbf{1}_{A_4^\varepsilon(r,R)}], \quad (3.7)$$

where $r \asymp s$, $R \asymp S$ are deterministic buffered square radii. To make this final transfer, first stop all the cell paths at their first hit of the square outer boundary and then trim their starts at their last hit of the inner boundary. Planar order and disjoint used site sets are preserved. The summation is over all fixed strands; no selection of a globally first strand is used when claiming that the remaining bits are free.

On G , the variable N has conditional moments of every fixed order, uniformly in the annulus ratio. Indeed its disjoint spanning strands have adjacent black banks. A gap between consecutive strands can account for at most the two black banks bordering it; black paths in different gaps cannot meet or cross a strand. This also separates their endpoint sites: a stopped strand ends inside a boundary edge between opposite-colored sites, and each black boundary site belongs to only one gap. Thus $N \geq 2k$ yields at least k black crossings with disjoint site witnesses, including their endpoints. A black crossing of the working annulus is incompatible with the white circuit in the fixed subannulus, so its conditional probability is at most $1 - \delta$. BK gives

$$\mathbb{P}^\eta(N \geq 2k) \leq (1 - \delta)^k \quad \text{on } G, \quad \mathbb{E}[\mathbf{1}_G N^q] \leq C_{q,\delta} \quad (q < \infty).$$

Keeping the indicator of G in (3.7), Hölder's inequality and (3.2) bound its second term by

$$C_q \mathbb{P}(A_4^\varepsilon(r, R))^{1-1/q} \leq C_q (s/S)^{(1+c_4)(1-1/q)}.$$

Choose q large enough that the displayed exponent is greater than one, and reduce c accordingly. Taking the microscopic limit proves the claim. No estimate of N on bad geometry is required. $\square$

3.5 Two separated returns of a rim

Lemma 3.7 (Near-return bound). *For some $c > 0$ and $C < \infty$, and the same fixed-scale convention,*

$$\limsup_{\varepsilon \downarrow 0} \mathbb{P} \left(\begin{array}{c} \text{some rim has two visits to } B(x, s) \text{ such that both} \\ \text{complementary rim arcs between the visits leave } B(x, S) \end{array} \right) \leq C(s/S)^{2+c}. \quad (3.8)$$

The rim can be part of a global spherical completion. The event is bounded by a local monochromatic-arm test, without a union over rims.

Proof. Suppose first that the rim P is black. For a large enough ratio S/s , take four nested primal cycles near radii $2s, 5s, S/5, S/2$. They are locally determined and pairwise disjoint on a good mesh event. The two visits and the excursions in both rim directions give four vertex-disjoint black journeys between the cycle near $2s$ and the cycle near $S/2$. Trim them properly and denote them by $P_1, \dots, P_4$ in their cyclic order. The margins at both ends ensure that the vertices between the middle cycles are interior vertices of the original rim journeys, with both incident rim edges still present. Retain these four complete journeys throughout the argument.

Each oriented journey has its right side in one of the two adjacent annular gaps. Since a gap has only two sides, at least two distinct gaps have a right-facing rim side. Fix one such gap. The vertices of the inner middle cycle block every connection between its annular ends. Apply the indicator-coloring assertion of [Theorem 3.4](#), coloring exactly those vertices black. It gives a transverse crossing between the two journey sides, using only vertices of that cycle. Trim this to a proper crosscut. Next work in the subdisk between this crosscut and the outer end. The outer middle cycle still separates the two ends, because the first crosscut lies strictly inside it. The same indicator argument gives a second proper transverse crosscut using its vertices. This successive construction orders the crosscuts even when the original journeys make extra visits to either middle cycle.

Consider the quadrilateral between these crosscuts. There is no actual black crossing between its two journey sides. Otherwise take a simple such crossing and start just after its last contact with the right-facing side. Both incident rim edges at that departure vertex belong to the original journey; the new edge therefore enters the open right sector of the whole rim. Follow the crossing only until its first subsequent contact with the *whole* rim, including any other rim portions in the quadrilateral. Such a contact exists because the other selected side is part of the rim. Simplicity makes the two contact vertices distinct. Until this contact the path stays in the right disk and has no rim vertices in its interior, so it is a bypass forbidden by [Theorem 2.4](#). This argument does not require the quadrilateral itself to be contained in the smaller annulus between the middle cycles. Hex in the actual coloring now gives a white crossing between the two transverse cuts. Trim it properly between the middle cycles.

Doing this in two distinct original gaps gives two disjoint white arms: the boundaries shared by original gaps are black journey vertices, which the white paths cannot use. These white arms divide the middle annulus into two interval quadrilaterals. Arbitrarily trim the original black journeys to obtain four disjoint black arms in this annulus. It remains to ensure that both white intervals contain a black arm; arbitrary trims alone need not ensure this if a journey revisits a middle boundary.

For this purpose let X be the set of vertices of all four *original* journeys, restricted to one of the white intervals. Every path between its two white sides must meet X . Otherwise it would join two distinct original gaps while avoiding all four paths that separate those gaps in the larger

working annulus. Apply [Theorem 3.4](#) with the artificial black set exactly X . It gives a crossing between the middle cycles using sites of X and hence genuinely black sites. The same argument works in the other white interval. If the four arbitrary black trims already occupy both intervals, retain them. If all lie in one, retain three and add the newly obtained black crossing in the other. The white arms separate the intervals, so the four chosen black paths are still vertex-disjoint. The new crossing need not be a subpath of a single original journey; membership of its sites in X is enough.

We now have four black arms and two white arms, with at least one black arm in each interval between the whites. Choose one black arm in each interval to obtain alternating four, add one of the two remaining black arms as the extra arm for A_5 , and reserve the last black arm for A_1 . These witnesses use disjoint site sets. Transfer all six paths to deterministic buffered square boundaries, stopping at first outer hits and last inner hits to preserve their order. The event implies $A_5^\varepsilon(r, R) \sqcap A_1^\varepsilon(r, R)$ for $r \asymp s$, $R \asymp S$.

Conditional on η , [Theorem 3.2](#) bounds its probability by $p_{5,\varepsilon}^\eta(r, R)p_{1,\varepsilon}^\eta(r, R)$. Averaging and applying Cauchy–Schwarz and (3.1) gives

$$\mathbb{E}[p_{5,\varepsilon}^\eta(r, R)p_{1,\varepsilon}^\eta(r, R)] \leq C\alpha_5^\varepsilon(r, R)\alpha_1^\varepsilon(r, R) \leq C(s/S)^{2+c_1}.$$

For a white rim interchange the colors, at a cost of a factor two. Remove the mesh exception and take the limit superior. The global rim was used only for no-bypass; the resulting arm event is local, so the same bound covers existence of any such rim in a bulk completion. $\square$

Taking the smaller of the positive exponents in [Theorems 3.6](#) and [3.7](#), we henceforth use one constant $c > 0$ for both estimates, reducing it further when a later Hölder inequality requires this.

3.6 Facing arms and two-site gates

A rim that visits a small mesh-good ball and reaches a much larger distance gives an arm of its own color. It also gives an arm of the opposite color from nearby to a buffered large distance. Indeed follow the facing vertices in the triangles along the relevant rim journey. By [Theorem 2.3](#) these form a connected chain in one opposing cluster, and the mesh bound keeps them close to the journey. Only that local part of the chain is needed.

Lemma 3.8 (Two-site gates). *Let a finite triangulated annulus have disjoint simple primal boundary cycles. If the maximal number of vertex-disjoint black paths joining its two boundary cycles, including disjoint endpoints, is at most two, then the closed annulus contains an essential simple primal circuit with at most two black vertices. The color-interchanged assertion also holds. If the working annulus has positive margins inside a prescribed collar, the resulting circuit lies within that collar.*

Proof. Vertex Menger gives a set of at most two black vertices meeting every black crossing. Boundary vertices are allowed in this cut. Temporarily recolor these vertices white; there is now no black crossing of the annulus. Attach a black cap vertex to each boundary cycle, fanning it onto that cycle. The two cap vertices are not black-connected.

An occupied dual circuit separates the caps. To see this directly, sites lying in the same component off all disagreement walls are joined in their color: a curve in that component, made transverse to dual edges, crosses only unoccupied dual edges and hence only equal-colored primal adjacencies. Since the caps are not connected, at least one wall separates them. The neighboring

white walk along that wall uses original annular sites and is deformable to the wall while avoiding the caps. It therefore has nonzero annular winding. Decompose the walk into simple cycles and retain one with nonzero winding; it is an essential white circuit. On restoring the original colors, only the cut vertices can become black, so the circuit has at most two black vertices. Applying the construction in a working annulus with the stated margins gives the last assertion. $\square$

4 Opening closed sums with local plugs

The closed estimate (2.7) does not directly control a proper subarc of a rim. We shall close such subarcs in controlled local experiments, compare the resulting closures, and eventually transfer the comparison back to the original configuration. The conclusion is the following bulk statement.

Theorem 4.1 (Bulk transport). *From every sequence $\varepsilon \downarrow 0$ one can extract a subsequence and a complex number a , depending only on that subsequence, such that $\operatorname{Re} a < 1/2$ and the following holds in every bulk patch. If $f \in C_c^\infty(B(x, R))$ and $\overline{B}(x, 6R)$ lies in the patch, then*

$$\max_{P: P \not\subset B(x, 4R)} |Q_P(f) - aV_P(f)| \xrightarrow[\varepsilon \rightarrow 0]{\mathbb{P}} 0. \quad (4.1)$$

As in the definition of a bulk patch, local tests may be set to zero on a geometry event whose probability tends to zero. The subsequence and scalar are common to all patches and tests.

The transport estimate is completed in Section 7; Proposition 8.1 proves the strict inequality for a . The output of this section is Proposition 4.11: for each fixed connector, its contribution is asymptotically a function of two retained states, and these functions have small cycle sums. Here and in Section 5, X denotes either Q or V .

Convention 4.2 (The order of limits). A comparison scale $h > 0$ is fixed before the microscopic limit. Every $o(1)$ and $o_{L^1}(1)$ in this section and the next is taken as $\varepsilon \downarrow 0$ at fixed h and fixed deterministic geometric pattern, through values for which the stated types are admitted. Later we use a countable sequence $h \downarrow 0$. Thus a finite pattern may have an arbitrarily small positive conditioning probability, and the number of patterns may depend on h . Bounds of the form Ch will instead be obtained from fixed normalized short templates, with a common constant.

4.1 Types, states, roots and gates

Normalized coordinates are multiplied by h and placed in a chosen oriented Euclidean frame. We usually retain the same symbols for the placed sets. The following definition is for a black path; exchanging the two colors gives the white types.

Definition 4.3 (A plug type). A type consists of the following deterministic data:

- (i) two concentric gate annuli $[a_1, b_1]$, $[a_2, b_2]$ and a cutoff radius m , with

$$10 < a_1 < b_1 < m/10, \quad 10m < a_2 < b_2;$$

- (ii) closed Jordan disks $O \subset \operatorname{int} H$, with $\overline{B}(0, b_2) \subset \operatorname{int} O$, and an orientation-preserving band chart

$$\Phi : [-12, 12] \times [-3, 3] \longrightarrow \mathbb{R}^2.$$

Writing its coordinates as (u, v) , membership in O and H within the chart is exactly $|u| \leq 3$ and $|u| \leq 8$, respectively; membership in the interiors is given by the strict inequalities. The full section $\Phi(\{0\} \times [-3, 3])$ is strictly inside $B(0, a_1)$;

(iii) one positive integer rank for each of the two windows

$$I_B = [-.3, .3], \quad I_W = [-1.8, -1.4].$$

No metric regularity of the Jordan disks or chart is imposed.

Let ν be the ordinary full-plane colored Poisson law at spacing ε . The state σ of a placed plug is the colored point data in O . Set

$$\rho_{\varepsilon, h} = h\sqrt{\varepsilon/h} = \sqrt{h\varepsilon}.$$

An edge is *retained* if it belongs to a Delaunay triangle with circumradius at most $\rho_{\varepsilon, h}$. Whether such a triangle incident to a vertex exists is determined by points within distance $2\rho_{\varepsilon, h}$ of that vertex. Thus retained-edge queries in a compact set use only a shrinking collar, on every configuration. By Lemma 3.3, all edges and triangles of a fixed local system are retained with probability tending to one.

For each color, take the connected components of the induced retained graph whose vertex positions belong to $\Phi([-2, 2] \times I_{\text{color}})$ and which meet both $u \leq -1.5$ and $u \geq 1.5$. Order them by the first site in chart lexicographic order. The component of the specified rank, when present, is its *root*. These graph queries lie compactly inside O . The test L^+ requires, for each color, a retained path from its root to $u \geq 6.7$, all of whose vertices, including its ends, lie in $\Phi([1, 7.5] \times I_{\text{color}})$. The test L^- uses $u \leq -6.7$ and $\Phi([-7.5, -1] \times I_{\text{color}})$.

Definition 4.4 (Usable states and admission). Fix positive thresholds β, β' . The state event G requires:

- (i) simple surrounding retained site circuits S_1, S_2 in the two gate annuli, with their edge segments in those annuli, enclosing the corresponding inner balls and containing at most two black sites each;
- (ii) both roots and the inequalities $\nu(L^+ \mid \sigma) \geq \beta$ and $\nu(L^- \mid \sigma) \geq \beta$.

The conditional probabilities use independent ordinary data on O^c ; in particular they average over the exterior point geometry as well as its colors. A type is *admitted* when $\nu(G) \geq \beta'$, and its state law is

$$\mu = \nu(\sigma \in \cdot \mid G).$$

The finite normalized menu and thresholds will be fixed in Section 6. Admission and μ may depend on h, ε .

Choose the two circuits by deterministic measurable rules on the state. They are compactly contained in O . The physical cutoff ball of a plug centered at z_s is the closed ball $B_s = \overline{B}(z_s, m_s h)$.

4.2 Connectors and their conditional success probability

Definition 4.5 (Connectors and simple systems). A directed connector $e : s \rightarrow t$ between same-color plugs with disjoint obstacles is an oriented rectangular band, with transverse parameter $v \in [-3, 3]$, attached by its full parameterized ends to the positive port $u_s = 8$ of H_s and the negative port $u_t = -8$ of H_t . Apart from these attachments it avoids both disks. Only the charts with $|u| \leq 8$ are prescribed at its ends.

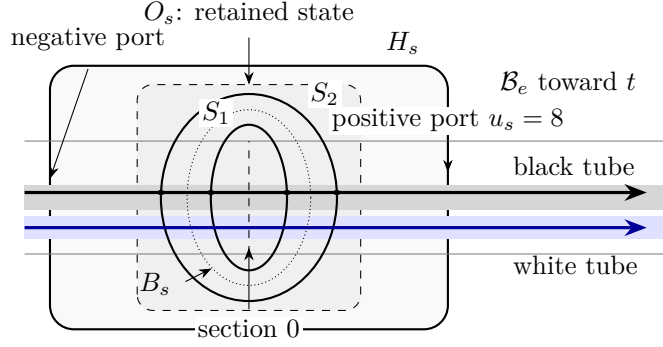


Figure 2: A topological schematic of a plug and its outgoing sector. The two gate circuits have only two black contacts with the displayed rim passage. The cutoff lies between the gate annuli. The state and the whole sector are retained in comparisons; the success tubes are strictly inset. The shapes and radii in this schematic are not to scale.

A *simple system* is a cyclic list of at least two distinct plugs and connectors. Its closed obstacles are pairwise disjoint, its exterior bands are pairwise disjoint and avoid all other obstacles, and the sectors from section 0 of one plug to section 0 of the next form an embedded annular band with the prescribed directed longitude and transverse parameter. Denote the whole closed sector of e by $\mathcal{B}_e$.

We also allow an orientation-preserving ambient homeomorphism g_s near each plug, supported in a preset bounded local region, replacing H_s and Φ_s by $g_s H_s$ and $g_s \Phi_s$. The state region, roots and tests remain unchanged. Choose the allowed uniform displacement small enough that $O_s \subset \text{int}(g_s H_s)$, that section 0 and a full central slab still lie strictly inside the first gate, and that positive and negative chart portions meeting either gate keep their signs. These conditions follow from the compact separation of portions away from the central slab. The tolerance is further reduced after any local stub and layer margins used below have been fixed.

The success event S_e asks for a retained root-to-root path of each color in its own tube of $\mathcal{B}_e$, using the actual end charts:

$$-.65 \leq v \leq .65 \quad \text{for black}, \quad -2.2 \leq v \leq -1 \quad \text{for white}.$$

Both tubes run longitudinally from $u_s = .5$ to $u_t = -.5$, and all path vertices, including the ends, must lie there. The exterior query sets for different connectors are disjoint. Indeed their whole sectors outside the closed states are separated; at a shared plug the common section is shielded inside O . The tube footprints and joining seams have room for the shrinking retained-geometry collars.

Figure 2 shows the state, gates and cutoff within a plug, together with the inset success tubes in its outgoing sector.

Lemma 4.6 (Conditional bridge). *For every fixed connector and admitted endpoint types, there is $c_e > 0$ such that, for all sufficiently small ε and almost every pair of usable endpoint states,*

$$\nu(S_e \mid \sigma_s, \sigma_t) \geq c_e. \quad (4.2)$$

The constants are uniform for rigid copies and for buffered families with a uniformly bounded number of fixed-shape crossing rectangles. Moreover, on good continuation geometry, the white

part of S_e has a positive quenched lower bound if the two white side tests each have conditional probability at least β given that geometry and the inner states.

Proof. Write $A = L_s^+ \cap L_t^-$ and $\sigma = (\sigma_s, \sigma_t)$. The endpoint exterior restrictions are independent, so $\nu(A \mid \sigma) \geq \beta^2$. For each color, continue through its open layer by a finite connected network of buffered crossing corridors. At each end include a transverse bar in the nominal slab $5 < |u| < 6$, crossing the entire corresponding window with margin; join these bars through the layer. All continuation queries lie strictly outside the state disks. A bar intersects every successful side path, since that path traverses the slab through the window. Retention and buffers allow passage between cell and primal crossings without changing this separation argument.

Here is the corridor construction in the form that will also be used for local preparations. Approximate a compact trace in an open corridor by finitely many horizontal and vertical polygonal legs on a sufficiently fine fixed square grid. Put thin crossing rectangles along the legs, with both horizontal and vertical crossings in successive overlap squares, and add the same overlaps at branch points. Crossing paths therefore join into the prescribed network. Slightly lengthening and shrinking the rectangles leaves room to convert them into retained primal paths. Along a charted simple trace the steps may be chosen with arbitrarily small longitudinal extent, so concatenation in order gives any prescribed positive bound on longitudinal backtracking.

Let E assert the retained-mesh and quenched rectangle lower bounds in these finitely many continuation footprints. Fixed collars keep its point queries outside $O_s \cup O_t$, independently of the states. Proposition 3.1 and Lemma 3.3 give

$$\nu(E^c \mid \sigma) \leq \delta_\epsilon \longrightarrow 0$$

uniformly over the usable states. No good-geometry requirement deep inside an arbitrary fixed state is needed: its successful side paths already use retained edges.

Condition now on all point geometry and on σ . On sites queried by black constraints order the bits toward black, and on the disjoint set queried by white constraints order them toward white. Both A and the continuation success K are increasing in this one product order; constraints of the same color may overlap. The two sets of colors have positive separation by the layer and chart-tolerance margins. On E , conditional positive association and the crossing input give

$$\nu(A \cap K \mid \eta, \sigma) \geq \nu(A \mid \eta, \sigma) \nu(K \mid \eta, \sigma) \geq q_e \nu(A \mid \eta, \sigma),$$

where $q_e > 0$ is fixed. Since $A \cap K \cap E \subset S_e$, integration before subtraction yields

$$\nu(S_e \mid \sigma) \geq q_e \nu(A \cap E \mid \sigma) \geq q_e(\beta^2 - \delta_\epsilon).$$

This proves (4.2). It does not infer any quenched black side propensity from an annealed one. If the two white side propensities are themselves bounded below after fixing geometry, the white-only version of the same product argument gives $q_e^W \beta^2$ without this annealed integration. For fixed normalized templates, finitely many buffered rectangle chains give common positive constants by scaling; rigid motions preserve the laws. $\square$

4.3 Cycle experiments and exact path agreement

Definition 4.7 (Cycle experiment). For a simple system, sample the states independently with their laws μ_s . Given the states, sample the exterior regions of the connectors independently, each conditioned on its success S_e , and sample the remaining plane data ordinarily. The exterior region of e may be the whole sector $\mathcal{B}_e$ with the closed endpoint state disks removed.

All queries of S_e are then in its region or its endpoint states, for small ε . The cycle law has density

$$\prod_s \frac{1_{G_s}}{\nu(G_s)} \prod_{e:s \rightarrow t} \frac{1_{S_e}}{\nu(S_e \mid \sigma_s, \sigma_t)} \quad (4.3)$$

relative to ν on these data; unused data remain free. Thus it is boundedly dominated by ν for a fixed system. In particular, the marginal of two endpoint states and one sector is exactly their ordinary law restricted by the three factors for G_s, G_t, S_e . Deterministic Borel restrictions suffice throughout: a Jordan boundary need not be smooth or have zero area.

Lemma 4.8 (A confined rim and its gate passages). *On a cycle experiment, outside an event of probability tending to zero, there is a rim P of the path color, oriented in the positive longitude, contained in the layer $-2.35 < v < .8$. At every selected gate circuit it has exactly two contacts, crossing from the negative side into the interior and then out through the positive side. Its outside path from the first-gate exit at s to the first-gate entry at t lies in $\mathcal{B}_e$. The inner gate paths occur in the prescribed cyclic order.*

Proof. The root components join negative to positive side vertices within their central windows. Concatenate these paths with the successful sector paths. For both colors one obtains a monochromatic closed walk of longitudinal index one, respectively in the disjoint layers $-.75 < v < .75$ and $-2.3 < v < -.9$. This index follows by lifting the longitude: root passages cross the central slabs, and sector passages go from one positive slab to the next negative one without passing the end section cuts. Arbitrary local backtracking is harmless. Fine mesh keeps the edge segments in the stated layers. Each walk contains an essential simple circuit. Lemma 2.5 supplies a black rim between them. The farther subbands on either side lie in the two exterior sides of this sandwich, so the chosen rim is confined to $-2.35 < v < .8$.

This rim visits both the inside and outside of either gate: it crosses section 0 deep inside and also reaches the other plugs. A black rim can meet a gate with at most two black sites only at those sites. By simplicity it must meet exactly two, once each, and cross at both. A tangency would not change side, leaving too few contacts. Following a gate edge between the two contacts is also impossible: the complementary rim arc would then have to visit both sides without another contact. Thus the incoming and outgoing edges are strictly in opposite gate fans.

All visits to the central section belong to the single inner gate arc. That arc lies in the end chart, since the gate and its bounded inside lie in O and all other bands avoid its obstacle. The full central slab is inside the first gate, and compact positive and negative chart portions meeting the gates are separated. The total positive longitudinal index one therefore forces negative entry and positive exit. This uses ordered passage, or equivalently lifted longitude, without requiring a transversal intersection with section 0. Outside its inner first-gate arc the rim cannot cross that section. Consequently the outside arc from s to the next t stays in their intervening sector. Since each inner arc contains all visits to its own cut and these chart slabs are disjoint, the inner arcs have the asserted cyclic order. $\square$

The following formulation isolates the precise deterministic comparison that will also apply to rims in different configurations.

Lemma 4.9 (Two-graph gate consistency). *Let P, P' be oriented simple monochromatic rims in their respective embedded triangulations. Suppose their directed subpaths γ, γ' have common distinct initial and terminal vertices v, w and satisfy these conditions:*

- (i) *each complete compared subpath is present and monochromatic in the other graph, with the same embedding;*
- (ii) *at v there is a shared gate circuit, both incoming rim rays lie strictly in one gate fan, and both outgoing rays lie strictly in the opposite fan;*
- (iii) *the rim orientations and direction of crossing that gate agree.*

Then $\gamma = \gamma'$ as directed edge paths. Each rim needs its no-right-bypass property only in its own graph.

Proof. Order rays in the outgoing fan counterclockwise between its two boundary rays. For any incoming ray in the opposite open fan, the counterclockwise sector from that ray to an outgoing ray contains the same initial portion of the outgoing fan. Hence, if the two outgoing rays differ, one enters the other's open right sector, independently of the differing incoming rays. Suppose, for instance, that γ' leaves to the right of P .

Follow γ' to its first subsequent contact with the *whole* rim P , not merely with γ . Such a contact exists no later than w . It is not v , because γ' is simple. As γ' is a path in P 's graph, planarity makes the first contact a vertex contact; a shared edge is first reached at its endpoint. Before this contact the path is connected and disjoint from P , so it stays in the open right component into which it departed. It is therefore a monochromatic right bypass between distinct vertices of P , contradicting Lemma 2.4. An earlier return on the complementary portion of P gives exactly the same contradiction. The other case uses P' in its own graph.

Thus the first edges agree. At any later first divergence the incoming edge is common, and cyclic order from that incoming ray again puts one outgoing edge into the other's open right sector. The same whole-rim first-return argument rules this out. The paths agree to w . $\square$

Lemma 4.10 (Deterministic collars for the compared paths). *For a fixed cycle pattern, every possible confined experiment rim has its inner second-gate path in a fixed compact subset of its state disk. Every possible first-exit-to-next-first-entry path lies in a fixed compact subset of*

$$D_e = O_s \cup O_t \cup \mathcal{B}_e.$$

Consequently, if two configurations retain these data, their complete incident triangulations along the compared paths agree on the mesh-good event, for all sufficiently small ε .

Proof. The second gate is in its prescribed outer ball, compactly inside O ; its bounded inside is in the same ball. This gives one deterministic envelope, also containing the gate neighborhoods. For an outside path use the closed inset footprint $-2.35 \leq v \leq .8$ in the whole sector. It has positive lateral margins. The chart-connector seams are interior product-band seams. At the two longitudinal ends the full section and a central slab are compactly inside the endpoint states, which shield the cuts. Hence this footprint is compactly contained in $\text{int } D_e$. Its collar is fixed before selecting a rim; there is no requirement that the rim follow a particular witness of S_e .

If a triangle is incident to a vertex in such an envelope and its circumradius is at most $\rho_{\varepsilon,h}$, its whole circumdisk is within $2\rho_{\varepsilon,h}$ of that envelope. For small ε the disk lies in the retained region, so its vertices and empty-circumdisk condition are unchanged. Mesh control in *both* configurations excludes additional large incident triangles on either side. Thus all incident triangles, not only the edges of a preselected path, agree. The required mesh probabilities follow from ordinary control and the fixed density bounds for the experiments. $\square$

For coupled cycle experiments sharing a state, choose the same deterministic gate circuits from that state. By Lemma 4.8, both rims visit the inside and outside of each chosen gate and can meet it only at its at most two path-color vertices. They therefore use the same two distinct vertices; negative entry and positive exit give the same ordering of these contacts. Thus the inner second-gate paths have common ordered endpoints, and the outside paths from the first-gate exit at s to the first-gate entry at t do also when both endpoint states are shared; these endpoints are distinct because the state disks are disjoint. At the start of an inner path, both incoming rays are in the exterior gate fan and both outgoing rays in the interior fan; at the start of an outside path these roles are reversed. All these inclusions are strict. Sharing the state preserves the colored collar of each complete inner second-gate path; sharing both endpoint states and the whole sector does the same for each complete outside path. On the mesh-good event, Lemma 4.10 therefore puts each compared path in the other graph with the same colors and preserves every incident triangle. Lemma 4.9 now identifies the directed paths. Their right incident triangles and local tagged sides consequently agree as well. The allowed chart changes preserve the sign and collar margins used in this argument.

4.4 Removing the exterior randomness

On a cycle experiment choose any measurable confined rim as above. For $X = Q, V$ define

$$\widehat{m}_s = X_P(\mathbf{1}_{B_s}), \quad \widehat{j}_e = X_P(\mathbf{1}_{B_e \setminus (B_s \cup B_t)}), \quad e : s \rightarrow t. \quad (4.4)$$

These are barycenter masks. Fix disjoint Borel conventions at sector boundaries; the rim is buffered away from lateral boundaries and the section cuts are inside cutoff balls. One may set the masses to zero on the exceptional geometry event. The deterministic-mask energy bound (2.4), ordinary triangle moments and bounded cycle density give finite L^2 bounds for each fixed pattern. For bounded normalized short templates with common density bounds these norms are at most Ch .

Every cutoff tag comes from the inner path through the *second* gates, because the cutoff is buffered inside that gate. The sector mass is precisely the contribution outside both cutoff balls along the path from the *first* exit at s to the first entry at t : the inner first-gate paths and cuts lie safely inside the balls. Thus, on the mesh-good event,

$$\sum_{e:s \rightarrow t} (\widehat{m}_s + \widehat{j}_e) = X_P(1).$$

Proposition 4.11 (The state cochain). *For every fixed connector, including its actual end charts, there are functions $i_e^\varepsilon(\sigma_s, \sigma_t)$ such that*

$$\begin{aligned} \widehat{m}_s + \widehat{j}_e - i_e^\varepsilon &= o_{L^1}(1) \quad \text{in every cycle experiment containing } e, \\ \sum_e i_e^\varepsilon &= o_{L^1}(1) \quad \text{on the independent state product of every simple system.} \end{aligned} \quad (4.5)$$

The same functions work in all completions. For a fixed collection of normalized short templates admitting uniform short completions,

$$\|i_e^\varepsilon\|_{L^2(\mu_s \otimes \mu_t)} \leq Ch$$

for all sufficiently small ε , after increasing a common constant.

Proof. For each type, fix a short unperturbed reference completion, for example using a sufficiently separated copy. Define $m_s^\varepsilon(\sigma_s)$ as the conditional mean of its cutoff mass given σ_s . Couple any other experiment to this completion sharing only that state and sampling the remaining data conditionally independently. Inner second-gate consistency identifies the cutoff mass on their common mesh-good event. Their L^2 bounds show that their difference is $o_{L^1}(1)$; indeed $\mathbb{E}[|Y|\mathbf{1}_E] \leq \|Y\|_2 \mathbb{P}(E)^{1/2}$ on a bad event E . Using two copies also in the reference completion and conditional Jensen shows that its mass is $o_{L^1}(1)$ from its own conditional mean. Consequently $\widehat{m}_s = m_s^\varepsilon + o_{L^1}(1)$ in every completion.

Next fix one two-plug completion of e and condition its sector mass on σ_s, σ_t and the exterior sector datum ξ_e . Call the result $j_e^\varepsilon(\sigma_s, \sigma_t, \xi_e)$. These retained data have the same marginal in every completion, by (4.3). Coupling and outside first-gate consistency, again with a second copy for the defining completion, give $\widehat{j}_e = j_e^\varepsilon + o_{L^1}(1)$ in every experiment containing e .

In a fixed cycle, conditional on its complete state vector, the data ξ_e are independent, each with the conditioned law determined by its two endpoint states. The closed estimate (2.7) and the cycle density show that

$$\sum_e (m_s^\varepsilon + j_e^\varepsilon) = o_{L^1}(1).$$

For Q the selected rim is contained in a fixed box and has a fixed positive minimum diameter, since it visits distinct plugs; for V its complete tangent sum is zero. Now resample one ξ_e independently given the states, leaving all other sectors fixed. Subtracting the two small cycle sums gives $j_e^\varepsilon - (j_e^\varepsilon)' = o_{L^1}(1)$. Conditional Jensen therefore gives

$$j_e^\varepsilon - \mathbb{E}[j_e^\varepsilon \mid \sigma_s, \sigma_t] = o_{L^1}(1).$$

The conditional law used in this last expectation depends only on those two states. Set

$$i_e^\varepsilon = m_s^\varepsilon + \mathbb{E}[j_e^\varepsilon \mid \sigma_s, \sigma_t].$$

This proves both statements in (4.5). The defining short completions and Jensen give the stated L^2 bounds. All replacements use only bounded second moments and high-probability path equality; uniform integrability of their squares is unnecessary. The choices are measurable because the locally finite graphs have countably many finite paths and walls, and different choices agree through the same path comparison. $\square$

4.5 Common closures, collars and changes of charts

Lemma 4.12 (Comparison by a common closure). *Suppose two directed paths of connectors from s to t admit a common complementary connector $c : t \rightarrow s$, making each a simple system. Their sums of cochain values differ by $o_{L^1}(1)$ on the product of all states, with shared states identified. Either path may be a single connector. For fixed end disks and charts, a connector value therefore depends, up to $o_{L^1}(1)$, only on the endpoint-fixed deformation class of its exterior simple centerline described below.*

Proof. Apply (4.5) to the two cycles and subtract the common i_c^ε . Adding independent unused states does not change an L^1 error. A complementary connector always exists: two disjoint closed Jordan disks joined by the attached rectangular band form a Jordan disk, with the unused ports on its boundary. In its exterior, the Jordan–Schoenflies theorem gives another handle band joining those ports. The transverse parametrizations match because departure and arrival sections have opposite induced boundary orders. The remaining deformation assertion follows from the collar construction and finite chaining below. $\square$

Lemma 4.13 (A band with prescribed full attachments). *Let a simple exterior arc join the middle points of two prescribed port sections. Suppose it continues for a short distance along the given product collars at its ends and otherwise avoids the closed end disks. Then a rectangular band about it, with those full parameterized attachments, fits in any neighborhood of the arc and attachments within the exterior. It can in addition avoid any fixed disjoint compact routes or unused obstacles.*

Proof. Extend a short piece of each given end collar and taper its transverse width to be very small at the far tip. The full port is usable initially: the compact middle arc is separated from it, while the given product collar handles its immediate neighborhood. Straighten a slightly extended middle arc by an orientation-preserving plane arc chart, with its left side above a horizontal segment.

From the upper and lower corners of each narrow collar tip, follow constant small signed transverse coordinates in that end collar for a short additional distance. In the straightened coordinates these side paths lie on the corresponding sides of the midline and progress along it with arbitrarily small error. Stop at their first hits of prescribed vertical lines, chosen in order at the two ends. Join the upper stopped paths, and separately the lower ones, by height graphs strictly on their side of the middle segment and as close to it as desired. The compact middle interval is separated from the initial collar pieces and disks, so these joins fit in the specified neighborhood and avoid all the specified compact sets. First-hit stopping keeps the side paths simple.

Each joined side, the appropriate half-tip segments and the centerline bound a Jordan rectangle. The two rectangles lie on opposite sides in the straightened coordinates and may be taken thin there. The initial disk-with-collar interiors abut their tips on the other side, so they cannot cross either boundary curve. Fill the rectangles by Jordan–Schoenflies, prescribing the centerline and boundary parametrizations, and glue them to the end collars. This gives the required band with the full original port attachments. $\square$

For completeness, the deformation in Lemma 4.12 is implemented as follows. Normalize the disjoint Jordan disks to round holes by an ambient homeomorphism. Inside an existing exterior band replace its centerline by a smooth proper arc, radial near its endpoints, retaining the same complementary connector. Indeed an attaching section has a relative exterior neighborhood;

polygonal approximation in this open set, loop erasure and smoothing give a simple arc with the prescribed short radial germs. The full port attachments are recovered by Lemma 4.13.

Consider now an endpoint-fixed deformation through smooth proper arcs, transverse at the hole boundaries. Product end collars are available throughout, adjusting their angular coordinate with the radial level near each endpoint. At a parameter value choose a narrow connector and a complementary band disjoint from it. By compact separation that complement also avoids all nearby centerlines; the collar lemma supplies connectors for them with the same full attachments. At overlapping parameter neighborhoods, narrow a connector to avoid both choices of complement. A finite subcover of the parameter interval and the common-closure comparison give the $o_{L^1}(1)$ equality along the whole deformation. Constants may depend on this finite chain, which is fixed before $\varepsilon \downarrow 0$.

Lemma 4.14 (Stability on buffered local templates). *Fix a buffered two-plug system e_0, c_0 . Suppose its parts in the preset neighborhoods where charts may move are fixed local stubs, and no other part of the system returns to these neighborhoods. For sufficiently small supported changes g of the two end charts and obstacles, let e_g, c_g be the corresponding images. Then*

$$i_{e_g}^\varepsilon = i_{e_0}^\varepsilon + o_{L^1}(1) \quad \text{on } \mu_s \otimes \mu_t.$$

The required tolerance depends only on the compact local stub and layer margins in the moved regions.

Proof. Consider the mixed experiment with successes S_{e_g} and S_{c_0} , with the nominal states and roots. Their exterior queries are still separated, and Lemma 4.6 applies. All allowed tubes lie in slightly widened, separated layers of the nominal annular band, with the same longitudinal order. This follows by the local tolerance where anything moved; elsewhere the system is unchanged. Thus the mixed experiment has the same confined-rim and gate conclusions, and we may use deterministic nominal masks for its masses.

Compare its e contribution with the pure e_g, c_g experiment by sharing the states and the exterior part of the *nominal* e_0 sector. Every possible pulled-forward inset e_g path outside the states, together with all queries of S_{e_g} , is buffered in that shared region. The queries of both S_{c_g} and S_{c_0} avoid it. Consequently the shared marginal in both experiments is the ordinary one conditioned on S_{e_g} at the two sampled states: the other success factor integrates to one given the states and affects only other exterior data. This also follows directly from the product density (4.3), irrespective of how unused points are assigned to the sampling regions.

The common collar and the two gate comparisons identify the first-gate path outside the cutoff balls and the starting cutoff contribution. Thus the extracted e value in the mixed experiment is $i_{e_g}^\varepsilon$ up to $o_{L^1}(1)$. The same argument identifies its other contribution with $i_{c_0}^\varepsilon$. Closing the mixed rim therefore gives $i_{e_g}^\varepsilon + i_{c_0}^\varepsilon = o_{L^1}(1)$, whereas the nominal system gives $i_{e_0}^\varepsilon + i_{c_0}^\varepsilon = o_{L^1}(1)$. Subtraction proves the claim. $\square$

The restriction to buffered local templates is deliberate. An arbitrarily thin fitted connector with changed charts is first deformed with those changed charts held fixed. Only its standard buffered representative is then compared to the nominal charts by Lemma 4.14. A narrow fitted band's conditioning constant is never used as the uniform constant of a short-template moment estimate.

5 The affine part and endpoint potentials

The state cochain now separates into a spatially affine term and two local endpoint terms. Short comparison experiments control the endpoint terms uniformly; the geometry of a long connector enters only a controlled deterministic remainder.

Fix a finite normalized menu. Actual placements may be its translates on the h -grid; auxiliary translated and rotated copies are also allowed. At fixed h , pass to a further subsequence on which the admissions stabilize. If any type is admitted, choose one reference type and its color flip. When the admitted menu is empty all comparisons at qualified endpoints are vacuous; we may set $a_h^V = 1$ and choose any bounded a_h^Q .

5.1 Fixed stubs and straight means

For each type fix two disjoint *stubs* outside its obstacle, taking the departure port to an eastward horizontal strip and the arrival port to a westward horizontal strip. Their exit sections lie at the midpoints of opposite sides of a sufficiently large square, with transverse coordinate v increasing upward. To construct them, use a Jordan annulus between the obstacle and the square, match the two attachment intervals in cyclic order, and draw disjoint product continuations. The transverse order is along the positive boundary direction at the east exit and against it at the west exit, precisely as required by the departure and arrival ports. Continue the strips straight outward.

Fix the local portions of the stubs and their positive margins in all regions where chart changes may be supported before placing other plugs. A later enlargement of the exit square uses only straight continuation. Rotate a type together with its stubs whenever needed. These choices meet the local-template hypotheses of Lemma 4.14.

Proposition 5.1 (Affine straight means). *For $X = Q, V$ there is a scalar a_h^X , uniformly bounded in h , such that every fixed straight transport in the countable family used below has mean*

$$\mathbb{E}_\mu i_e^\varepsilon = a_h^X(z_t - z_s) + o(1). \quad (5.1)$$

The same scalar applies to translations, common rotations and color flips of the reference type. All choices can be made simultaneously for the two fields and countably many h .

Proof. Join identical unrotated reference copies on a horizontal line, using their fixed stubs and a straight gap. Their center separation is ℓh , where $\ell > D$ for a fixed sufficiently large D . Write $A_\varepsilon^X(\ell) = \mathbb{E}_\mu i_e^\varepsilon$. A remote complementary connector through the unused stubs closes both this transport of length $(\ell + r)h$ and the two transports obtained by inserting a middle reference copy. Lemma 4.12 gives

$$A_\varepsilon^X(\ell + r) = A_\varepsilon^X(\ell) + A_\varepsilon^X(r) + o(1), \quad \ell, r > D.$$

Translation and rotation of the entire experiment, including the root ordering, preserve its law and rotate both fields as vectors. If a conditional-average or gate tie-breaking convention was chosen separately at the translated placement, its transported version is an alternative defining completion. Gate consistency with common gates in the shared states and Proposition 4.11 compare the two choices to $o_{L^1}(1)$. Thus covariance to this accuracy is sufficient. Color exchange preserves the law as well.

The ratios $A_\varepsilon^X(\ell)/h$ are bounded uniformly for $D + 1 \leq \ell \leq D + 4$, in the limit superior: these transports have bounded normalized size and uniformly buffered short completions. For each fixed larger length there is likewise a finite bound, not necessarily uniform in that length.

Only countably many lengths are required: the standard routes below are fixed by grid terminal pairs, finitely many types and integer twists. Include the positive integers and take the rational span of these lengths, a countable additive subgroup $\mathcal{L} \subset \mathbb{R}$. Extract limits of $A_\varepsilon^X(\ell)/h$ for all $\ell \in \mathcal{L}$ with $\ell > D$. Their function A is additive whenever both arguments exceed D . Extend it to $x \in \mathcal{L}$ by

$$\tilde{A}(x) = A(u+x) - A(u), \quad u, u+x > D.$$

The value is independent of a sufficiently large u : comparing two choices u, v uses $A(u+x)+A(v) = A(u+v+x) = A(v+x) + A(u)$. Choosing all offsets large similarly proves additivity on $\mathcal{L}$. For small x , take a fixed rational u in the interior of $[D+1, D+4]$; the interval bound gives a common bound on $\tilde{A}(x)$. Additivity then gives continuity at zero: if $|x| < \delta/N$, a bound for $\tilde{A}(Nx)$ gives the same bound divided by N for $\tilde{A}(x)$. Since $\mathcal{L}$ contains the rationals, continuity implies $\tilde{A}(x) = a_h^X x$. The interval bound also bounds $|a_h^X|$ independently of h . Covariance yields (5.1) in every direction. Countable diagonal extraction makes these choices simultaneously as asserted. $\square$

5.2 Normalization of the tangent field

Lemma 5.2. *The scalar of the tangent field is*

$$a_h^V = 1. \tag{5.2}$$

Proof. We give the mask argument also for curvilinear end charts. In an unperturbed straight experiment choose a deterministic polygonal Jordan quadrilateral R_e approximating the sector from section 0 to section 0 with transverse interval $[-2.7, 1.5]$. Its boundary is close in the ordered parametrization. The transverse end cuts remain in the first gate balls, and its top and bottom sides lie beyond the possible rim layer $[-2.35, .8]$ on its left and right, respectively. Permit a small longitudinal enlargement only inside the cutoff balls. By compact margins, outside slightly smaller cutoff balls the mask contains exactly the relevant sector side of every confined rim, with room, and excludes the complementary side.

Here the required polygonal approximation follows directly from Jordan simplicity. Trace a polygonal closed walk uniformly close to the boundary in order, split any crossings in general position and retain a simple subcycle with positive winding about a fixed inner point. Dropped parameter gaps have endpoints tending together, and therefore are short by injectivity of the limiting Jordan parametrization. They cannot be almost a whole turn: the retained cycle would then be local and have zero winding about the inner point. This preserves the ordered approximation and the four chosen sides. Fix one such quadrilateral in a sufficiently long template. For larger ℓ , stretch only its straight middle region horizontally, keeping both end regions rigid. The margins persist for each fixed ℓ , and the total length of the two end cuts is $O(h)$ uniformly in ℓ .

On good geometry, polygonal intersection with the left side of P gives

$$\int_{P \cap R_e} dz = \ell h + O(h).$$

Indeed this integral is minus the integral over the portion of ∂R_e on that left side. The entire top side contributes with reversed orientation, the bottom side does not contribute, and the end

cuts cost $O(h)$. Their endpoint locations differ from the plug centers by $O(h)$. A fixed cut has probability zero of an accidental exact overlap with a random primal edge or vertex degeneracy.

We next compare this geometric integral with the barycenter mask $V_P(\mathbf{1}_{R_e})$. On fine mesh a discrepancy can involve only rim edges whose tagged triangles meet a cut segment. Put $s = \varepsilon^\alpha$ with $\alpha < 1$ sufficiently close to one and cover these finitely many segments by $O_e(s^{-1})$ boxes of side s . In any one enlarged box the L^2 norm of the total candidate edge length is $O(s^2/\varepsilon)$, by the scaled local triangle moments. The presence of a candidate rim edge implies a monochromatic arm from scale s to a fixed positive distance for this pattern, since the rim also visits a different plug. Its ordinary probability is $O_e(s^c)$, apart from a negligible mesh event. Cauchy–Schwarz and summation thus bound the expected discrepancy by

$$C_e s^{-1} \frac{s^2}{\varepsilon} s^{c/2} = C_e \varepsilon^{\alpha(1+c/2)-1} \longrightarrow 0$$

when $\alpha > 1/(1+c/2)$. Bad mesh events are discarded using the same local moments. The bounded density of this fixed experiment transfers the assertion to its law. Alternatively, convergence in probability and the Bessel L^2 bound suffice for the corresponding mean limit.

Outside the two cutoff balls, R_e gives exactly $\hat{j}_e$. Its difference from $\hat{m}_s + \hat{j}_e$ consists of deterministic masks inside the endpoint cutoff balls. Couple at either endpoint state to a fixed short completion. The inner second-gate paths and their triangles agree, so these correction means are $O(h) + o(1)$ by the short second-moment estimate. Their constant is independent of ℓ : the end masks were fixed before the middle strip was stretched. By Proposition 4.11 and (5.1),

$$|a_h^V \ell - \ell| \leq C$$

for arbitrarily large positive integers ℓ . This proves (5.2). $\square$

5.3 The two endpoint twists

We next compare an arbitrary fixed connector to a controlled standard representative. Its class is understood with the end disks, port points and charts fixed.

Lemma 5.3 (Classification by endpoint twists). *After normalizing the two disjoint Jordan obstacles to round holes, every exterior simple centerline can be replaced, by common-closure comparisons, by a smooth proper arc radial at its ends. Such arcs with fixed endpoints are classified, for the comparisons needed here, by two integer endpoint twists. Relative to a standard route with bounded endpoint turns, their total size k is bounded by a constant plus a constant times the sum of the absolute net argument changes about the two centers along the exterior arc.*

Proof. The smoothing and full-port collars were established after Lemma 4.13. Identify each hole boundary with the circle of directions at its center. Fix a radial identification of the plane with the two punctures blown up and the exterior of the circular holes: in disjoint neighborhoods of the centers, send radius zero to the hole radius by an increasing radial coordinate, agreeing with the identity outside those neighborhoods. Apply its inverse to each radial-ended proper arc and fill in the punctures. The result is a smooth embedded compact interval, radial near its ends, and the fixed identification recovers exactly the original proper arc after blowing up. Any two such smooth intervals, with ordered endpoints matched, are carried one to the other by an orientation-preserving plane diffeomorphism. To obtain this smooth extension, first map smooth strip-and-cap neighborhoods of the two intervals by an orientation-preserving diffeomorphism,

preserving their ordered axes. Choose smooth disk boundaries inside these neighborhoods. The prescribed Jordan–Schoenflies theorem extends this map across their complementary surfaces, retaining its smooth values on collars of those boundaries. Hatcher’s smoothing theorem for surface homeomorphisms, in its form relative to a boundary neighborhood [8, Theorem B and the following refinement, p. 2], replaces the exterior homeomorphism by a diffeomorphism without changing these collars. Gluing with the retained neighborhood map gives the required plane diffeomorphism and preserves the two intervals and their endpoints exactly.

An orientation-preserving plane diffeomorphism has a locally smooth deformation to the identity. After subtracting its value at a base point, the maps $x \mapsto F(tx)/t$ tend on compact sets to its positive differential as $t \downarrow 0$; restore the translation and join the positive linear map to the identity. During the resulting deformation of the interval, postcompose by the unique direct similarity restoring the two current ordered endpoint positions to the two fixed punctures. This correction is the identity at both ends of the deformation. Thus the punctures stay fixed throughout.

Blowing up their directions gives a deformation of proper arcs whose ends move on the two hole circles. Lift both moving endpoint angles to real-valued continuous functions. Compensate each angle in a disjoint boundary collar, tapering the compensating rotation to zero away from that boundary. The deformation now fixes the port points. At its end the two compensations are integral full twists, since the original and final port points agree. These are the only possible differences between the fixed-end deformation classes. The compensated arcs are transverse to the hole circles. Compactness of the parameter interval provides uniformly small product collars in which all these transverse germs can be straightened to radial germs, continuously in the parameter. Take the straightening correction to be the identity at both terminal times, where the germs are already radial. The resulting deformation therefore has the full fixed-end collar form required for the common-closure comparisons.

They are detected separately by net arguments. For the starting puncture, lift the argument along the deforming arc. The difference of its final and initial net argument changes is minus the net turn of the starting direction loop: the direction at the other puncture has zero net argument change about the starting puncture. For the arriving puncture the analogous identity has the opposite sign. Hence equal net argument changes at both centers make both compensations trivial at the end, giving an endpoint-fixed deformation of the kind used in Lemma 4.12. Conversely integer full turns at the ends realize the corresponding changes; the explicit disjoint representatives are constructed below.

Choose the signs of the two integers so they record the additional argument turns in the orientation from s to t . Their absolute sum is bounded by the displayed net changes divided by 2π , plus the bounded turns of the reference route. Normalizing the Jordan disks does not change winding differences between two arcs with the same ends: these differences are integer winding numbers of their joined closed curves about the respective centers. For allowed perturbed charts, pull the route back first to the nominal disks. The arc stays outside the fixed central balls, and the small supported change has only a bounded effect on the endpoint argument choices. This gives the same bound for k up to a fixed additive constant. $\square$

5.4 Quantitative standard representatives

For each terminal type choose a fixed short start insertion: go east along its departure stub to a horizontally aligned reference copy at a fixed large distance. Choose a fixed short end insertion from a reference copy west of the arriving terminal into its arrival stub. Their distances are

large enough for the full local structures and the supports of chart changes; all these lengths are nevertheless fixed in normalized units. They have uniformly bounded short standalone completions and hence the Ch cochain bounds. The first and last insertions will remain fixed when the long route changes.

Lemma 5.4 (Standard spirals and short corner templates). *Fix nominal terminal placements, their types and two twist integers n_s, n_t , and put $k = |n_s| + |n_t|$. There is a constant C such that, if*

$$|z_t - z_s| > Ch(1 + k),$$

the class has a buffered representative beginning and ending with the fixed insertions above. Between them it consists of straight transports and at most $C(1 + k)$ short pieces. The short pieces belong, up to rigid motion, to uniformly buffered compact families of bounded normalized size, and satisfy

$$\|i_j^\varepsilon\|_{L^2} \leq C_{\text{short}} h$$

for small ε , with a constant independent of the terminal separation, the twists and the particular long connector.

A common complementary connector closes both the inserted path and its unsplit representative. All auxiliary placements and straight lengths can be fixed from the nominal terminal data and integers before making allowed chart changes. In the applications they form a countable list.

The three features of this representative have different roles in the cochain comparison. Its fixed first and last insertions determine the endpoint state functions. Its straight pieces, however long, have the exact limiting means in Proposition 5.1. Only the $O(1 + k)$ short pieces contribute an $O(h(1 + k))$ deterministic error. Figure 3 shows how the used route and a common complement can turn around the terminals without meeting. We first derive the endpoint formula from these properties; the full geometric construction follows after that proof.

5.5 The affine rule and its uniform terms

Proposition 5.5 (Endpoint potentials). *For every admitted departure or arrival placement and type there are centered state functions $\phi_s^{X,+}$ and $\phi_t^{X,-}$, allowed to depend on ε , with*

$$\|\phi_s^{X,+}\|_{L^2(\mu_s)} \leq C_{\text{short}} h, \quad \|\phi_t^{X,-}\|_{L^2(\mu_t)} \leq C_{\text{short}} h.$$

They depend only on the indicated endpoint data, and not on the long connector or its allowed small chart changes. For each fixed connector $e : s \rightarrow t$ of class size k , provided $|z_t - z_s| > Ch(1 + k)$,

$$\begin{aligned} i_e^\varepsilon(\sigma_s, \sigma_t) &= a_h^X(z_t - z_s) + \phi_s^{X,+}(\sigma_s) + \phi_t^{X,-}(\sigma_t) + b_e^\varepsilon + \rho_e^\varepsilon, \\ \|\rho_e^\varepsilon\|_{L^1(\mu_s \otimes \mu_t)} &\longrightarrow 0, \quad \limsup_{\varepsilon \rightarrow 0} |b_e^\varepsilon| \leq C_{\text{short}} h(1 + k). \end{aligned} \tag{5.3}$$

Here b_e^ε is deterministic. The constants depend on the fixed normalized menu and short templates, but not on a thin fitted connector, its length or a finite cover of such connectors.

Proof. If e uses changed charts, first hold those charts fixed and compare it, by Lemma 5.3 and the common-closure rule, to the corresponding image of its standard representative. Apply Lemma 4.14 only to this buffered representative to return to the nominal charts. Finally compare

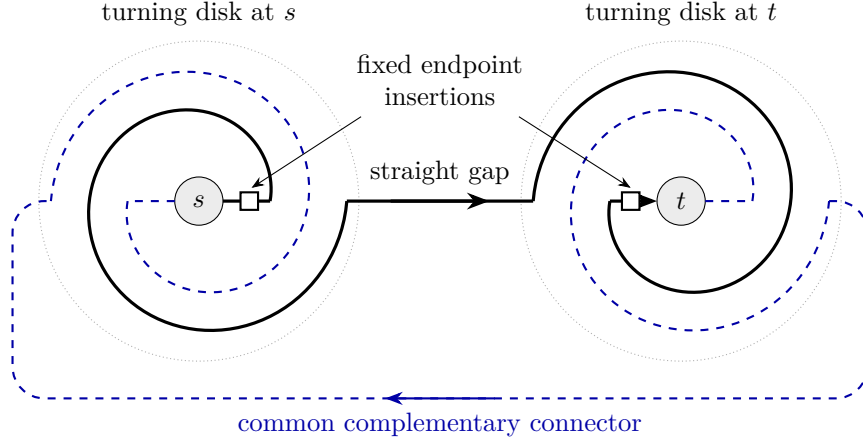


Figure 3: A standard representative, shown schematically with one full turn at each terminal. The solid route runs from s to t ; the dashed opposite strands and remote arc form its common complement. The small squares mark the reference copies used by the fixed endpoint insertions. The proof uses polygonal spirals and inserts further reference copies before and after their bends; these are omitted here. The turning disks have radius $O(h(1+k))$, while the number of short pieces is $O(1+k)$.

the nominal unsplit representative to the inserted system of Lemma 5.4, using its common complement. Thus, on the product of endpoint and auxiliary state laws,

$$i_e^\varepsilon = \sum_j i_j^\varepsilon + o_{L^1}(1).$$

Each chain of comparisons is finite with its pattern fixed. Narrow bands, small positive conditioning probabilities and the length of a finite deformation chain may enter the rate of this vanishing error; they do not enter the short-template constant.

Average over every auxiliary state, leaving σ_s, σ_t fixed. Only the fixed first and last insertions still have a nonconstant part. Define

$$\begin{aligned} \phi_s^{X,+}(\sigma_s) &= \mathbb{E}_\mu[i_{\text{first}}^\varepsilon \mid \sigma_s] - \mathbb{E}_\mu i_{\text{first}}^\varepsilon, \\ \phi_t^{X,-}(\sigma_t) &= \mathbb{E}_\mu[i_{\text{last}}^\varepsilon \mid \sigma_t] - \mathbb{E}_\mu i_{\text{last}}^\varepsilon. \end{aligned}$$

Conditional expectation followed by centering is an orthogonal projection in L^2 , so the fixed short-completion bounds give the stated norms. The definitions use the nominal fixed insertions; this is why neither the long route nor the allowed chart changes affect them. Conditional Jensen preserves the comparison's $o_{L^1}(1)$ error.

Let J be the short pieces and T the straight ones, with oriented center displacements d_j . Their total displacement is $z_t - z_s$. For each fixed $j \in T$, Proposition 5.1 gives $\mathbb{E}_\mu i_j^\varepsilon - a_h^X d_j \rightarrow 0$. Put the sum of these finitely many deterministic deviations into ρ_e^ε , together with the preceding comparison error. Define the remaining deterministic term by

$$b_e^\varepsilon = \sum_{j \in J} (\mathbb{E}_\mu i_j^\varepsilon - a_h^X d_j).$$

There are at most $C(1+k)$ such pieces. Each has cochain norm at most $C_{\text{short}}h$, displacement $O(h)$ and the same bounded scalar a_h^X . Enlarging C_{short} once gives the bias bound in (5.3). In

particular a large total length of straight legs creates no length-proportional bias: their limiting means have already been included exactly in the affine displacement. $\square$

5.6 Geometry of the standard representatives

It remains to construct the representative used in the endpoint formula. We build the two terminal turning regions, verify their separation, and then insert reference copies to isolate uniformly bounded corner pieces.

Proof of Lemma 5.4. All lengths until the last rescaling are normalized. Let D_{st} be the lower length threshold for a straight transport. Continue the two stubs at each terminal along opposite east and west spokes beyond all local structures. Set $\delta = 2\pi/16$. To make a turn Θ , including its bounded directional adjustment and integer full turns, write

$$|\Theta| = N\delta + \alpha, \quad N \in \mathbb{Z}_{\geq 0}, \quad 0 \leq \alpha < \delta.$$

Choose a large constant D_2 and take $D_1 = D_2$. Starting at radius D_1 , use N polygonal legs of angular increment δ in the sign of Θ , increasing radius by D_2 at each leg. If $\alpha > 0$, append one leg of angle α , also increasing radius by D_2 . The second strand is the image under rotation by π . Both finish with radial outward continuations.

At s choose the final used spoke toward t . At t choose its final used spoke, viewed outward, back toward s ; the connecting arc traverses this latter spiral inward. In the integer convention of Lemma 5.3, the outward constructions therefore use

$$\Theta_s = \theta_s + 2\pi n_s, \quad \Theta_t = \theta_t - 2\pi n_t, \quad |\theta_s| + |\theta_t| \leq C.$$

This specifies both used strands and their unused opposite companions.

We verify the geometric margins, including a last partial leg. A leg from radius r to $r + D_2$ with angular increment $0 < \gamma \leq \delta$ has length at least D_2 , since

$$|(r + D_2)e^{i\gamma} - r|^2 = D_2^2 + 2r(r + D_2)(1 - \cos \gamma).$$

Its direction relative to its initial radial ray is

$$\frac{\gamma}{2} + \arctan\left(\frac{2r + D_2}{D_2} \tan \frac{\gamma}{2}\right).$$

This lies between 0 and $\pi/2 + \delta/2$; relative to the final radial ray the incoming direction lies between 0 and $\pi/2$. Thus all bends, including radial joins and a partial last leg, have absolute turning angle at most

$$\beta_* = \pi/2 + \delta/2 < \pi.$$

Reflection gives the same assertion for the opposite sign of turn.

For a full leg its radius at angle $0 \leq \theta \leq \delta$ is

$$F_r(\theta) = \frac{\sin \delta}{\sin(\delta - \theta)/r + \sin \theta/(r + D_2)}. \quad (5.4)$$

Uniformly for $r \geq D_2$,

$$\cos \delta r \leq F_r(\theta) \leq r + D_2, \quad |\partial_\theta F_r(\theta)| \leq C_\delta r, \quad \partial_r F_r(\theta) \geq \cos(\delta/2).$$

For the last inequality put $A = \sin(\delta - \theta)$ and $B = \sin \theta$; Cauchy–Schwarz gives

$$\partial_r F_r = \sin \delta \frac{A/r^2 + B/(r + D_2)^2}{(A/r + B/(r + D_2))^2} \geq \frac{\sin \delta}{A + B} \geq \cos(\delta/2).$$

In the sixteen fixed angular sectors, two distinct full legs in one sector have starting radii differing by at least $8D_2$: a return of one strand is offset by sixteen steps and the opposite strand by eight. Their radial graphs are therefore separated by at least $8D_2 \cos(\delta/2)$. For legs in adjacent sectors, the endpoint radii at the common ray differ by at least $8D_2$, unless they are consecutive legs of one strand.

These radial gaps give Euclidean clearance $c_0 D_2$ between nonconsecutive full legs, for a fixed $c_0 > 0$. Indeed, if two points were within cD_2 , their radii would be comparable and their angular separation at most CcD_2/r . For small c , their sectors must be equal or adjacent. Compare their radii on a common ray, or on the intervening step ray for adjacent sectors. The angular derivative bound in (5.4) changes the radii by only $C'cD_2$, contradicting the preceding radial gaps.

A partial last leg need not have a uniform angular derivative bound. Instead let $P_{r,\alpha}(\theta)$ be its radial graph. For $0 \leq \theta \leq \alpha < \delta$ it lies outside the hypothetical full leg:

$$P_{r,\alpha}(\theta) \geq F_r(\theta).$$

To check this, write

$$\frac{1}{P_{r,\alpha}(\theta)} = \frac{\cos \theta}{r} + \frac{\sin \theta}{r} \frac{r/(r + D_2) - \cos \alpha}{\sin \alpha}.$$

For $0 < a < 1$, the derivative of $(a - \cos \gamma)/\sin \gamma$ is $(1 - a \cos \gamma)/\sin^2 \gamma > 0$. This proves the comparison. Every earlier nonincident leg in the same or an adjacent sector is on the inner side of the hypothetical full leg, with the established gap. The close-points argument thus uses only the angular derivatives of that hypothetical leg and the earlier full leg, not of the partial one. The simultaneous partial legs of the two strands lie in opposite sectors and are separated on the scale D_2 .

The initial radial continuations have radius at most $D_1 = D_2$. All turning legs after the first have radius at least $2D_2 \cos \delta > D_2$, giving a gap of order D_2 ; the first is incident to its own initial spoke and angularly separated from the opposite one. At the outer end, each radial continuation begins at the largest vertex radius, whereas all earlier legs except its last incident leg have radius at least D_2 less. The opposite final leg is separated angularly. The central stub neighborhoods retain their fixed buffers, and the turning legs lie beyond them when D_2 is sufficiently large.

The number of legs and bends is $O(1 + |\Theta|)$, and their maximum radius is $O(1 + |\Theta|)$. Thus the two terminal turning regions lie in disks of radii $Ch(1 + k)$ after rescaling. Under the separation assumption they are disjoint. Join the two used final spokes by their straight gap. The unused spokes point away from this gap; continue them outward and join them by a remote arc outside the two turning disks and the used gap. Its narrow band gives a complementary connector avoiding all subsequent insertions.

We next fix the short corner experiments. Let S bound the reference copy and its fixed stub structures, and choose E_0 large compared to S . At a corner p with incoming and outgoing unit directions u_-, u_+ , place aligned reference copies at $p - E_0 u_-$ and $p + E_0 u_+$. The non-reversal bound gives their separation at least $2E_0 \cos(\beta_*/2)$. Outside their stub exit squares, place a narrow strip around the intervening two-leg path, with brief straight continuations matching the exits. Round the corner within a fixed small neighborhood if desired. The angle bound leaves positive margins from both copies, their back stubs and adjacent straight transports. A

standalone completion joins the two unused back stubs outside a regular neighborhood of this bounded configuration, still in a ball of radius CE_0 .

The bend angles range over the compact interval $[-\beta_*, \beta_*]$. A buffered experiment at one angle persists in a neighborhood of it: move the terminal copy together with both its stubs by a small rigid motion, taper this motion to the identity in a disjoint collar and deform the strip slightly. Its fixed continuation corridors keep their margins, so Lemma 4.6 supplies a common positive bound in this neighborhood. A finite subcover gives common bounds on normalized size and density over all corner angles. Proposition 4.11 now supplies one C_{short} . Include the fixed first and last insertions and bounded straight-size templates in the same finite collection of compact families.

Finally take $D_2 = D_1$ much larger than E_0, D_{st} , the central local scales and c_0^{-1} . The corner neighborhoods are disjoint. Place copies before and after every bend as just described; after reserving their bounded neighborhoods, all remaining aligned gaps have length exceeding D_{st} . The first and last fixed insertions fit on the initial radial parts. Increasing the constant in the terminal separation condition ensures the same bound for the middle gap. Narrow strips along all gaps give the inserted simple system. The unchanged remote complementary connector also closes the unsplit representative.

There are only $O(1+k)$ short pieces, each with displacement $O(h)$. No linear bound on the total arclength is asserted or needed: the spiral's full legs can have total length of order $(1+k)^2$ in normalized units. For each nominal terminal pair, type pair and pair of integers, fix this construction once. In the support of allowed chart changes it uses only fixed local stubs, with no return of remote pieces. The auxiliary placements and straight lengths can therefore be chosen before those changes. Grid terminal pairs and twist pairs are countable, the menu is finite, and the scale list is countable. Hence the required straight lengths are countable and may be included in the simultaneous extraction of Proposition 5.1. $\square$

All limits of straight means may be fixed jointly at countably many scales for both fields. Subsequent comparisons of individual fixed connectors, including the finite fitted covers in Section 7, use only these limits and the vanishing cochain comparison errors. They require no additional subsequence depending on the sampled rim. At each fixed h a finite sum of pattern-dependent failure probabilities still vanishes as $\varepsilon \downarrow 0$; only afterward is h decreased. The constants in the potential and bias terms remain the common short-template constants throughout.

6 A finite menu of plugs and dense pins

We construct a finite menu of local endpoint models that occurs densely along every relevant macroscopic rim: each sufficiently large subarc contains a usable endpoint from that menu. These are the models used in Proposition 5.5. Its geometric constants and the thresholds $\beta, \beta' > 0$ will be independent of the plug scale h . Placements are translates by $h\mathbb{Z}^2$, in one common frame, of the scaled members of this menu. Both color choices are included. Throughout this section a probability assertion is interpreted with $\varepsilon \rightarrow 0$ first, at fixed h , and subsequently $h \rightarrow 0$. Put

$$r_h = h^{1-\kappa}, \quad 0 < \kappa < 1.$$

We will choose κ sufficiently small in Section 7.

6.1 The qualifications at a pin

Definition 6.1 (Qualified pin). Let P be an oriented rim in a bulk patch. A qualified pin consists of a vertex $p \in P$, a grid center z with $|p - z| \leq h$, a placed menu type, its root ranks, and the following data. We describe the black case; interchanging colors gives the white case.

- (i) In the original configuration there are three surrounding gate circuits, in prescribed annuli $[a_k h, b_k h]$, $k = 1, 2, 3$. The first two are the annuli of the plug. The third has $a_3 > 100b_2$ and isolates an ordered central subarc Δ through p : all visits of P to $B(z, a_3 h)$ belong to Δ , and $\text{diam } \Delta \leq 2b_3 h$.
- (ii) In the nominal chart the local ordered traversal of P follows the horizontal midline from before $u = -8$ to after $u = 8$, with transverse error and longitudinal backtracking less than 0.1. The arc Δ lies in $|u| < 1$, and its marked point is near $u = 0$. There are no other visits to H . The fit test has the closed-limit property of Lemma 6.3 below.
- (iii) A prescribed local preparation, applied to the original sample using an auxiliary seed attached to this chart, placement, and color, only recolors existing sites in the opposing strip from black to white, keeping the point configuration fixed. It leaves the fitted black traversal unchanged. Its inner state σ' belongs to G , and the chosen type is admitted. The black root contains vertices of the *actual traversal* near $u = -1.7$ and $u = 1.7$, with error less than 0.05, joined along that traversal in the root window. The two black side passages are also supplied by that traversal. All these paths are retained paths.
- (iv) For the white color separately, each of the two ordinary side-test probabilities, conditional on σ' and the *actual point geometry*, is at least β .

The prepared state is shared whenever the same candidate choice is used; preparations for other choices start again from the original sample.

The last qualification is stronger than the annealed propensity condition in G . It will permit an exterior white forcing while retaining the actual point geometry. We will not need a corresponding quenched lower bound for the black side passages, since those passages are already present on P .

6.2 Three gates and local inverse-diameter tests

Work in units of h , centered at one grid point. Choose integers n and a large constant M , and set

$$A_1 = 30, \quad B_i = M A_i, \quad A_{i+1} = 10^4 B_i, \quad L = B_n.$$

In each trial annulus take slightly inset primal boundary cycles and test for four disjoint arms with both colors represented. The tests use retained graphs in separated annular collars; local density or mesh failure is included as a positive test. Thus different trials use independent restrictions of the marked Poisson process. By Lemma 3.6, each test has limiting probability at most $C M^{-1-c}$, for some $c > 0$.

Suppose a rim has a vertex within distance one of the center and extends well beyond the trials. Besides its own color arm it supplies an opposing arm: the facing neighboring vertices near the mark and beyond the trial belong to one opposing cluster by Lemma 2.3. Trim its connection to the trial annulus. This is a local arm even when the rim is completed on the sphere.

If there are three disjoint black crossings of a trial annulus, this opposing arm gives the positive four-arm test. A negative test therefore permits the two-gate construction of Lemma 3.8.

Call a trial clean when its test is negative. The probability that fewer than three trials are clean, allowing either rim color, is bounded in the microscopic limit by

$$Cn^2(CM^{-1-c})^{n-2}. \quad (6.1)$$

Indeed at least $n - 2$ prescribed trials must then be positive, and there are at most Cn^2 ways to choose the remaining indices. From three clean trials choose gate circuits in their increasing order. Denote the corresponding inset annular radii by a_k, b_k . The large separation allows a cutoff m satisfying all the inequalities in Section 4; it also gives $a_3 > 100b_2$.

A simple black rim visiting the inside and outside of the third circuit meets it at its two black vertices, once at each and nowhere else. Its inner arc Δ contains the marked vertex, is contained in $\overline{B}(0, b_3)$, and contains every visit to $B(0, a_3)$. This description remains true after adding whites away from the traversal: the circuit still has at most two black vertices, and its two contacts with the unchanged traversal remain black.

Extend Δ in both directions to the nearest hits of $\partial B(0, 10L)$ and call the resulting ordered arc K . Mark its two endpoints, the two endpoints of Δ , and the vertex p . The complementary rim travels beyond radius $100L$ for all sufficiently small h in the application. Let E be its portion in $\overline{B}(0, 50L)$, together with $\partial B(0, 50L)$. Include the two endpoints of K in E . This compact set is connected: each component of the truncated complementary path meets the added circle, or is joined to it along that path. A *profile* is (K, E) with these marks and the finite trial-index data. We use marked Fréchet distance for K and Hausdorff distance for E . A Fréchet comparison allows an increasing reparametrization matching the marks.

Figure 4 separates the ordered arc from the complementary data. Adding the outer circle keeps E connected, while the third gate isolates the middle interval Δ from every other visit to the inner region. The inverse-diameter tests below will preserve this separation when profiles converge.

Fix $\eta_j \downarrow 0$, with $\eta_1 \leq 1$. Choose $\delta_j > 0$ successively so small that the following forbidden-return tests have a summably small total probability. Two visits in $\overline{B}(0, 60L)$ at distance at most δ_j must not have departures to distance η_j along *both* intervening directions of the rim. To flag violations use only the local necessary arm events in Lemma 3.7, on a δ_j -grid inside $B(0, 100L)$, together with the relevant retained-mesh tests. Their limiting total probability at level j is at most

$$C(L/\delta_j)^2(\delta_j/\eta_j)^{2+c} = CL^2\delta_j^c\eta_j^{-2-c}. \quad (6.2)$$

It can be made arbitrarily small. The events, unlike the profiles themselves, depend only on this bounded local Poisson restriction. Only a finite initial list of levels will ever be imposed at a fixed h .

Lemma 6.2 (Compactness of profiles). *Every sequence of the above profiles satisfying increasingly long initial lists of the inverse-diameter constraints has a subsequence converging in marked Fréchet and Hausdorff distance to a simple marked arc K and a compact connected set E . The required marks remain distinct. In the limit all visits of K to $\overline{B}(0, 4b_2)$ occur in the interior of its marked middle interval $\Delta \subset \overline{B}(0, b_3)$. The set E misses $\overline{B}(0, 4b_2)$ and misses $K \cap B(0, 6L)$. The family of all such limiting profiles is compact.*

Proof. The complementary portion of the rim makes a fixed large departure. Consequently, if two points of K are sufficiently close, the portion between them in K has small diameter;

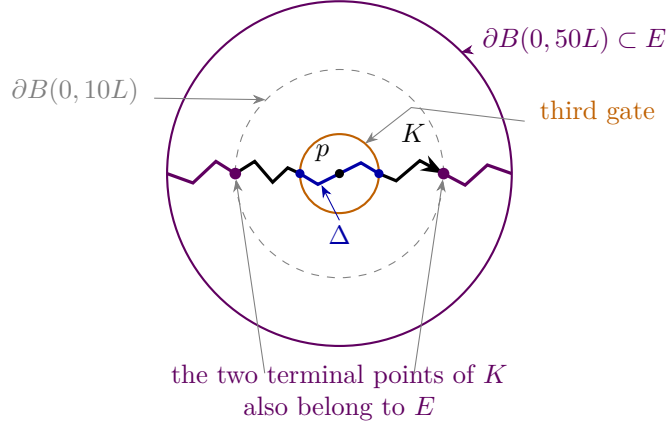


Figure 4: A schematic profile in units of h , with radial scales not to scale. The blue interval Δ is the part of K inside the third gate. The purple set E consists of the complementary rim inside the outer circle, including the two endpoints of K , and the entire outer circle. The omitted complementary rim reaches beyond radius $100L$. Additional excursions are allowed; no smoothness of a profile is assumed.

otherwise both directions would violate one of the imposed constraints. More precisely, for a fixed j , closeness at scale δ_j forces this portion to have diameter at most $4\eta_j$, after requiring $\delta_j \leq \eta_j$. For tests initially formulated at vertices, use the buffered test and mesh smaller than a fixed fraction of each of the finitely many used δ_j 's to obtain the same implication for points on the polygonal arc. The harmless factor four can be absorbed by starting with a correspondingly smaller sequence η_j .

Here is a finite-net argument that does not assume length bounds. Fix a displacement scale $d > 4\eta_j$. Starting at the first endpoint, stop successively when the curve first reaches distance d from the preceding stop, also inserting the marked times. Apart from a bounded number of stops adjacent to inserted marks, different stop starts are δ_j -separated: an intervening displacement d rules out their closeness by the preceding inverse-diameter property. Packing in the fixed disk bounds the number of stops. Every portion between stops is contained in a d -ball about its start. Joining the stops by segments and rounding their positions to a sufficiently fine finite grid gives a finite marked Fréchet net, with error tending to zero as $d \rightarrow 0$.

Choose a rapidly Cauchy subsequence in these nets and align successive parametrizations by increasing homeomorphisms matching the marked times. The aligned curves converge uniformly to a continuous curve. If two of its times have the same image, the intervening portion has arbitrarily small diameter by the inverse constraints, and hence is constant. Every point fiber is thus an interval. Collapse these constant intervals. The quotient is a nontrivial compact connected metrizable linear order, with its order topology: order rays have open preimages, and compactness identifies the quotient topology. Such an order is an interval, as is seen by embedding a countable dense suborder in an interval and completing the order. The induced parametrization is injective and continuous, so the resulting arc is simple. A nondecreasing parametrization with constant intervals can be uniformly approximated by increasing homeomorphisms; with finitely many distinct marks this can be done separately between consecutive marks. More explicitly, after choosing the quotient parametrization so that its nondecreasing map q fixes the normalized mark times, the maps $(1-t)q + t \text{id}$, $0 < t < 1$, are increasing homeomorphisms fixing those times

and converge uniformly to q . Thus the convergence is in the asserted marked Fréchet topology.

The radial requirements keep the marks distinct. For example, if the ends of Δ approached one another, the path through the core would make a fixed departure in one direction and the complementary path would do so in the other, contradicting an inverse constraint. The same argument treats the endpoints of K and every pair of nonadjacent marks; adjacent marks have the radial separation furnished by the core and the two circles. In particular the limit is not constant.

Pass also to a Hausdorff subsequence for E . Connectedness is preserved under Hausdorff limits of nonempty compact connected sets. Before the limit, all visits to $B(0, a_3)$ belong to Δ , and $a_3 > 100b_2$; this gives the assertion about $\overline{B}(0, 4b_2)$ with a margin. If a complementary-rim point approached a point of K in $B(0, 6L)$, both intervening rim portions would pass through a terminal hit at radius $10L$. A fixed inverse-diameter constraint excludes this approach. It excludes a limiting intersection as well. The added circle at radius $50L$ is irrelevant in these inner regions.

Finally take any sequence of limiting profiles. Approximate its j th member closely by a prelimit profile satisfying at least the first j constraints, with the same finite trial data after passing to a subsequence. The argument just proved gives a convergent subsequence of these approximants, and therefore of the limiting profiles. This proves compactness of the limiting family. $\square$

6.3 Models that fit exactly after a small ambient change

Lemma 6.3 (Exact fitting at closed limits). *Each limiting profile in Lemma 6.2 has a nominal plug model, fixed local stubs, a positive allowed chart-change tolerance, and an open fit neighborhood with the following properties. Every fitting prelimit traversal has the strict tracking properties in Definition 6.1. For every simple profile in the closure of this neighborhood there is an orientation-preserving ambient homeomorphism g , supported in $B(0, 50L)$ and within the allowed tolerance, such that its ordered arc agrees exactly with $g\Phi(u, 0)$ through gH and continues along it to $|u| = 9.5$. There are no other visits of the arc or its unwanted set E to gH . A finite collection of such neighborhoods contains all profiles satisfying a sufficiently long finite initial list of constraints.*

Proof. Extend K to a proper line by outward rays at its two distinct endpoints on $\partial B(0, 10L)$. The rays are disjoint and meet the closed disk only at those endpoints. In the one-point compactification of the plane, this proper line together with infinity is a Jordan curve on the sphere. Apply the sphere Jordan–Schoenflies extension theorem [10, Theorems 10.2–10.3, p. 71] to a correspondence from the compactified horizontal axis that fixes infinity. Choose the correspondence in the given longitudinal order and match the two complementary sides to preserve orientation. The extension fixes infinity and hence restricts to an orientation-preserving plane homeomorphism Ψ taking the ordered axis to this line. Prescribe its longitudinal parametrization in the marked order, with p at $u = 0$ and all of Δ strictly within $|u| < 1$. Assign $u = -16$ and $u = 16$ to the first extensions of Δ in the two directions that reach radius $2L$. The central axis segment through these parameters lies in $\overline{B}(0, 2L)$; by continuity, for some $d > 0$, $\Psi([-16 - d, 16 + d] \times \{0\}) \subset B(0, 3L)$. This compact segment has positive distance from E . Its parameter interval contains the chart interval $[-12, 12]$, the later cuts at $u = \pm 11$, and their longitudinal buffers. The rest of the line follows the remaining parts of K and, beyond its endpoints, the prescribed outward rays to infinity. Outside the marked middle interval the line misses $\overline{B}(0, 4b_2)$.

For sufficiently small $\tau > 0$, the chart

$$\Phi(u, v) = \Psi(u, \tau v), \quad (u, v) \in [-12, 12] \times [-3, 3],$$

lies in $B(0, 4L)$ and misses E ; its section at zero is strictly inside $B(0, a_1)$. We describe the state and obstacle disks to ensure the prescribed exact slab intersections. In straight coordinates take nested polygonal Jordan neighborhoods approximating the preimages of the disks of radii $2b_2$ and $3b_2$, inside the preimage of $B(0, 4b_2)$. They contain the preimage of $\bar{B}(0, b_2)$ with a margin and avoid a thin strip about the axis for $|u| \geq 2$. Adjoin thin horizontal rectangles with longitudinal intervals $[-3, 3]$ and $[-8, 8]$, respectively, and fill bounded complementary components. The polygons and rectangle widths can be chosen strictly nested and with transverse intersections of boundaries. Their unions have connected interiors and disjoint simple polygonal boundary circuits; filling the holes leaves a single outer boundary. They are therefore nested Jordan disks.

Choose 3τ smaller than these rectangle half-widths. Beyond the specified vertical cuts the remaining thin axis strip escapes to infinity without meeting either union. Hole filling consequently does not change its intersections with this strip. Their images O, H have precisely the slab intersections $|u| \leq 3$ and $|u| \leq 8$ required in Section 4. They miss E : before filling they miss this connected set, which reaches the outer circle, and hence E lies in the unbounded complementary component. In physical coordinates the unions lie in $B(0, 6L)$, and their filled hulls remain there because the exterior of that disk is connected. Fix stubs and a small positive chart-change tolerance as in Sections 4 and 5. Decrease the tolerance to preserve all compact window margins; in particular nominal coordinates in the terminal windows change by less than 0.1. The support of the ambient change is allowed inside $B(0, 50L)$.

It remains to justify exact fitting, rather than only close tracking. Consider simple profiles converging to the nominal profile in the marked topology. In Ψ -coordinates take the subarc from its last hit of $u = -11$ before the core mark to its first hit of $u = 11$ after the mark. For close enough profiles this is a proper left-to-right crosscut of a fixed thin rectangle. Its interior lies between the vertical sides, and it stays away from the horizontal sides. Fréchet convergence controls its order and backtracking, so these crosscuts converge in parametrized boundary distance to the axis crosscut.

There are homeomorphisms of this rectangle converging uniformly to the identity that take its midline, in order, to the new crosscuts. Here is the extension argument. Split the rectangle along the old midline and the new crosscut. On the boundaries of the two old half-rectangles prescribe correspondences to the two new halves, using the matching crosscut parametrizations, the identity on the horizontal sides, and piecewise linear increasing height maps on the vertical sides. These maps agree on the common cut and converge uniformly to the identity. Each new half is a Jordan domain, and its parametrized boundary converges uniformly to that of the old half. Choose a fixed interior point of each old half, which belongs to every sufficiently close new half, and normalize the conformal maps at that point with positive derivative. Radó's uniform convergence theorem for Jordan domains with Fréchet-convergent boundaries [14, §§1–2, pp. 182–183] gives convergence of these maps on the closed disk. Write them as F_n, F , and write h_n for the prescribed boundary correspondence. The circle homeomorphisms $\chi_n = F_n^{-1}h_nF$ converge uniformly to the identity: for $\zeta \in \partial\mathbb{D}$,

$$|F(\chi_n(\zeta)) - F(\zeta)| \leq \|F_n - F\|_\infty + \|h_n - \text{id}\|_\infty,$$

and the inverse of the fixed closed Jordan chart F is uniformly continuous. Extend these circle homeomorphisms radially by $re^{i\theta} \mapsto r\chi(e^{i\theta})$; these are disk homeomorphisms with the same

uniform convergence. Conjugation by the conformal charts gives the desired half-rectangle homeomorphisms, with the prescribed exact values on the common cut and uniform convergence to the identity. The cut parametrizations can match all three interior marks: interpolate the small changes of their terminal parameters by increasing piecewise linear maps fixing those marks. Glue them along the cut. Interpolate the small vertical-side height changes to the identity in thin exterior collars, keeping the top and bottom fixed. The result extends by the identity to a plane homeomorphism. After conjugating by Ψ , its support can be kept in $B(0, 50L)$ and its displacement tends uniformly to zero.

The portions of the ordered arc omitted by this crosscut correspond, for close enough profiles, to nominal axis positions with $|u| > 10.5$. The original compact margins separate them from the changed obstacle and the central track through $|u| \leq 9.5$. The unwanted set E is separated there as well. Thus every sufficiently close simple profile admits the required change within any prescribed positive tolerance. First obtain this conclusion on a neighborhood, and then choose a smaller neighborhood whose closure among simple profiles is contained in it. This gives the stated closed-limit property and the strict prelimit tracking requirements.

Cover the compact family of limiting profiles by finitely many such open neighborhoods. If no finite initial constraint list forced membership in their union, one could choose a profile outside the union satisfying the first j constraints for every j . Lemma 6.2 would give a subsequential limit inside the union, contradicting openness. Thus one finite list suffices. The construction uses only normalized profile geometry and the finite trial data, so the same finite menu works at every sufficiently small h . $\square$

On fine retained mesh a fitting traversal supplies vertices near $u = \pm 1.7$ joined along that rim within $\Phi([-2, 2] \times I_B)$, and supplies both side passages in the nominal windows. The transverse and backtracking margins, rather than a smoothness or transversality assumption on the rim, give these conclusions. All visits near the second gate annulus belong to Δ .

6.4 Local preparations, ranks, and propensities

For each of the finitely many chart and color choices at a grid center, apply a separate preparation to the original points and colors. In the black case it acts only in a narrow substrip of $\Phi([-7.5, 7.5] \times I_W)$. It forces a retained white walk from about $u = -7.1$ to $u = 7.1$ with longitudinal backtracking less than 0.2. Choose a buffered chain of crossing rectangles along the substrip, using overlapping progress steps whose chart-longitudinal widths are much smaller than 0.2. Crossings in both directions in the overlaps join the successive walks. They can be concatenated in order, with the desired width and progress margins, through both side-test terminals. The corridor argument of Section 4 and the quenched rectangle bounds give, on a local geometry event with probability tending to one, a conditional success probability at least $p_* > 0$. Since there are finitely many normalized models, take a common positive lower bound. The good event can be the retained-mesh event and this lower-propensity event itself.

The preparation uses the finite ordered-coupling principle; see [7, Theorem 4.1 and its finite-network discussion]. For completeness, the finite-spin construction needed here is as follows. If A is increasing under the white order for a finite product measure π , positive association implies $\pi(B \mid A) \geq \pi(B)$ for every increasing set B . There is therefore a coupling (ω, ω') with laws $\pi, \pi(\cdot \mid A)$ and $\omega \leq \omega'$. One may obtain it by a finite flow network: supply $\pi(\omega)$ at each initial state, demand $\pi(\omega' \mid A)$ at each final state, and allow an edge exactly when $\omega \leq \omega'$. The inequalities for increasing sets are precisely the cut inequalities, and the max-flow criterion

supplies the coupling. Its conditional transition law is a kernel that only recolors existing black sites white, with the point geometry fixed. Fix a deterministic choice of flow for each finite ordered truth table and probabilities; this also ensures measurability when the point configuration varies. Each bounded local region contains finitely many sites almost surely.

Activate this kernel solely according to its local geometry event, using an independent seed, and otherwise do nothing and flag the location. In particular its activation does not condition on a fitting rim. Given the geometry its output law is bounded by $\max(1, p_*^{-1})$ times ordinary spinning. Writing σ^{prep} for the full prepared coloring, its joint law with the unchanged point configuration consequently satisfies

$$\mathcal{L}(\eta, \sigma^{\text{prep}}) \leq \max(1, p_*^{-1})\nu,$$

as measures, uniformly in h . Different candidate outputs need not have a jointly dominated law; we use the bound for each individual preparation, and for two disjoint preparations with independent seeds. The flipped construction recolors existing white sites black in the opposing strip for white rims, again fixing the point geometry.

At a fit the preparation avoids the rim. It keeps both gate circuits with at most two black vertices and supplies a single white traversal whose root portion and side portions are compatible. Together with the original black traversal, some pair of root ranks therefore has both side successes. The ranks can be truncated uniformly with arbitrarily small error. Indeed distinct crossing components of a root window supply vertex-disjoint same-color paths across its middle slab. On good geometry an opposing transverse bar has quenched probability at least $q > 0$. Thus the probability of one such monochromatic crossing is at most $1 - q$, and the conditional BK inequality bounds the probability of k disjoint crossings by $(1 - q)^k$. Transfer this bound to the prepared output using its density bound, and discard the negligible geometry failure. A fixed large rank cutoff works for the finite collection of windows.

We next discard actual successes whose ordinary conditional propensities are too small. For any event A and sigma field $\mathcal{F}$,

$$\nu(A \cap \{\nu(A \mid \mathcal{F}) < t\}) = \mathbb{E}_\nu[\mathbf{1}_{\{\nu(A \mid \mathcal{F}) < t\}} \nu(A \mid \mathcal{F})] \leq t. \quad (6.3)$$

Apply this first to L^+ and L^- conditional on the inner state, and then to each white side success conditional also on the local exterior point geometry. Transfer each estimate by the preparation density and take a finite union over the retained ranks and models. Choosing $\beta > 0$ small makes the total cost arbitrarily small. Finally, for each resulting type discard its actual prepared occurrence of G when $\nu(G) < \beta'$. Its probability is at most $C\beta'$ per type. Choose $\beta' > 0$ after the finite rank truncation. The remaining prepared states have exactly the qualifications in Definition 6.1.

All discard flags are local. A side-test propensity conditional on the inner state is a measurable function of that state and fixed local test; its quenched counterpart additionally uses only the exterior point restriction needed by its retained paths. The preparations, rank counts, actual side successes, and admission indicators use no hypothetical long connector. Their range is bounded by CL in normalized units. Stubs are not queried by these flags.

6.5 From rare flags to dense pins

Proposition 6.4 (Density of qualified pins). *There exist a finite normalized menu and fixed $\beta, \beta' > 0$ such that the following holds in every bulk patch with $\overline{B}(x, 6R)$ available. With probability*

tending to one in the ordered limit $\varepsilon \rightarrow 0$, then $h \rightarrow 0$, every subarc of diameter at least r_h in $\overline{B}(x, 2R)$ on a rim reaching outside $B(x, 4R)$ contains a qualified pin. The statement holds simultaneously for all such subarcs and both colors. It also holds for the coupled bulk settings on their patch-agreement event.

Proof. For either color, the preceding local tests have the following deterministic implication on patch agreement and good mesh: every eligible marked rim vertex within distance one of a grid center fits some qualified choice unless that center is flagged. The trial part of its limiting flag probability is (6.1). For fixed L all other costs can be made arbitrarily small. The order of choices is essential. Choose M, n first. Allocate a summable inverse-test budget, which determines the finite profile menu and its finite constraint list. For that menu fix the preparation corridors and their positive lower bounds, then truncate ranks, and finally choose the two propensity thresholds. Every choice is in normalized units and independent of h . Admission itself need not stabilize here.

Tile the plane by blocks of side Lh . A block is bad if its enlargement by distance $2Lh$ contains at least c_0L flagged grid centers, where $c_0 > 0$ is a sufficiently small absolute constant. There are $O(L^2)$ candidate centers in that enlargement, so the first-moment bound gives

$$\limsup_{\varepsilon \rightarrow 0} \mathbb{P}(\text{a specified block is bad}) \leq CL[Cn^2(CM^{-1-c})^{n-2} + \zeta], \quad (6.4)$$

where ζ is the sum of the adjustable additional costs. Since $L = 30M(10^4M)^{n-1}$, first taking M large and then n large makes the contribution $Ln^2(CM^{-1-c})^{n-2}$ as small as desired. Choose ζ afterward. The dependence range of these block events is an absolute constant in block units, including the independently assigned preparation seeds.

An eligible subarc of diameter at least r_h without a qualified pin forces a connected chain of bad blocks of length at least $cr_h/(Lh)$. Choose two points of the subarc at distance at least $r_h/2$ and follow the portion between them. Since $r_h/(Lh) \rightarrow \infty$, at least one of these endpoints lies outside the $2Lh$ enlargement of each visited block. From a visit to the block, follow the subarc toward that endpoint until it first leaves the enlargement. Even if the visit only clips a corner of the block, this portion travels distance at least $2Lh$. On fine mesh, its initial passage over distance Lh supplies vertices within distance h of at least c_0L distinct grid centers, all still inside the enlargement. Every such center is flagged by the deterministic implication above. Thus every visited block is bad. The visited blocks connect the two chosen endpoints; erasing repetitions gives a chain of distinct neighboring bad blocks, whose length is at least $cr_h/(Lh)$ by their separation.

There are exponentially many block chains of a specified length, and only $O(h^{-2})$ possible starting blocks in the fixed patch. From any chain of length k choose at least k/C_0 blocks whose flag-data regions are pairwise disjoint. Their bad events are independent under the full-plane marked Poisson law with the local seeds. If the bound in (6.4) is sufficiently small, exponential path counting therefore gives an upper bound of the form

$$C_R h^{-2} \exp\{-cr_h/(Lh)\} \rightarrow 0.$$

At each fixed h the chain length and all local scale lists are finite; take the microscopic limit before this estimate. Discarding the vanishing patch-agreement and mesh failures completes the proof. $\square$

7 Transport along actual arcs

We first transfer the fixed-connector rule to arcs in the original random configuration, retaining their assigned prepared endpoint states. We then pass from arc contributions to smooth weights. The second step requires a separate control of signed sums and completes bulk transport.

Fix a countable sequence $h \downarrow 0$. Starting with any microscopic sequence, make the subsequence choices of Section 5 simultaneously for these h 's. The grid translates of the menu, their representatives, and their integer winding classes form countable lists; admissions can be stabilized for each h because there are finitely many types. All microscopic limits below are taken on this one further subsequence. In particular the scalars a_h^Q are bounded uniformly in h and $a_h^V = 1$. Comparisons involving an empty set of admitted types are vacuous.

We continue to work in a bulk patch containing $\overline{B}(x, 6R)$. Every qualified pin is used with a specified fitting test, type, ranks, and the prepared state assigned in Section 6. No comparison below changes that assigned state.

For two ordered pins s, t on P , write $D_s = B_s = \overline{B}(z_s, m_s h)$ for the previously defined placed cutoff ball at s , and similarly $D_t = B_t$ at t . Define

$$I_X(P; s, t) = X_P(\mathbf{1}_{D_s}) + \sum_{\substack{e \text{ on the forward arc } p_s \rightarrow p_t \\ \text{right tag of } e \notin D_s \cup D_t}} X_P(e), \quad X \in \{Q, V\}.$$

Here $X_P(e)$ denotes the contribution of the oriented edge with its right tagged triangle. The first term is the *entire* start cutoff mass, not only the portion after p_s . This convention will make concatenation exact.

7.1 Two conditional-measure and locality facts

Lemma 7.1 (A dominated forcing kernel). *Let ν be a probability measure on configurations, let Z be retained data, and let ν_z be a regular conditional law given $Z = z$. Suppose H is a measurable set of retained data and $p(z) = \nu_z(W) \geq q > 0$ on H . Let K fix Z and satisfy $\nu_z K_z = \nu_z(\cdot \mid W)$ there. If an unnormalized subprobability λ satisfies $\lambda \leq C\nu$ and is supported over H , then*

$$\lambda K \leq (C/q)\nu.$$

For the additional monotone assertion, suppose that, conditional on $Z = z$, the remaining coordinates have law $\nu_z = \pi_z \otimes \tau_z$, where π_z is a product spin law on a finite, Z -measurable set $I(z)$ of exterior sites and τ_z is the law of the complementary data. Suppose also that W depends only on Z and the spins in $I(Z)$ and is increasing in white on those spins. Then K can be chosen to recolor only existing black sites in $I(Z)$ white and to fix Z and every complementary coordinate.

Proof. For every nonnegative measurable function g , positivity of a kernel and disintegration give

$$\begin{aligned} \int \lambda(d\omega) K g(\omega) &\leq C \int \mathbf{1}_H(Z(\omega)) \nu(d\omega) K g(\omega) \\ &= C \int_H \nu^Z(dz) \frac{1}{p(z)} \int \mathbf{1}_W(\omega') g(\omega') \nu_z(d\omega') \\ &\leq (C/q) \int g d\nu. \end{aligned}$$

No assumption is made about the conditional law of λ given Z . Under the additional hypothesis, association of π_z shows that $\pi_z(\cdot \mid W)$ stochastically dominates π_z in the white order. The finite ordered coupling construction in Section 6, applied on each retained-data fiber, gives a measurable white-adding kernel for this block. Independence of the complementary data lets us use the identity there: the output law is $\pi_z(\cdot \mid W) \otimes \tau_z = \nu_z(\cdot \mid W)$. $\square$

Lemma 7.2 (Retention with collars). *Fix h and finitely many deterministic path envelopes and state disks. Suppose the point data are kept on a deterministic Borel set containing an open collar of every envelope and of the state portions under comparison. Set $\rho_\varepsilon = \sqrt{h\varepsilon}$. For small ε , every Delaunay triangle of circumradius at most ρ_ε meeting these envelopes, together with its incident edge and tag data there, is determined by the kept point restriction. Thus two configurations with the same restriction have the same such local triangulations.*

In a fixed two-plug comparison one can keep the two endpoint states and the whole exterior sector of the used connector. The inset paths of the cycle experiment and any sufficiently closely tracking actual arc, including their inner second-gate paths, then have the common triangulations just described on the ordinary mesh-good events. The complementary connector can be resampled without altering them.

Proof. A Delaunay triangle is determined by its three vertices and emptiness of its circumdisk, apart from the null degeneracies excluded in our model. If its circumradius is at most ρ_ε and it meets a fixed envelope, its circumdisk lies within a constant multiple of ρ_ε of that envelope. This is eventually inside its kept collar. The same observation determines all retained triangles incident to its local edges. On mesh-good events all triangles needed for the ordinary local paths and their right tags satisfy this bound, by Lemma 3.3. This proves the first assertion without any assumption on the area of the boundaries of the kept sets.

For the second assertion, the paths in the cycle experiment lie in fixed inset layers of the embedded annular band. Outside states these layers, the tube-test footprints, and the joining seams have positive margins to the lateral boundaries of the used sector. Inside states, the relevant second-gate paths and cutoff triangles lie in a compact subset of O , since the gate annuli and cutoff balls have the prescribed radial margins. The same envelopes contain the actual paths when the track fit is sufficiently close. Their exterior portions and state portions therefore have collars within the kept union, as also in Lemma 4.10. The other sector queries disjoint exterior restrictions. Deterministic Borel allocation of boundaries suffices; no smooth-boundary convention is needed. The first assertion now applies in both configurations. $\square$

7.2 Winding and a finite deterministic family of comparisons

Lemma 7.3 (Winding bound). *Except on events of probability tending to zero in the ordered limit, every subarc of a rim in $B(x, 3R)$ that stays at distance at least h from a grid center in use has absolute net change of continuous argument about that center at most $C_R(1 + |\log h|^2)$.*

Proof. Fix a ray from the center and divide its relevant distances into dyadic bins $d \asymp 2^j h$, up to scale $O(R)$. In each bin a bounded number of balls of radius a small fixed fraction of d cover possible ray passages. Give each ball a buffered outer ball of comparable radius, all within the bulk patch. A passage off the ray followed by a quarter turn leaves one of these outer balls, and so gives a monochromatic crossing of its fixed-ratio annulus. Successive full turns contributing to a net argument change can be selected by first hits of successive levels of a lifted argument.

Their portions between the relevant passage and departure times are disjoint on the simple rim; trimming with the fixed margins yields vertex-disjoint annular crossings on fine mesh.

On crossing-good geometry an opposite-color circuit in such a test annulus has probability at least a fixed $q > 0$. Hence one monochromatic crossing has probability at most $1 - q$, and conditional BK bounds the probability of k disjoint crossings by $(1 - q)^k$. There are $O_R(h^{-2})$ centers and $O(1 + |\log h|)$ bins per center. At each fixed h the geometry exceptions for this finite list vanish as $\varepsilon \rightarrow 0$. A union bound with $k = C_R(1 + |\log h|)$ leaves probability tending to zero as $h \rightarrow 0$. Summing over the bins bounds the net number of turns by $C_R(1 + |\log h|^2)$, and the incomplete initial and final turns add only a constant. $\square$

Fix h for the moment and an arbitrarily small probability budget $\zeta > 0$. Impose the local inverse-diameter tests from Section 6 throughout $\overline{B}(x, 3R)$, to a finite number of accuracy levels to be chosen below, at total limiting cost less than ζ . Track a qualified arc from s to t together with its extension from the start of K_s to the end of K_t , saving both marked endpoint profiles. Assume the original arc lies in $\overline{B}(x, 2R)$, $|z_s - z_t| \geq R/3$, and its complementary rim reaches outside $B(x, 4R)$. For small h the endpoint neighborhoods of size $10Lh$ are disjoint and occur in the required order, so the extended arc lies in $B(x, 5R/2)$ and its complement still makes a fixed large departure.

The proof of Lemma 6.2, in this fixed physical region, applies to these extended marked arcs. Along sequences satisfying increasingly many constraints it gives simple Fréchet limits, simultaneously with their endpoint profile limits. At fixed h the placements, templates, and retained ranks lie in finite lists. Pass to fixed choices. The endpoint limits are in the closures of their assigned fit neighborhoods. A visit near an end obstacle not belonging to its prescribed K is included in that profile's E : the relevant neighborhood is well inside the radius $50Lh$ truncation.

Lemma 6.3 supplies independently supported permitted changes of the two end charts, making the limiting extended arc the exact centerline through both changed obstacles and for a further collar on either side. It avoids the obstacles elsewhere. The band-and-collar construction of Section 4 now fits a connector to the intervening simple arc, with the full prescribed port sections, inside the bulk patch. A complementary connector exists in the plane; it need not lie in the patch. The two-twist classification applies: a net argument along a limiting arc separated from a center persists under close Fréchet approximation. Thus Lemma 7.3 bounds the two winding integers by $C_R(1 + |\log h|^2)$, up to a bounded contribution from the fixed stubs and allowed chart changes. Since $|z_s - z_t|/h \geq R/(3h)$, the distance requirement in Proposition 5.5 holds for small h .

This fixed connector works on an open neighborhood of the limiting track data. In particular the actual root vertices near nominal $u_s = 1.7$ and $u_t = -1.7$ are joined by the actual rim within its black tube. The corresponding compact portion of the limiting centerline is interior to that tube and is well beyond the longitudinal cutoffs $u_s = 0.5$ and $u_t = -0.5$. Small allowed chart changes preserve the nominal root positions and windows. The product-band seams are interior, so a sufficiently close ordered fit and fine mesh preserve all these margins, including retention of the required paths.

There is a further gate-order issue. Use the first and second circuits selected from the prepared states. Every contact with the actual rim lies on its isolated central traversal and remains black after preparation. Each selected circuit has at most two black vertices in that prepared state, so it has at most two contacts with the original rim. Passage from inside to outside requires two; there are therefore exactly two contacts, traversed once. This argument applies even if a selected circuit differs from the original gate choice. A full chart slab about zero lies in the inner gate

ball. The limiting track enters this slab from the negative side and leaves on the positive side; its contacts with a surrounding gate lie away from the slab. On a sufficiently close ordered fit their signs cannot change. Thus the first exit at s and the first entry at t are the designated positive and negative gates. The intervening path remains in the used sector, with room in its interior together with the endpoint states, and avoids the opposing layer. It cannot recross a central section while outside the corresponding first gate. The inner second-gate paths also track the local centerline inside their states. This argument uses the progress through a slab, not a transversal crossing of a discrete edge with a curve, and is uniform over the possible locations of the circuits in their annuli.

Compactness and the finite-list argument in Lemma 6.3 now give a finite family of these deterministic patterns whose open neighborhoods cover all tracks satisfying a sufficiently long finite constraint list and the winding bound. Patterns are chosen from limiting deterministic track data before the random comparison; a new band is not selected after observing the colors. Only finitely many placed patterns, endpoint types, and rank choices occur at this fixed h . The open track tests and existential path conditions are measurable: graph paths are countable given the locally finite graph, and the marked fits use the usual Fréchet open sets.

7.3 The comparison with a cycle experiment

Proposition 7.4 (Transport on a real arc). *Simultaneously for $X = Q, V$ and every qualified ordered arc from s to t contained in $\overline{B}(x, 2R)$, with $|z_s - z_t| \geq R/3$ and complementary rim reaching outside $B(x, 4R)$, one has*

$$\left| I_X(P; s, t) - a_h^X(z_t - z_s) - \phi_s^{X,+}(\sigma'_s) - \phi_t^{X,-}(\sigma'_t) \right| \leq C_R h(1 + |\log h|^2) \quad (7.1)$$

outside events whose probability tends to zero in the ordered limit. The constant is independent of the complexity and conditioning constants of the individual fixed comparison patterns.

Proof. We construct one comparison for all qualifying arcs of each pattern in the finite deterministic cover. Write ξ for the original input, with law P , including the bulk/full-plane coupling and all assigned local preparation seeds. Let E_0 be the common input event on which patch agreement and the covering and winding conclusions hold, intersected with the original mesh and continuation-geometry conditions of all patterns. These additional conditions have total failure probability $o(1)$ at fixed h . The covering exception will be paid once, through $P(E_0^c)$.

Fix a placed connector $e : s \rightarrow t$, a complementary connector c , and their endpoint choices. Let $\mathcal{C}_e(\xi)$ be the collection of all qualifying ordered rim arcs of this pattern, including their assigned preparation outcomes. Put

$$A_e = \{\mathcal{C}_e(\xi) \neq \emptyset\}, \quad A_{e,0} = A_e \cap E_0.$$

We retain the original input throughout and work with the *unnormalized* restriction $P|_{A_{e,0}}$; its mass is never divided by the probability of a fit. We describe the construction for black arcs; for white arcs exchange the colors.

Prepare the endpoints and force the white connection. Let $F_e(\xi)$ be the underlying plane configuration after the two specified local white-adding preparations, with their already assigned seeds. Their supports are disjoint. Before restriction, independence of the seeds conditional on

geometry and the local preparation bounds give a configuration marginal bounded by $C_{\text{prep}}\nu$. Hence

$$\lambda := F_{e*}(\mathbf{P}|_{A_{e,0}}) \leq C_{\text{prep}}\nu.$$

Restriction only deletes mass, even when it inspects actual arcs, other preparations, seeds, and patch data. The endpoint states of $F_e(\xi)$ are precisely the assigned $\sigma'_s(\xi), \sigma'_t(\xi)$.

Define Z on the ordinary configuration space to be the coordinate map consisting of the full point geometry η and the colored configuration restrictions to the two state regions O_s, O_t . On a prepared output these last two coordinates have the assigned values σ'_s, σ'_t . Thus $\nu(\cdot | Z)$ refers to ordinary coordinate restrictions, not to conditioning on the outcome of a preparation algorithm under its input law. Let W be the white part of S_e . Under the ordinary law conditional on Z , the exterior colors are independent fair bits. Each needed white side event has probability at least β by qualification. These two endpoint events use disjoint exterior bits. The white continuation network uses the buffered exterior corridors of Lemma 4.6. On its good point geometry, association in the white order therefore gives

$$\nu(W | Z) \geq q_{\text{cont}}\beta^2 =: q_0 > 0. \quad (7.2)$$

The local qualification is unchanged by additionally conditioning on the rest of η , since its retained side tests query only their specified local point restrictions. The continuation-geometry conditions were included in E_0 and have vanishing failure probability; the preparations keep the point geometry fixed.

Let H be the resulting set of Z values on which (7.2) holds. For this fixed pattern let $I = I(\eta)$ contain *all potential exterior sites* in its buffered white footprint on which W can depend. This finite block is selected from the geometry before colors or witnesses are inspected; it is not an adaptive query transcript. Every site in it is separated from the permitted black track layer. Conditional on Z , its fair product spin law is independent of all complementary exterior spins, and W depends only on Z and this block. Equivalently, with $\tilde{Z} = (Z, \sigma_{\text{ext} \setminus I})$, these spins remain independent fair bits and $\nu(W | \tilde{Z}) = \nu(W | Z) \geq q_0$ over H . The monotone clause of Lemma 7.1 therefore applies to λ , which is supported over H . Use its kernel K_e^W to add white only in I , fixing every other coordinate. The lemma yields

$$\lambda K_e^W \leq (C_{\text{prep}}/q_0)\nu.$$

This conclusion does not assume that the selected input has ordinary conditional exterior spins. Its conditional law may be arbitrary; the subprobability domination is what permits the forcing. The kernel only removes black sites in the separated white footprint. It preserves every track in $\mathcal{C}_e(\xi)$, with its buffered collars and fixed states. Thus the output satisfies S_e .

Complete the configuration while retaining the used sector. Keep the endpoint states and the whole exterior e -sector datum, denoted together by M_e . Under the ordinary kept-data law ν_{M_e} , the corresponding marginal ρ_{M_e} of the two-plug cycle experiment has density

$$\frac{d\rho_{M_e}}{d\nu_{M_e}} = \frac{\mathbf{1}_{G_s} \mathbf{1}_{G_t} \mathbf{1}_{S_e}}{\nu(G_s)\nu(G_t)\nu(S_e | \sigma'_s, \sigma'_t)}. \quad (7.3)$$

The denominator is positive by admission and (4.2), and is at most one. The post-forcing marginal is supported on the three events in the numerator and dominated by $(C_{\text{prep}}/q_0)\nu_{M_e}$; it is therefore dominated by $(C_{\text{prep}}/q_0)\rho_{M_e}$. Apply to both measures the same completion kernel:

sample the other exterior connector region, geometry and colors, conditional on S_c and the two states, and sample the remaining plane ordinarily. This keeps M_e and the collars of the compared paths unchanged. Denote the composition of K_e^W with this completion kernel by $\mathcal{K}_e$, extending it arbitrarily off the relevant support. On the original input and final configuration together, define

$$\Pi_e(d\xi, d\omega') = \mathbf{1}_{A_{e,0}}(\xi) P(d\xi) \mathcal{K}_e(F_e(\xi), d\omega').$$

Its input marginal is exactly $P|_{A_{e,0}}$. The construction gives the two properties needed below:

$$\begin{aligned} (\text{pr}_{\text{out}})_* \Pi_e &\leq D_e \rho_e, \\ (\sigma_s(\omega'), \sigma_t(\omega')) &= (\sigma'_s(\xi), \sigma'_t(\xi)) \quad \Pi_e\text{-almost surely.} \end{aligned}$$

Here ρ_e is the two-plug cycle law and $D_e < \infty$ is fixed at fixed h and pattern. The first follows by applying the same completion kernel to the dominated kept-data marginals; the second holds because both kernels fix the endpoint states. No product domination of the joint law is required.

Compare every fitting arc to one output rim. Choose one experiment rim $P'_e = P'_e(\omega')$ by a measurable rule depending only on the output. There is one event $\mathcal{G}_e$ in this joint space, independent of a choice of $\gamma \in \mathcal{C}_e(\xi)$, such that $\Pi_e(\mathcal{G}_e^c) = o(1)$ and the following path comparisons hold. Use the mesh and shielding conditions on the deterministic state and sector envelopes in both configurations, and the mesh conditions for the confined cycle-rim construction. Their original failure mass is bounded by the original mesh probability; their final failure mass is bounded by D_e times the cycle mesh probability. The strict track and gate-order margins hold for every member of $\mathcal{C}_e(\xi)$ by the one fixed pattern's fit test. Thus the good event and the output kernel are not chosen separately for different arcs.

On $\mathcal{G}_e$, the first-exit to first-entry path of P'_e and its inner second-gate paths are black in the original graph. Colors only changed from black to white before completion, and completion resamples outside their kept envelopes. Conversely, the corresponding portions of every input arc in $\mathcal{C}_e(\xi)$ remain black in the final graph: the preparations and white forcing avoid their permitted track layer, and completion keeps the whole used sector and endpoint states. Lemma 7.2 gives the common complete incident triangulations and gate fans. Since $b_2 < a_3$, the first and second gates lie strictly inside the unchanged third circuit. Third-gate isolation puts every original-rim contact with a gate selected from the prepared state on the untouched central traversal. The selected gate has at most two path-color vertices, while each fitting rim visits both sides, so every such rim uses precisely those two contacts. The output rim does too, and ordered passage through the central slab gives the same signs; this also covers a selected gate differing from the originally exhibited gate. Lemma 4.9 now identifies the inner second-gate paths and the outside path, using the inside and outside starting fans, respectively. At a first divergence a mutually black compared path would give a forbidden bypass up to its first subsequent contact with the whole other rim. This argument applies in each graph separately and to every input arc with the same $P'_e(\omega')$; the original rim need not remain a complete rim after modification.

Set

$$H_e^X(\omega') = \widehat{m}_s(P'_e(\omega')) + \widehat{j}_e(P'_e(\omega')), \quad X = Q, V.$$

All cutoff tags belong to the common inner second-gate paths. The marks and inner first-gate paths lie inside the cutoffs, and the outside-cutoff mass lies on the common outside path. Thus on the same event $\mathcal{G}_e$,

$$I_X(\gamma; \xi) = H_e^X(\omega') \quad \text{for all } \gamma \in \mathcal{C}_e(\xi), \quad X = Q, V, \quad (7.4)$$

where $I_X(\gamma; \xi)$ is the original $I_X(P; s, t)$. In particular the right side is one output random variable, not a separately coupled variable for each input arc.

Return the estimate to the original input. For endpoint states $\sigma = (\sigma_s, \sigma_t)$ write

$$\Theta_e^X(\sigma) = a_h^X(z_t - z_s) + \phi_s^{X,+}(\sigma_s) + \phi_t^{X,-}(\sigma_t).$$

Separate the cycle-mass error from the retained-state remainder:

$$\begin{aligned} C_e^\varepsilon(\omega') &= \max_X |H_e^X(\omega') - i_e^\varepsilon(\sigma_s(\omega'), \sigma_t(\omega'))|, \\ R_e^\varepsilon(\sigma) &= \max_X |i_e^\varepsilon(\sigma) - \Theta_e^X(\sigma) - b_e^{X,\varepsilon}|. \end{aligned}$$

Here i_e^ε in each term is the function for field X , and $b_e^{X,\varepsilon}$ is its deterministic bias. Equations (4.5) and (5.3) give

$$\mathbb{E}_{\rho_e} C_e^\varepsilon = o(1), \quad \mathbb{E}_{\mu_s \otimes \mu_t} R_e^\varepsilon = o(1).$$

The input law of its two prepared states, restricted to $A_{e,0}$, is bounded by $C_{\text{prep}} \mu_s \otimes \mu_t$. Indeed the endpoint marginal of λ is bounded by C_{prep} times the independent ordinary state product and is supported on $G_s \cap G_t$; restricting that product gives $\nu(G_s) \nu(G_t) \mu_s \otimes \mu_t \leq \mu_s \otimes \mu_t$. The pathwise retained-state equality makes $\Theta_e^X(\sigma'_s(\xi), \sigma'_t(\xi))$ exactly the target computed from the output states.

Let $B_e^\varepsilon = \max_X |b_e^{X,\varepsilon}|$ and fix $u > 0$. On $\mathcal{G}_e$ the error of every fitting input arc is at most

$$B_e^\varepsilon + C_e^\varepsilon(\omega') + R_e^\varepsilon(\sigma'_s(\xi), \sigma'_t(\xi)).$$

Consequently the original-input probability, restricted to $A_{e,0}$, that any such arc has error greater than $B_e^\varepsilon + 2u$ is at most

$$\Pi_e(\mathcal{G}_e^c) + D_e \rho_e(C_e^\varepsilon > u) + C_{\text{prep}}(\mu_s \otimes \mu_t)(R_e^\varepsilon > u) = o(1).$$

This is a projection of a pointwise assertion on one joint space; no union over input arcs is taken.

On the common event E_0 , every eligible arc belongs to at least one of the finitely many patterns. Apply the preceding estimate to each pattern on the same original input space and take a finite union. The total input failure probability is bounded by $\mathbb{P}(E_0^c)$ plus the sum of the three displayed error terms over the fixed N_h patterns. The common covering exception is therefore paid once; no term counts individual arcs. All pattern constants and N_h are fixed before the microscopic limit. Finally $\limsup_{\varepsilon \rightarrow 0} B_e^\varepsilon \leq Ch(1+k)$ and $k \leq C_R(1 + |\log h|^2)$. Taking $u = h$ absorbs the residual tolerance into (7.1).

At fixed h , the lower constants q_0 , the cycle domination, and the size of the cover can be arbitrarily small or large positive constants. They multiply errors that vanish as $\varepsilon \rightarrow 0$ at this fixed h . The deterministic $Ch(1+k)$ bias and the Ch state norms, in contrast, come from the fixed normalized short templates of Section 5 and have uniform constants. Since the extra finite-cover budget ζ is arbitrary, it can be chosen tending to zero with h . The winding exceptions also tend to zero. This proves (7.1) in the stated order of limits. $\square$

7.4 Additive chunks and the endpoint-potential sum

Let f be a fixed smooth function supported in $B(x, R)$. For each rim P reaching outside $B(x, 4R)$, consider its maximal excursion intervals in $B(x, 2R)$ that meet $B(x, 3R/2)$. On the density event choose pins $s_0, \dots, s_l$ in order in each interval so that the initial and final unmarked portions have diameter $O(r_h)$ and every consecutive marked subarc has diameter between r_h and Cr_h . Here is an explicit choice. Take a pin on an initial portion reaching displacement $2r_h$. From a chosen pin, if the remaining excursion reaches displacement $4r_h$, apply Proposition 6.4 to the portion between its first hits of displacement $2r_h$ and $4r_h$ and choose the next pin there. Otherwise stop. Density applies to that portion because its diameter is at least $2r_h$. Closure of $B(x, 2R)$ may be used for the endpoints. The construction gives all the asserted diameter bounds. Set $F_i = f(z_{s_i})$. Then $F_0 = F_l = 0$, $|F_{i+1} - F_i| \leq C_f r_h$, and a nonzero change places both pins near the support, a fixed macroscopic distance from s_0 .

For distinct chosen pins within one interval, both directions along the rim include a departure of diameter at least r_h , or a departure outside the bulk disk. Thus neither mark belongs to the other's third-gate arc Δ , whose diameter is at most $2Lh$. Their cutoff balls are disjoint. Indeed if two overlapped, choose the plug with larger b_2 . Since $m < b_2/10$ for each type and $a_3 > 100b_2$, the other mark would lie strictly inside this plug's third circuit and hence on its short trapped arc, a contradiction.

At an intermediate pin all its cutoff-tagged edges occur between the preceding and following pins: their vertices belong, on fine mesh, to the trapped Δ containing that pin, and this arc avoids both neighbors. At either end the corresponding statement holds on the one neighboring side. The first-gate paths lie inside the cutoffs, so the convention defining I_X gives the exact identity

$$\sum_{i=0}^{j-1} I_X(P; s_i, s_{i+1}) = I_X(P; s_0, s_j), \quad 1 \leq j \leq l. \quad (7.5)$$

In particular, the full start cutoff and the omitted terminal cutoff are essential parts of this equality.

Assign coefficient F_i to the chunk $I_X(P; s_i, s_{i+1})$, and coefficient zero to all other tags; denote the resulting weighted sum by $\mathcal{A}_X(P)$. Near the support every central isolating arc lies in the same excursion, so full-cutoff reassignment cannot mix two excursion intervals. At end collars the coefficients are zero and overlaps do not matter. Every support tag is covered, and its assigned coefficient differs from its value of f by $O_f(r_h)$, using the chunk diameter bound and fine mesh.

At active coefficient changes there is no repeated grid location on one rim, even between different excursion intervals. A repeat would put the two marks inside the third circuit at that location. Within an interval the preceding two-departure argument applies; between intervals both directions reach the outer excursion collar. Either possibility contradicts the single short trapped arc. Allowing finitely many types at a location changes ensuing counts only by a menu-dependent constant.

Summation by parts using (7.5) gives, on each excursion,

$$\sum_{i=0}^{l-1} F_i I_X(P; s_i, s_{i+1}) = \sum_{j=1}^l (F_{j-1} - F_j) I_X(P; s_0, s_j).$$

For active changes the hypotheses of Proposition 7.4 hold. Subtract a_h^Q times the identity for V from that for Q . The displacement terms cancel since $a_h^V = 1$, and the base potentials cancel

because $\sum_j (F_{j-1} - F_j) = F_0 - F_l = 0$. Only terminal potentials and the bounded comparison errors remain.

We record the aggregate bounds in the original probability space, with the jointly assigned local preparations. Let N count all grid locations near the support whose fixed Ch -neighborhood meets any macroscopically reaching rim; set it to zero on the geometry exception allowed for a bulk patch. This count dominates all possible candidate mark locations, independently of the later pin selection or fitting events. Each counted visit gives a one-arm event from scale h to a fixed macroscopic distance. Thus

$$\limsup_{\varepsilon \rightarrow 0} \mathbb{E}N \leq C_R h^{-2+c}.$$

For each location and admitted type define $\psi^- = \phi^{Q,-} - a_h^Q \phi^{V,-}$ on that type's usable-state event G_s , and zero outside G_s . Admission means the deterministic inequality $\nu_s(G_s) \geq \beta'$. Its prepared-state square expectation is at most Ch^2 : the individual preparation law is dominated by $C\nu$, and the $L^2(\mu)$ norm of each centered potential is at most Ch . Consequently

$$T_h := \sum_{\substack{\text{locations and types} \\ \text{in the fixed region}}} |\psi^-(\sigma')|^2, \quad \limsup_{\varepsilon \rightarrow 0} \mathbb{E}T_h \leq C_R.$$

This uses no independence of different candidate preparations. For every adaptive pin choice on a single rim the sum of squared selected potentials is bounded by T_h , by non-repetition and the finite type list. Cauchy–Schwarz therefore controls the maximum over all rims by the same random bound. Off the preceding probability exceptions, the limiting upper expectation of the aggregate error is at most

$$C_{f,R} \left[r_h h^{-1+c} (1 + |\log h|^2) + r_h h^{-1+c/2} \right]. \quad (7.6)$$

The first term is $Cr_h h(1 + |\log h|^2) \mathbb{E}N$; the second follows from $Cr_h \mathbb{E} \sqrt{NT_h}$. The long-connector forcing densities from the proof of Proposition 7.4 do not enter this estimate: the potentials here are evaluated in the original assigned local preparations, whose uniform density constants were fixed in Section 6.

7.5 Signed fields and mixed atoms

Pointwise closeness of assigned coefficients to f does not itself control a signed field. We now prove the needed replacement.

Lemma 7.5 (Mixed-atom replacement). *Use the preceding excursion and pin assignments, and choose $0 < \kappa < c/2$, decreasing the positive exponent c if necessary. Then, for $X = Q, V$,*

$$\max_P |X_P(f) - \mathcal{A}_X(P)| \longrightarrow 0$$

in probability in the ordered limit. The maximum is over the same macroscopically reaching rims as above, on the preceding good events.

Proof. Let $\mathcal{G} = \mathcal{G}_{\varepsilon,h}$ be the common event on which the preceding qualified-pin assignments, density, winding bounds, finite-cover conclusions, and simultaneous actual-arc comparisons hold,

together with bulk-patch agreement. Choose the finite-cover budget tending to zero with h . The preceding results give

$$\limsup_{\varepsilon \rightarrow 0} \mathbb{P}(\mathcal{G}^c) \leq q(h), \quad q(h) \rightarrow 0.$$

This event is distinct from the local usable-state events G_s . Extend the adaptive coefficient assignment arbitrarily off $\mathcal{G}$; all restricted counts and errors below are set to zero there. We restrict the original probability measure by $\mathbf{1}_{\mathcal{G}}$, without dividing by $\mathbb{P}(\mathcal{G})$. Any additional mesh tests at the finitely many levels used below have failure probability tending to zero at fixed h and fixed depth.

All discrepancies occur in a fixed compact neighborhood of $\text{supp } f$. Tile it by h -squares and dyadically refine to $s_j = 2^{-j}h$. At every level intersect the squares with the deterministic partition recording membership in *all candidate cutoff balls*, with fixed disjoint Borel boundary conventions. A square meets only a bounded, menu-dependent number of these balls: their radii are fixed multiples of h , and their centers lie on the h -grid. Thus there are at most a fixed number of atoms and children per square. The partitions are deterministic, regardless of which pins were selected.

For a fixed rim call an occupied atom *mixed* if two of its tag centers have different assigned numerical coefficients. Stop unmixed atoms and refine mixed ones. We first establish the geometric implication that makes this refinement summable. A mixed atom has a nonzero coefficient involved, so its tags are near the support. Neither of two differently labeled tags can be in any cutoff ball assigned to that rim. Membership in every candidate ball is constant throughout the atom, and an assigned ball near the support assigns its entire rim mass a single coefficient. Thus if one tag belonged to it, both would have that coefficient. This argument concerns assigned balls; it asserts nothing about unused candidates.

Choose a differently labeled tag with nonzero coefficient. Since it is outside all assigned cutoffs, it lies on a traversal chunk between consecutive pins p_i, p_{i+1} . The other tag cannot lie on that same chunk. In one direction along the rim between the two visits the path must therefore pass p_i , and in the other direction it must pass p_{i+1} . This remains true if the second coefficient is zero, or if its visit belongs to another excursion. Each pin core is within h of its center, whereas the first tag is outside both corresponding radius- mh cutoffs. Its distance from each core is at least $(m_{\min} - 1)h - o(h)$, with $m_{\min} > 100$. The two visits in a common s_j -square are at distance $O(s_j)$. They thus have two distinct departures of order h , as illustrated in Figure 5.

For sufficiently fine fixed levels, Lemma 3.7 gives the local necessary near-return arm event from scale $O(s_j)$ to a fixed multiple of h . It can be tested inside, for example, $B(y, 10h)$ about the square center y , taking the outer radius of the arm test below $8h$. The same macroscopically reaching rim supplies a one-arm event through a separated outer annulus, from scale $32h$ to a fixed $\delta > 0$, whose retained queries can be confined outside $B(y, 24h)$. The numerical constants can be changed to fit the fixed buffers. At fixed h, j the mesh exceptions vanish and these tests use disjoint deterministic marked-Poisson restrictions. For each color b , the precise necessary-event inclusion is

$$\begin{aligned} & \{\mathcal{G}; \text{ atom mixed for some color-}b \text{ rim}\} \\ & \subset E_{\text{in},b}(s_j, h) \cap E_{\text{out},b}(h, \delta) \cup \{\text{mesh failure}\}. \end{aligned}$$

The events on the right do not select a rim. We multiply their ordinary probabilities, not probabilities conditioned on adaptive labels or on the good event. The near-return gain and the one-arm bound give, after decreasing $c > 0$ to a common value,

$$\mathbb{P}(E_{\text{in},b} \cap E_{\text{out},b}) \leq C(s_j/h)^{2+c}h^c.$$

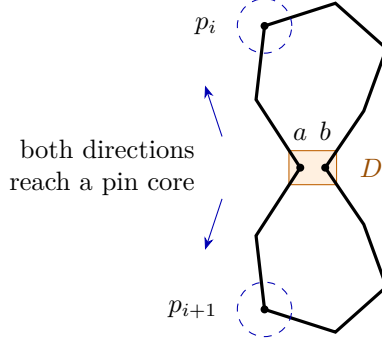


Figure 5: The deterministic implication for a mixed atom D , shown in orange. The tag a lies on the chunk between consecutive pins p_i, p_{i+1} ; the other tag cannot lie on that chunk. Both intervening rim portions must therefore reach a distant pin core. Dashed circles depict the two cutoffs. The drawing is schematic and makes no regularity assertion about a rim.

The two colors cost only a factor two. At the bounded number of coarse levels where these numerical buffers do not fit, use the one-arm estimate alone.

On $\mathcal{G}$, let N_j count the geometric atoms to be considered at level j because their parent is mixed for at least one relevant rim. At level zero count the occupied atoms in use. Each atom is counted once, regardless of how many rims visit it. There are $O_R(s_j^{-2})$ candidate squares, with a bounded number of atoms and children each. Define $\hat{N}_j = \mathbf{1}_{\mathcal{G}} N_j$, with value zero off $\mathcal{G}$. For $j \geq 1$, apply the preceding inclusion to the mixed parent at scale $s_{j-1} = 2s_j$; its bounded number of children and the factor two in scale change only the constant. Summing the ordinary arm probabilities yields

$$\limsup_{\varepsilon \rightarrow 0} \mathbb{E} \hat{N}_j \leq C_R h^{-2+c} 2^{-cj}. \quad (7.7)$$

The same restricted bound at level zero follows from one arm.

Choose f_D to be the value of f at the center of the level- j square containing D . Define the nonnegative square sum

$$B_j = \sum_{D, \lambda} \left(|X_\lambda(\mathbf{1}_D)|^2 + s_j^{-2} |X_\lambda(\mathbf{1}_D(f - f_D))|^2 \right). \quad (7.8)$$

The sum includes every deterministic atom at this level and every circuit-component label λ . Since these are disjoint deterministic masks, $|f - f_D| \leq C_f s_j$, and all masks have support in a fixed compact region, the Bessel bound (2.4) and the local triangle moments give

$$\limsup_{\varepsilon \rightarrow 0} \mathbb{E} \hat{B}_j \leq C_{f,R}, \quad \hat{B}_j = \mathbf{1}_{\mathcal{H}_\varepsilon} B_j.$$

Here $\mathcal{H}_\varepsilon$ is the color-independent geometry event of bulk-patch agreement, and is the whole space in the full-plane setting. We have $\mathcal{G} \subset \mathcal{H}_\varepsilon$. The full deterministic atom and label sum defining B_j is never restricted according to the adaptive coefficients; its geometric indicator is allowed in the conditional Bessel bound. The bound is uniform in j when that finite level is fixed before the microscopic limit.

For one rim P let $\mathcal{S}_j(P)$ be its stopped atoms at level j , and write $\lambda(P)$ for its circuit-component label. On such an atom its constant coefficient error relative to f_D has magnitude at

most $C_f r_h$. The other error uses $f - f_D$. Because $s_j \leq h \leq r_h$, Cauchy–Schwarz gives, on $\mathcal{G}$,

$$\begin{aligned} & |\text{level-}j \text{ contribution to } X_P(f) - \mathcal{A}_X(P)| \\ & \leq C_f r_h \sqrt{\#\mathcal{S}_j(P)} \left[\sum_{D \in \mathcal{S}_j(P)} (|X_{\lambda(P)}(\mathbf{1}_D)|^2 + s_j^{-2} |X_{\lambda(P)}(\mathbf{1}_D(f - f_D))|^2) \right]^{1/2} \\ & \leq C_f r_h \sqrt{\widehat{N}_j \widehat{B}_j}. \end{aligned}$$

After extending the level error by zero off $\mathcal{G}$, the last bound holds everywhere and bounds the maximum over all rims. In particular no extra number-of-rims factor appears: $\widehat{N}_j$ counts geometric atoms, while the second factor already sums over *all* field labels. Taking expectation and applying Cauchy–Schwarz once more, (7.7) and (7.8) bound the limiting upper expectation by

$$C_{f,R} r_h h^{-1+c/2} 2^{-cj/2}.$$

This is summable over levels, with total $O_{f,R}(r_h h^{-1+c/2})$.

To justify stopping, fix h and a finite depth $J \geq 1$. A mixed level- J atom has a mixed parent for the same rim, so it is itself counted by $\widehat{N}_J$ on $\mathcal{G}$. Therefore

$$\limsup_{\varepsilon \rightarrow 0} \mathbb{P}(\mathcal{G} \cap \{\text{some level-}J \text{ atom is mixed}\}) \leq C_R h^{-2+c} 2^{-cJ}.$$

On $\mathcal{G}$ when no terminal atom is mixed, the stopped atoms through depth J account for the whole discrepancy. The preceding finite sum is bounded by its geometric series, independently of J . For each $t > 0$, Markov’s inequality and the last display give

$$\begin{aligned} & \limsup_{\varepsilon \rightarrow 0} \mathbb{P} \left(\max_P |X_P(f) - \mathcal{A}_X(P)| > t \right) \\ & \leq q(h) + C_R h^{-2+c} 2^{-cJ} + \frac{C_{f,R}}{t} r_h h^{-1+c/2}. \end{aligned}$$

For a prescribed tolerance $\tau_h \downarrow 0$, choose a finite $J = J(h)$ making the middle term at most τ_h , before taking $\varepsilon \rightarrow 0$. Only finitely many mesh tests are needed at that fixed h, J , so their failure terms vanish in this microscopic limit. Finally let $h \rightarrow 0$: both $q(h) + \tau_h$ and $r_h h^{-1+c/2} = h^{c/2-\kappa}$ tend to zero. This proves the claim. $\square$

7.6 Completion of bulk transport

Take $\kappa > 0$ smaller than half the common positive exponent used above. Both terms in (7.6) tend to zero, since they equal, up to constants, $h^{c-\kappa}(1 + |\log h|^2)$ and $h^{c/2-\kappa}$. Propositions 6.4 and 7.4, followed by Lemma 7.5, show that for every $\xi > 0$,

$$\lim_{h \rightarrow 0} \limsup_{\varepsilon \rightarrow 0} \mathbb{P} \left(\max_P |Q_P(f) - a_h^Q V_P(f)| > \xi \right) = 0. \quad (7.9)$$

All limits are on the common microscopic subsequence chosen at the start of the section. The auxiliary finite covers and couplings for a given patch and test require no further subsequence.

Pass along the h sequence so that $a_h^Q \rightarrow a \in \mathbb{C}$. The Bessel bound for V implies

$$\limsup_{\varepsilon \rightarrow 0} \mathbb{P} \left(\max_P |(a_h^Q - a) V_P(f)| > \xi \right) \leq C_f |a_h^Q - a|^2 / \xi^2.$$

Combine this with (7.9), taking the microscopic limsup at fixed h and then $h \rightarrow 0$. This proves (4.1) for the resulting bounded scalar a , for every fixed bulk patch and test f . The scalar choices used only the countable menu representatives and mean limits, and do not depend on the later test or on points of evaluation of an observable. The remaining assertion $\operatorname{Re} a < 1/2$ in Theorem 4.1 is established in Proposition 8.1.

8 Nondegeneracy of the transport scalar

Section 7 extracts a limit a of the bounded scalars a_h^Q and uses the energy bound to obtain (4.1) with this single scalar. Its choice depends only on the bulk subsequence, and does not depend on any probes used in Section 9. We now prove the asserted restriction on a . The strict area imbalance between the two Fourier fields supplies the positive quantity; contour estimates and uniform integrability justify passing it through the transport limit.

Proposition 8.1 (Nondegeneracy). *Every scalar obtained from the preceding transport construction satisfies*

$$1 - 2 \operatorname{Re} a > 0.$$

In particular, $a \neq 1$.

Fix a nonzero $f \in C_c^\infty(\mathbb{C})$ and work in the full plane. Write $R = Q(f) - aV(f)$. The consequence of bulk transport needed here is the averaged pairing limit

$$\mathbb{E}\langle R, V(f) \rangle \longrightarrow 0. \quad (8.1)$$

Here the inner product is in the random label space $\ell^2(\{\lambda\})$, with the second factor conjugated. Once (8.1) is proved, the exact energy identities for U and Q will force $1 - 2 \operatorname{Re} a > 0$.

Bulk transport gives convergence in probability on macroscopic rims; it does not by itself justify this sum over all labels or its expectation. We therefore replace $V(f)$ by contour integrals. Their squared norm on small rims is negligible, while area multiplicity and wall-count moments control their macroscopic sum. These two estimates will make the pairing uniformly integrable and establish (8.1). For a label $\lambda = (L, C)$ with polygonal rim P , put

$$Y_\lambda = Y_P = \int_P f(z) \, dz;$$

for an isolated-edge component put $Y_\lambda = 0$. All rims are oriented as in Lemma 2.3; norms below use the same random label space.

8.1 Replacing the tagged tangent field by a contour integral

We first establish the two estimates

$$\mathbb{E} \sum_\lambda |Y_\lambda - V_\lambda(f)|^2 \longrightarrow 0, \quad (8.2)$$

$$\lim_{b \downarrow 0} \limsup_{\varepsilon \rightarrow 0} \mathbb{E} \sum_{\operatorname{diam}(P) < b} |Y_P|^2 = 0. \quad (8.3)$$

Choose a compact box K_0 containing $\operatorname{supp} f$ in its interior; enlarge it by a fixed amount whenever a mesh estimate requires a collar.

For a tagged triangle t , its clockwise missing side is the corresponding oriented rim edge. Comparing $f(z_t)$ with f along that edge costs at most $C_f \text{diam}(t)^2$. An isolated edge has two opposite oriented edge integrals, which cancel, and the same estimate applies to its two tags. Consequently there are nonnegative budgets

$$B_\lambda = C_f \sum_{\substack{t \text{ tagged by } \lambda \\ t \cap \text{supp } f \neq \emptyset}} \text{diam}(t)^2 \quad \text{such that} \quad |Y_\lambda - V_\lambda(f)| \leq B_\lambda.$$

Triangles in a vanishing mesh collar may be included in this sum without changing the argument. A triangle has at most one tag, so $S_\varepsilon = \sum_\lambda B_\lambda$ has uniformly bounded moments of every fixed finite order by Lemma 3.3.

We claim that $M_\varepsilon = \max_\lambda B_\lambda \rightarrow 0$ in probability. First fix $b > 0$ and tile a neighborhood of $\text{supp } f$ by squares of side $s \ll b$. If a rim of diameter at least b has a tag in such a square, then, on fine mesh, its monochromatic path gives an arm from a fixed enlargement of the square to distance of order b . The local diameter-square sum has L^2 norm at most Cs^2 in the microscopic upper limit. Cauchy–Schwarz and the one-arm estimate therefore bound the expected budget in that square belonging to these rims by

$$C_f s^2 (s/b)^{c/2}$$

in that upper limit. There are $O(s^{-2})$ squares. Thus the total budget of the rims of diameter at least b vanishes in probability after first taking $\varepsilon \rightarrow 0$ and then $s \downarrow 0$.

A rim of diameter less than b has all its relevant tags in an enlarged box of side $O(b)$ on fine mesh. The same is true of an isolated tag pair. Cover the fixed support neighborhood by $O(b^{-2})$ such boxes. If D_B denotes the diameter-square sum in an enlarged box, then, for every fixed $p > 1$,

$$\limsup_{\varepsilon \rightarrow 0} \mathbb{P} \left(\max_B D_B > u \right) \leq C_p u^{-p} b^{2p-2}, \quad u > 0,$$

using the local L^p bound $\limsup_{\varepsilon \rightarrow 0} \|D_B\|_p \leq C_p b^2$. Letting $b \downarrow 0$ proves the claim. Finally,

$$\sum_\lambda |Y_\lambda - V_\lambda(f)|^2 \leq M_\varepsilon S_\varepsilon.$$

The right side tends to zero in probability and is uniformly integrable: $M_\varepsilon \leq S_\varepsilon$ and the latter has bounded moments of order greater than two. This proves (8.2).

8.2 Distinct walls and area multiplicity

We need two bounds on these contours: a small-diameter estimate for their expected square sum, and higher moments of their absolute sum above a fixed diameter cutoff. Both follow by counting distinct walls and controlling how many of one wall's component interiors cover a point.

Stokes' theorem, applied with either orientation, gives

$$|Y_P| \leq C_f \int_{K_0} \mathbf{1}_{\text{int}_b(P)}(z) \, d^2 z, \quad (8.4)$$

where $\text{int}_b(P)$ is the bounded geometric inside of P . Also, $Y_P = 0$ unless P meets K_0 .

For each fixed wall L , at most two of its rims enclose a generic point. Indeed, the bounded components of the untouched triangles have disjoint disk interiors by Lemma 2.3; in addition,

there is only one unbounded component. Its rim can contribute one further bounded inside. Polygonal boundaries have area zero, so this pointwise assertion is sufficient for all area integrals below. It is an area-multiplicity statement, and does not bound the number of rims associated with a wall.

We record the wall-count estimate used at both small and fixed scales. At any scale $t > 0$, the expected number of rims of diameter in $[t, 2t]$ enclosing a specified generic point is bounded uniformly in t and ε . Such a rim lies within distance $2t$ of the point. On mesh fine relative to t , its wall visits the triangles facing the rim edges and therefore has a spanning arc in one of finitely many fixed-ratio annuli centered at a grid of mesh a small fixed multiple of t . Nested primal boundary cycles with room between them may be chosen as in Section 3. For each such annulus, let N count the proper spanning wall strands obtained by cutting at all boundary contacts and stopping at the boundary edges. Use the bank extraction from the proof of Lemma 3.6: every strand has an adjacent black bank, each gap between consecutive strands accounts for at most two such banks, and banks selected from distinct gaps give vertex-disjoint black crossings after proper trimming. This includes their boundary endpoints, by the stopped-edge argument in that proof: each black boundary site belongs to only one gap. In particular, $N \geq 2k$ supplies k crossings with disjoint site witnesses. On the crossing-good geometry event G , a fixed buffered subannulus has quenched white-circuit probability at least $\delta > 0$, supplied by finitely many fixed-aspect rectangle tests. Such a circuit excludes a black spanning crossing. BK, applied conditionally on the point geometry, therefore gives

$$\mathbb{P}^\eta(N \geq 2k) \leq (1 - \delta)^k \quad \text{on } G.$$

Consequently the spanning-strand counts have bounded quenched moments of every fixed order on G .

Here and below the exceptional geometry is controlled quantitatively. For t/ε large, the probability that either the needed mesh or crossing tests fail is at most $C_\gamma(t/\varepsilon)^{-\gamma}$, for any prescribed finite γ , by Proposition 3.1 and Lemma 3.3. Without these tests, count distinct walls by the number J of incident triangles in the relevant $O(t)$ region. Every counted wall has a tagged incident triangle there, and the moments of J grow at most polynomially in $1 + t/\varepsilon$. For example, if

$$\|J\|_{2m} \leq C(1 + t/\varepsilon)^{k_m},$$

then

$$\mathbb{E}[J^m \mathbf{1}_{\text{bad}}] \leq \|J\|_{2m}^m \mathbb{P}(\text{bad})^{1/2} \leq C(1 + t/\varepsilon)^{mk_m} (t/\varepsilon)^{-\gamma/2}.$$

Choose $\gamma > 2mk_m$. For bounded t/ε , the incidence bound alone suffices. The two-per-wall area multiplicity gives the asserted uniform expected rim count.

Since $|Y_P| \leq C_f t^2$ for $\text{diam}(P) \in [t, 2t]$, this count and (8.4) imply

$$\mathbb{E} \sum_{t \leq \text{diam}(P) < 2t} |Y_P|^2 \leq C_f t^2 \int_{K_0} \mathbb{E} \sum_{t \leq \text{diam}(P) < 2t} \mathbf{1}_{\text{int}_b(P)}(z) \, d^2 z \leq C_f t^2.$$

Dyadic summation over scales below b proves (8.3).

For fixed $b > 0$, let $N_{\varepsilon, b}$ count the distinct walls represented by rims with $\text{diam}(P) \geq b$ and $Y_P \neq 0$. Cover K_0 by finitely many grid balls whose radii are a sufficiently small fixed fraction of b . A rim of diameter at least b meeting one of these balls must leave a fixed enlargement of it, so

its wall has a spanning arc in a corresponding fixed-ratio annulus. The preceding spanning-arc and bad-geometry estimates, now at fixed scale, prove

$$\sup_{\varepsilon \text{ small}} \mathbb{E} N_{\varepsilon, b}^m < \infty \quad (b > 0, m < \infty).$$

The constants may depend on b and m . Put $M_f = C_f \text{Area}(K_0)$. Applying area multiplicity before summing labels gives the precise bounds

$$\begin{aligned} \sum_{\text{diam}(P) \geq b} |Y_P| &\leq 2M_f N_{\varepsilon, b}, \\ \sum_{\text{diam}(P) \geq b} |Y_P|^2 &\leq 2M_f^2 N_{\varepsilon, b}. \end{aligned} \tag{8.5}$$

In the second line we used $|Y_P| \leq M_f$ individually. Thus the absolute sum, and the square sum, have all fixed higher moments at fixed b , without a count of all rims belonging to one wall.

8.3 Uniform integrability and the positive area identity

Put $A_\varepsilon = \|R\|_{\ell^2}$. The energy identity gives $\sup_\varepsilon \|A_\varepsilon\|_2 < \infty$. At fixed b , split f into finitely many smooth tests supported on disks small compared with b . Every relevant rim leaves the fourfold disk of each such test that it meets. Hence (4.1) gives

$$\delta_{\varepsilon, b} := \max_{\substack{\text{diam}(P) \geq b \\ Y_P \neq 0}} |R_P| \longrightarrow 0 \quad \text{in probability.}$$

By (8.5),

$$\left| \sum_{\text{diam}(P) \geq b} R_P \overline{Y_P} \right| \leq \delta_{\varepsilon, b} \sum_{\text{diam}(P) \geq b} |Y_P| \longrightarrow 0 \quad \text{in probability.}$$

For the required passage to expectation, define $B_{\varepsilon, b} = (\sum_{\text{diam}(P) \geq b} |Y_P|^2)^{1/2}$. For any fixed $q > 2$ the wall moments give $\sup_\varepsilon \|B_{\varepsilon, b}\|_q < \infty$. With $p = 2q/(q+2) > 1$, Cauchy–Schwarz in labels and Hölder in probability give

$$\left\| \sum_{\text{diam}(P) \geq b} R_P \overline{Y_P} \right\|_p \leq \|A_\varepsilon B_{\varepsilon, b}\|_p \leq \|A_\varepsilon\|_2 \|B_{\varepsilon, b}\|_q \leq C_{b, f}. \tag{8.6}$$

This is uniform integrability, so the macroscopic pairing tends to zero in L^1 . The two errors left in replacing this pairing by the full pairing with $V(f)$ satisfy

$$\begin{aligned} \mathbb{E} |\langle R, V(f) - Y \rangle| &\leq (\mathbb{E} \|R\|^2)^{1/2} (\mathbb{E} \|V(f) - Y\|^2)^{1/2}, \\ \mathbb{E} \left| \sum_{\text{diam}(P) < b} R_P \overline{Y_P} \right| &\leq (\mathbb{E} \|R\|^2)^{1/2} \left(\mathbb{E} \sum_{\text{diam}(P) < b} |Y_P|^2 \right)^{1/2}. \end{aligned}$$

The first vanishes as $\varepsilon \rightarrow 0$ by (8.2); the second vanishes after the microscopic upper limit and then $b \downarrow 0$ by (8.3). This proves the pairing limit (8.1).

Proof of Proposition 8.1. Since $V = U + Q$, the cross terms in the following real part cancel exactly. By (2.4) and the triangle area identity,

$$\begin{aligned} \mathbb{E} \Re \sum_{\lambda} (V_{\lambda}(f) - 2Q_{\lambda}(f)) \overline{V_{\lambda}(f)} &= \mathbb{E} \sum_{\lambda} (|U_{\lambda}(f)|^2 - |Q_{\lambda}(f)|^2) \\ &= \sqrt{3} \mathbb{E} \sum_t \text{Area}(t) |f(z_t)|^2 \longrightarrow \sqrt{3} \int_{\mathbb{C}} |f(z)|^2 d^2z > 0. \end{aligned}$$

The final approximation follows from fine mesh and the diameter moments. On the other hand, (8.1) writes the left side as

$$(1 - 2 \operatorname{Re} a) \mathbb{E} \|V(f)\|^2 + o(1).$$

The norm is nonnegative and has bounded expectation. A nonpositive coefficient is incompatible with the strictly positive limit. Therefore $1 - 2 \operatorname{Re} a > 0$, as claimed. $\square$

9 The observable and the crossing limit

We construct an observable with a boundary coefficient equal to the desired crossing probability, and use bulk transport to prove that its limit is holomorphic.

We first identify the free crossing limit on disks approximating a polygonal Jordan quadrilateral. We then compare those crossings with the closed-cell event in an arbitrary Jordan domain. All random discrete domains in this section are chosen from the point geometry, independently of the colors.

9.1 Discrete disks and the closing star

Lemma 9.1 (Ordered polygonal approximation). *Let D be a polygonal Jordan domain with finitely many marked boundary points in their cyclic order. With probability tending to one, the Delaunay triangulation contains a simple primal cycle whose boundary and selected marked vertices converge to the marked boundary of D in order and in Fréchet distance. The triangulation inside this cycle agrees with the full-plane triangulation on every fixed compact subset of D for all sufficiently fine good approximations.*

Proof. On fine mesh in a fixed collar of ∂D , record the successive Voronoi cell sites met during one traversal of ∂D and join successive sites by primal edges. A fixed polygon meets cell switches generically almost surely. Equivalently, one can use walks through the nearby triangles. This produces a closed primal walk, with sites and edges uniformly close to the boundary position at the corresponding traversal times. Its winding about a fixed interior point is one.

Extract return loops and contract them to obtain the oriented simple-cycle decomposition of this walk. Winding is additive, so retain a cycle with positive winding about the interior point, retaining the original cyclic times of its edges. Every dropped interval of times has endpoints whose physical positions tend together. Boundary simplicity then forces its parameter length to tend to zero, unless it tends to a whole turn. The latter possibility would put all retained times, and thus the retained cycle, in a shrinking neighborhood of one boundary point, contradicting its positive winding about the fixed interior point. Hence the retained cycle follows the boundary in its order, and marked vertices can be selected at the desired parameters. Finite deterministic tie-breaking makes the choice measurable. Filling the cycle by its interior triangles gives the required disk. The mesh and local-determination estimates ensure agreement on interior

compacts. Taking a deterministic sequence of decreasing tolerances gives the claimed convergence in probability. $\square$

Fix such a disk, and initially retain only the three marked vertices v_0, v_1, v_2 approximating a, c, d , respectively, in positive order. Indices in the fan construction are taken modulo three. Add an exterior vertex s_i fanned to the inclusive boundary arc $v_i v_{i+1}$, add the junction triangles $s_{i-1} s_i v_i$, and close the exterior by a triangle t_* with vertices s_0, s_1, s_2 . This is a simple triangulation of the sphere. Give the added vertices independent fair colors, as for the original disk sites. No Euclidean weights on the exterior triangles are needed.

At the closing star t_* , let n_i be a point on the dual spoke crossing $s_{i-1} s_i$, and assign it the label $w_i = \omega^i$. This order is *opposite* to the positive cyclic order at t_* . For a probe m in the interior of a dual edge, define, conditionally on the geometry η ,

$$F_m^\eta = 4 \sum_{i=0}^2 w_i K(m, n_i) = 4 \mathbb{E}_{\text{sp}}[w_* H_{\lambda_*}(m)]. \quad (9.1)$$

In the second expression, if t_* is occupied by an ordinary wall, let i be its missing port and let λ_* be that wall together with the component reached through the missing port; set $w_* = w_i$. Set the entire contribution to zero if t_* is unused. To verify the equality, observe that a wall not using t_* has identical component indicators at all three n_i , and its contribution is killed by $\sum_i w_i = 0$. The observable is constant along the interior of a dual edge, so we also write F_e^η for its value at a primal edge e . In the physical disk the dual probe can be chosen at the intersection with that side.

Figure 6 shows the completion and the boundary pairing used below. The kernel definition is adapted to orthogonality; its alternative description as a mean partner label will give positivity and identify the crossing coefficient.

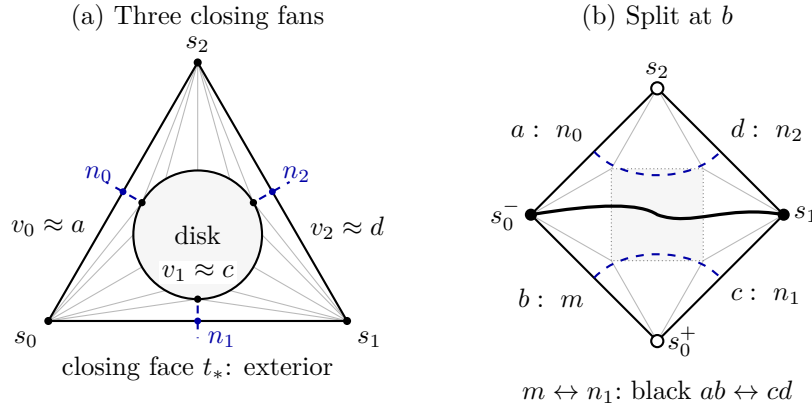


Figure 6: The closing construction and one of the two boundary pairings. In (a), the unbounded face of the drawing is the closing triangle on the sphere; the dashed segments indicate its dual spokes. Thus the labels n_0, n_1, n_2 run opposite to the positive order at its dual star. In (b), splitting s_0 produces black fans on ab, cd and white fans on bc, da . The labels indicate the attachments of the four boundary switches. Dashed strands show the pairing $m-n_1, n_0-n_2$; the solid black path shows the corresponding connection of the black fans. In (b), only the exterior colors are prescribed; all original disk vertices remain free. The drawings are topological schematics.

Proposition 9.2 (Triangular observable and the crossing coefficient). *The value F_m^η belongs to the closed triangle with vertices w_0, w_1, w_2 . At a fan probe $s_i u$, where u is interior to the boundary arc $v_i v_{i+1}$, it belongs to the side $[w_i, w_{i+1}]$. For $i = 0$ and u the marked vertex approximating b , the coefficient of w_1 on this side is exactly the free black vertex crossing probability between the arcs approximating ab and cd in the original disk.*

Proof. Start with the uniform edge-pattern ensemble having oddness at t_* and m , and move the star defect to n_i as in the proof of (2.1). If the star has degree one, the loop count is independent of the chosen port. If it has degree three, the extra loop count is the indicator that the exterior partner of m is port i : joining the other two star ports creates exactly that extra loop. Degree three has probability $1/4$. Indeed, toggling a fixed defect path bijects to the even ensemble, whose four local even patterns at the star are equiprobable. The port-independent terms cancel in the weighted sum, and the factor four in (9.1) cancels this probability. Thus F_m^η is the mean partner label conditional on degree three, proving the triangular convex-combination assertion.

For a fan probe $s_i u$, split s_i into two exterior vertices, fanned respectively to the two successive subarcs, both containing u , and add the triangle formed by these two vertices and u . Remove t_* and prescribe alternating colors, in one fixed convention, on the four exterior vertices now present. Leave all original disk spins free. The four boundary edges joining consecutive exterior vertices are switches. Their disagreement strands have the preceding conditional pairing law.

Here is the parity correspondence explicitly. The two half-edge values at the old probe, with oddness there, have exactly one occupied half. They become the two spoke values entering the added triangle, which has even parity because its split exterior boundary edge is occupied. The other three occupied exterior boundary edges are the three forced switches at the old degree-three star. This gives a bijection from the conditional edge-pattern ensemble to parity-even disagreement data on all triangles of the augmented disk, with the four boundary switches fixed. Triangle parity gives cycle parity, so the data recover colors up to a global flip. Fixing the alternating exterior convention removes that flip, leaving exactly uniform independent colors on all original disk vertices.

The four switches have a noncrossing pairing. The split switch can therefore be paired only with port i or port $i + 1$, proving the side constraint. For $i = 0$ at $u \approx b$, prescribe the exterior vertices along ab and cd to be black, and the other two to be white. Connecting the two black exterior vertices is equivalent to a black crossing in the original disk: any connection between the two black fans contains a path of original black vertices between their respective arcs, and conversely such a path joins the fans. Pairing the split switch with port 1 gives this connection along the neighboring black sites. The other pairing joins the opposite white pair, and the two possibilities are exclusive by planarity. This proves the stated coefficient identity. $\square$

9.2 Annealed compactness for geometry-selected probes

On failed disk-approximation events, extend all observables by arbitrary bounded values. Those events have vanishing probability and will not affect a limit. Original disk spins are always coupled to the fair full-plane site colors.

The admissible probes in this subsection and below are original disk edges, located at their physical midpoints, and fan edges $s_i u$, located at their disk attachments u . A geometry-selected probe tends to a fixed point when its location converges in probability to that point.

Lemma 9.3 (Probe compactness). *From every microscopic sequence one can select a subsequence, still satisfying the bulk transport conclusion, on which the annealed observables converge to a*

continuous function F in D and on its three open boundary arcs. The convergence holds for every geometry-measurable choice of admissible probes tending to a specified point off the three marks. The limit takes values in the closed triangle and has the corresponding closed side values on the open boundary arcs.

Proof. We use the ordinary-wall representation in (9.1). A nonisolated selected component has a rim using the exterior edge $s_{i-1}s_i$, followed by a path of original disk sites between attachments to the two different boundary arcs of those fans. The third exterior vertex has the opposite color and cannot lie on this rim. The disk path can degenerate to the common endpoint of its two arcs. The component indicator tests the component side of the rim, including its rim edges: an open dual edge belongs to that component exactly when it is incident to an untouched triangle on the filled disk side. An isolated selected component contributes zero at the probes in use.

For two nearby bulk probes, different indicators force a selected-rim visit between them. Such a visit gives a monochromatic path of original disk sites from their neighborhood to a fixed positive distance, since the disk portion reaches the boundary. Near a boundary point off the three marks, compare a fan probe first with its attached disk vertex. Unless that vertex lies on the rim, following its incident fan edge to the vertex tests the same side; if it does lie on the rim, the required visit already occurs. Uniform closed-disk Jordan charts join nearby disk positions tending to the same boundary point by paths of vanishing diameter inside the closed approximating disk. Thus a difference again forces a nearby visit by the disk portion of the selected rim. Its two attachments lie on different marked arcs and cannot both approach this nonmark boundary point. One direction of the disk path therefore reaches a fixed positive distance *among original disk sites*. The exterior fan edges do not replace this arm.

More precisely, on a fixed compact set off the marks choose a positive distance ρ valid for these continuations. For probes in a small neighborhood of diameter s , choose a deterministic enlarged neighborhood containing their short comparison paths on good geometry. Let $q_{s,\rho}(\eta)$ be the conditional probability of a monochromatic arm of either color from that neighborhood to distance ρ , under the full-plane independent site colors. Then, for every pair of admissible geometry-selected probes in that neighborhood, the same test gives

$$|F_m^\eta - F_{m'}^\eta| \leq 4q_{s,\rho}(\eta). \quad (9.2)$$

There is no union bound over the chosen edges. The disk and chart control needed for this statement follows along every deterministic sequence of good marked Jordan approximations by closed Jordan convergence; mesh control transfers edges to their sites. Taking expectations and using the ordinary one-arm estimate gives an upper limit tending to zero as $s \downarrow 0$. The geometry errors vanish first at fixed s .

Choose near-position representative edges at a countable dense set of interior points, and fan representatives at a countable dense set of each open boundary arc. These choices exist with vanishing location error on good geometry. Boundedness permits a diagonal subsequence of their annealed expectations. Estimate (9.2) extends the resulting function continuously and identifies the same limit for all geometry-selected probes converging to any such point. The triangle and side constraints pass to this limit by Proposition 9.2. All choices are made after the universal bulk subsequence, so the transport conclusions remain valid. $\square$

9.3 Holomorphy without separating geometry and colors

Proposition 9.4 (Holomorphy). *Every limit F in Lemma 9.3 is holomorphic in D .*

Proof. For a smooth compactly supported bulk test f , orthogonality against the closing star, with its reversed labels, gives the exact conditional identity

$$\mathbb{E}_{\text{sp}}[w_* U_{\lambda_*}(f)] = 0.$$

Indeed, at every bulk triangle the weighted sum $\sum_j \omega^j F_{m_j(t)}^\eta$ is zero by (2.1). The selected nonisolated rim reaches the exterior and therefore leaves every sufficiently small fixed test collar. An isolated selected pair tags only exterior triangles. Apply (4.1), partitioning f into local disk tests when needed. Its residual at the selected label tends to zero in probability and in mean: its absolute value is bounded by the full residual ℓ^2 norm, which is bounded in L^2 by (2.4). Since

$$U = (1 - a)V - (Q - aV), \quad a \neq 1,$$

we obtain $\mathbb{E}[w_* V_{\lambda_*}(f)] \rightarrow 0$. On high-probability geometry events the discrete disk agrees with the full triangulation throughout the test collars; the conditional identities hold on each such graph. The energy moments remove the omitted geometry.

Using (9.1) and $\sum_j d_{tj} = 0$, we have therefore proved

$$\mathbb{E} \left[\sum_t f(z_t) \sum_j d_{tj} F_{m_j(t)}^\eta \right] \rightarrow 0. \quad (9.3)$$

The edge coefficients and the conditional observable both depend on the point geometry, so their product cannot be replaced by a product of expectations. We instead express the edge coefficients as a random measure and prove that this measure has a deterministic limit.

Group the two contributions of each shared edge $e = t \cap t'$. If $d_{t,e}$ is its clockwise side vector for t , set

$$\mu_\varepsilon = \sum_e d_{t,e} (f(z_t) - f(z_{t'})) \delta_{m_e},$$

where now m_e denotes the physical midpoint of the side. This weight is independent of which adjacent triangle is called t . All edges involved are buffered inside the disk. Let $T_\varepsilon(B) = |\mu_\varepsilon|(B)$. Because f is smooth,

$$|d_{t,e}| |f(z_t) - f(z_{t'})| \leq C_f (\text{diam}(t)^2 + \text{diam}(t')^2).$$

Consequently, for each sufficiently small fixed square B of side s in a compact bulk neighborhood,

$$\limsup_{\varepsilon \rightarrow 0} \|T_\varepsilon(B)\|_2 \leq C_f s^2. \quad (9.4)$$

The triangles incident to an edge assigned to B lie in a fixed enlargement on fine mesh; the exceptional contribution is controlled by their higher moments.

For a deterministic smooth function G , midpoint integration on clockwise triangle boundaries gives

$$\mu_\varepsilon(G) \rightarrow - \int_D f \, dG \wedge dz = \int_D G \, df \wedge dz \quad \text{in } L^1 \text{ of the geometry.}$$

To check the approximation, the midpoint error on a triangle is $O_{f,G}(\text{diam}(t)^3)$. The sum of these errors tends to zero in L^1 , using the maximum mesh diameter times the bounded diameter-square sum. Stokes' theorem, the clockwise orientation, and area approximation give the displayed sign.

Replacing $f(z_t)$ by f in each triangle has the same vanishing cubic-error bound. The resulting limiting complex measure is

$$\mu = df \wedge dz,$$

which has a bounded compactly supported density.

We need the stronger conclusion for fixed-square indicators. If B is such a square, choose a smooth g_τ with $0 \leq g_\tau \leq 1$ agreeing with $\mathbf{1}_B$ outside a boundary strip U_τ of width $O(\tau)$. Then

$$|\mu_\varepsilon(B) - \mu(B)| \leq |\mu_\varepsilon(g_\tau) - \mu(g_\tau)| + T_\varepsilon(U_\tau) + |\mu|(U_\tau).$$

Cover U_τ by $O(\text{perimeter}(B)/\tau + 1)$ squares of side $O(\tau)$. The first-moment consequence of (9.4), after the microscopic limit, makes the expected variation on this strip $O_f(\text{perimeter}(B)\tau + \tau^2)$. The limiting variation tends to zero there as well. First take $\varepsilon \rightarrow 0$, then $\tau \downarrow 0$, obtaining

$$W_\varepsilon(B) := \mu_\varepsilon(B) \longrightarrow w(B) := \mu(B) \quad \text{in } L^1. \quad (9.5)$$

Now fix a deterministic tiling of the test neighborhood by squares of side s , and choose a conditional representative F_B^η on an edge near the center of each square. The same ordinary arm test in (9.2) controls every edge assigned to B :

$$|F_e^\eta - F_B^\eta| \leq 4q_B(\eta), \quad \limsup_{\varepsilon \rightarrow 0} \mathbb{E}q_B \leq C(s/\rho)^c, \quad 0 \leq q_B \leq 1.$$

Here $\rho > 0$ is fixed by the distance of the support neighborhood to the boundary. Thus $\mathbb{E}q_B^2 \leq \mathbb{E}q_B$, and without an independence assumption Cauchy–Schwarz gives

$$\limsup_{\varepsilon \rightarrow 0} \mathbb{E}[T_\varepsilon(B)q_B] \leq C_{f,\rho}s^{2+c/2}.$$

Summing over $O(s^{-2})$ squares bounds the mean replacement error by $C_{f,\rho}s^{c/2}$. For fixed s , omitted geometry contributes a vanishing mean by boundedness of the observable and the variation moments.

At this fixed tiling, (9.5) can be multiplied by the correlated representative because $|F_B^\eta| \leq 1$:

$$|\mathbb{E}[F_B^\eta W_\varepsilon(B)] - w(B)\mathbb{E}F_B^\eta| \leq \mathbb{E}|W_\varepsilon(B) - w(B)| \longrightarrow 0.$$

Lemma 9.3 identifies $\mathbb{E}F_B^\eta \rightarrow F(\text{center}(B))$. Hence (9.3), followed first by $\varepsilon \rightarrow 0$ at fixed tiling and then by $s \downarrow 0$, yields

$$\int_D F df \wedge dz = 0 \quad (f \in C_c^\infty(D)). \quad (9.6)$$

Since $df \wedge dz = 2i \partial_{\bar{z}} f d^2z$, this is the distributional Cauchy–Riemann equation. Local mollification and continuity of F show that F is holomorphic. $\square$

9.4 The three boundary arcs determine the conformal map

The three closed sides of a nondegenerate triangle have empty common intersection. The side constraints therefore show that F is nonconstant. The open mapping theorem puts its interior values in the open equilateral triangle. Conformal charts for the polygon and the triangle, with their Carathéodory boundary extensions, reduce the remaining uniqueness issue to the following lemma.

Lemma 9.5 (Boundary uniqueness). *Let $G : \mathbb{D} \rightarrow \mathbb{D}$ be nonconstant and holomorphic. Let $\zeta_0, \zeta_1, \zeta_2$ and η_0, η_1, η_2 be positively ordered triples of distinct points of the unit circle. Suppose that G extends continuously to each open arc (ζ_i, ζ_{i+1}) and takes its values there in the corresponding positively oriented closed arc $[\eta_i, \eta_{i+1}]$. Then G extends continuously to the closed disk, maps ζ_i to η_i , and is a conformal bijection of the disk.*

Proof. On each open source arc, Schwarz reflection across the unit circle extends G holomorphically across the boundary. Near each point of that arc, G is nonzero, so $u = \log |G|$ is harmonic, negative on the disk side, and zero on the circle. The Hopf boundary lemma and the Cauchy–Riemann equations give

$$\frac{d}{dt} \arg G(e^{it}) = \frac{\partial}{\partial r} \log |G(re^{it})| \Big|_{r=1} > 0.$$

The proper target arc has a bounded real argument lift. Thus the boundary argument on each open source arc is strictly increasing and has finite one-sided limits at its endpoints.

We prove that the two limits at a source mark agree, without assuming interior continuity there. Rotate that mark to 1 and denote the limits as $t \uparrow 0$ and $t \downarrow 0$ by L_- and L_+ , respectively. Let $g(e^{it})$ be the known boundary values, with arbitrary values assigned at the three marks. For fixed $z \in \mathbb{D}$, Cauchy’s formula on a circle of radius $R > |z|$ is

$$G(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{G(Re^{it})}{1 - (z/R)e^{-it}} dt.$$

As $R \uparrow 1$, the numerator converges away from the three marks and is uniformly bounded, while the denominator stays uniformly away from zero for this fixed z . Dominated convergence therefore gives

$$G(r) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{g(e^{it})}{1 - re^{-it}} dt, \quad 0 < r < 1.$$

For a fixed small $\alpha > 0$, write $\ell_r = \log(1/(1-r))$. The logarithmic primitive of the kernel gives

$$\int_0^\alpha \frac{dt}{1 - re^{-it}} = \frac{\ell_r}{i} + O_\alpha(1), \quad \int_{-\alpha}^0 \frac{dt}{1 - re^{-it}} = -\frac{\ell_r}{i} + O_\alpha(1).$$

For example, a continuous branch of $i^{-1} \log(e^{it} - r)$ is a primitive on either interval. Moreover, the kernel is bounded in modulus by $C/(1-r+|t|)$ there. Replacing g by its corresponding one-sided limit therefore gives an $o(\ell_r)$ error: for every $\delta > 0$ choose a smaller fixed interval where $|g - L_\pm| < \delta$, use the logarithmic kernel bound on that interval, and note that the complementary fixed interval has a bounded integral as $r \uparrow 1$. Then let δ tend to zero. The part away from the mark is also bounded. Consequently,

$$G(r) = \frac{L_+ - L_-}{2\pi i} \log \frac{1}{1-r} + o\left(\log \frac{1}{1-r}\right). \quad (9.7)$$

Boundedness of G forces $L_+ = L_-$. The common value belongs to the two adjacent target arcs, whose intersection is their prescribed common endpoint; hence G has the required continuous circle trace g .

To obtain continuity for all interior approaches, apply the same radial-circle limiting argument to the Poisson formula for the bounded harmonic real and imaginary parts of G . They are the Poisson extensions of the continuous circle trace g . The Poisson extension of continuous circle

data is continuous on the closed disk. In particular, this proves full continuity at the marks, not merely radial or nontangential convergence.

On each open arc the boundary argument increases strictly between the two prescribed consecutive endpoints. The boundary curve therefore traverses the circle once in positive order. For any $w \in \mathbb{D}$, continuity on the closed disk gives uniform convergence of $G(re^{it}) - w$ to $g(e^{it}) - w$ as $r \uparrow 1$. The winding number is one for all sufficiently large r . The argument principle gives exactly one zero of $G - w$ in $|z| < r$, counted with multiplicity. Letting $r \uparrow 1$ proves that every w has exactly one preimage, of multiplicity one. Thus G is a conformal bijection. $\square$

It follows that F is the conformal map to the equilateral triangle with a, c, d mapped to w_0, w_1, w_2 . Normalize a conformal map to the upper half-plane by

$$(a, b, c, d) \mapsto (0, x, 1, \infty), \quad 0 < x < 1.$$

The Schwarz–Christoffel integral for three angles $\pi/3$ gives the proportion along the side from w_0 to w_1 at b as

$$\mathcal{C}(x) = \frac{\int_0^x u^{-2/3}(1-u)^{-2/3} du}{\int_0^1 u^{-2/3}(1-u)^{-2/3} du}. \quad (9.8)$$

By Proposition 9.2, this is the limiting annealed coefficient of w_1 at the fan probe for b , and therefore the limiting free black vertex crossing probability. Every microscopic sequence has a further subsequence with this same limit. Since the crossing probabilities are bounded, their full sequence converges to $\mathcal{C}(x)$ on the discrete polygonal disks.

9.5 The closed-cell event in every Jordan domain

Proposition 9.6 (Fixed-buffer Jordan comparison). *Let D be any bounded Jordan domain with distinct positively ordered boundary marks a, b, c, d . For the full-plane Poisson tessellation with independent fair cell colors, the annealed probability of a connected crossing in the closed black-cell union intersected with $\overline{D}$, from ab to cd , converges to $\mathcal{C}(x)$ with the marked conformal parameter of D .*

Proof. Choose an orientation-preserving plane Jordan–Schoenflies homeomorphism Φ taking $\overline{D}$ to $[0, 1]^2$, with the target arcs ab and cd going to the left and right sides. For fixed $0 < \delta < 1/8$, define in these coordinates the harder and easier rectangles

$$R_\delta^- = [-\delta, 1 + \delta] \times [\delta, 1 - \delta], \quad R_\delta^+ = [\delta, 1 - \delta] \times [-\delta, 1 + \delta].$$

The first rectangle lengthens a horizontal crossing while restricting its height; the second shortens the crossing and allows more height. Figure 7 displays these two comparisons. Approximate their inverse images under Φ by deterministic polygonal marked quadrilaterals, in their boundary order and much more accurately than the buffer margins. For each of the two polygons use the discrete disks from Lemma 9.1. Their original vertices inherit the full-plane free colors, so the crossing results already proved apply to them.

Keep all these choices fixed while taking $\varepsilon \rightarrow 0$. Work on the high-probability event of the needed disk approximation and mesh control in a fixed compact neighborhood. Both Φ and Φ^{-1} are uniformly continuous on the relevant compact sets. Thus any fixed coordinate margin is preserved by taking the physical polygonal and mesh errors sufficiently small. No Lipschitz estimate for either chart is required.

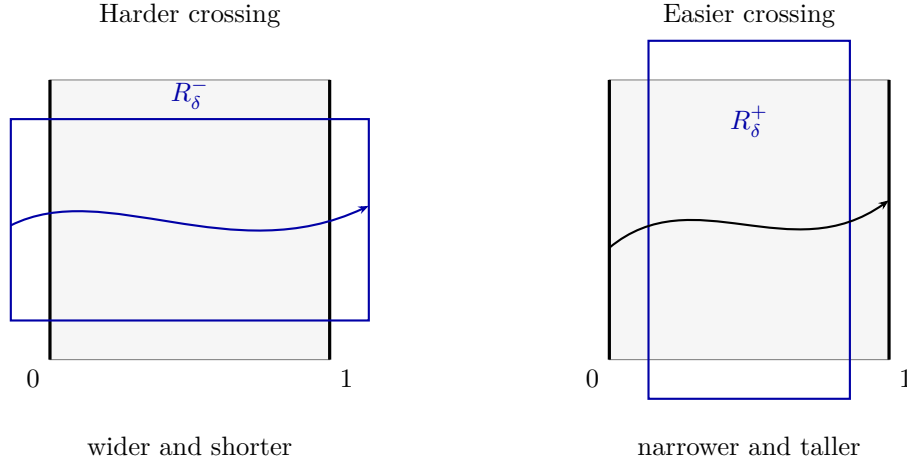


Figure 7: Fixed buffers in the coordinates of Φ . The gray square is $\Phi(\overline{D})$, with its two target sides in bold; the blue boundary is the comparison rectangle. A path crossing the harder rectangle contains a crossing of the square, while a path crossing the square contains a crossing of the easier rectangle. Strict margins preserve these implications under the polygonal and discrete approximations. The picture is in square coordinates and imposes no metric regularity on the physical Jordan boundary or on the chart.

The harder crossing implies a closed-cell crossing. A black primal vertex crossing of the harder disk gives a continuous black-cell path: for each successive pair of neighboring sites, join them through their shared Voronoi side, with the two segments lying in the two closed black cells. Mesh control keeps this path uniformly close to the primal path. In square coordinates it stays in a height band strictly inside $(0, 1)$, starts to the left of $x = 0$, and ends to the right of $x = 1$. Take its first contact with $x = 1$ and the last preceding contact with $x = 0$. Between those times its first coordinate lies in $[0, 1]$, and its height remains in $(0, 1)$. This clipped path is therefore a crossing of $\overline{D}$ in the closed black-cell union.

A connected closed-cell crossing gives a finite primal chain. First recall local finiteness, including cells whose nuclei lie outside the region of interest. On a fixed compact neighborhood, the distance to the nearest site has a finite maximum: a finite cover, or comparison with any fixed site, gives such a bound. A cell meeting that neighborhood has its nucleus within the resulting bounded enlargement. Poisson local finiteness therefore leaves only finitely many such cells.

If a connected crossing set is not closed, take its closure. The black cell union is closed locally, being a locally finite union of closed cells. The closure consequently remains black, connected, and contained in $\overline{D}$, and retains its contacts with both target arcs. Denote this compact connected set by K . For every black cell C_j meeting K , let $E_j = K \cap C_j$. These finitely many nonempty relatively closed sets cover K . Their incidence graph is connected: a partition into two nonempty collections with no intersecting members would express K as the disjoint union of two relatively closed sets, a separation. Hence there is a finite touching chain of black cells from a left-contact cell to a right-contact cell.

Cells meeting along a Voronoi edge have neighboring sites. If two cells meet only at a Voronoi vertex, Poisson general position gives exactly three cells there, with nuclei forming a Delaunay triangle; those two sites are again neighbors. Thus the touching chain yields a black

full-plane primal vertex chain, also for the vertex-only contacts permitted by the closed-cell event. Fine mesh makes its nuclei and edges arbitrarily close to K . For an edge of the chain, use its cell-contact point in K : both neighboring nuclei lie close to that point, and so does their straight Delaunay edge. This justifies the proximity without any convexity assumption on D .

The chain contains a crossing of the easier disk. In square coordinates the chain stays in a small enlargement of $[0, 1]^2$, starts strictly to the left of the easier disk, and ends strictly to its right. Its allowed height band is disjoint from the easier disk's entire top and bottom boundary arcs. A fixed vertical middle strip within that band lies inside the approximating disk; its left and right boundary-arc neighborhoods lie on opposite sides of this strip. These assertions have strict margins, so hold for the polygonal and discrete approximations already fixed.

The chain and the disk boundary are paths in the same planar primal graph. They can meet only at vertices or along shared edges. The chain has a left boundary contact before its first right boundary contact: to reach the right side from its starting position it must pass through the interior middle strip, and the top and bottom arcs are inaccessible. Choose that first right contact, and then the last left contact preceding it. Between these contacts there are no boundary visits. Any shared boundary edge would have boundary endpoint vertices and would also be such a visit. The intervening subpath crosses the interior middle strip, since its endpoints lie on opposite sides. A connected path avoiding a simple boundary cycle cannot move between its interior and exterior. This subpath is therefore in the interior, with its two endpoints on the left and right vertex arcs of the closed disk. Removing repeated vertices if necessary gives its black crossing.

Limit order. Write $p_{\varepsilon, \delta}^-$ and $p_{\varepsilon, \delta}^+$ for the free crossing probabilities of these fixed harder and easier discrete disks, and $p_\varepsilon(D; a, b, c, d)$ for the requested closed-cell probability. The two pathwise comparisons, with the vanishing geometry exception, imply

$$p_{\varepsilon, \delta}^- - o(1) \leq p_\varepsilon(D; a, b, c, d) \leq p_{\varepsilon, \delta}^+ + o(1),$$

where δ and the deterministic polygons are fixed during the microscopic limit. The outer probabilities converge to $\mathcal{C}(x_\delta^-)$ and $\mathcal{C}(x_\delta^+)$ by the polygonal result. Now shrink the buffers, choosing their polygonal approximation errors to vanish faster than their coordinate margins. The marked boundaries converge in order and in Fréchet distance to the original Jordan boundary. Closed Jordan conformal convergence gives $x_\delta^-, x_\delta^+ \rightarrow x$. Since $\mathcal{C}$ is continuous on $(0, 1)$, the displayed inequalities squeeze the requested limit to $\mathcal{C}(x)$. $\square$

Finally,

$$\int_0^1 u^{-2/3}(1-u)^{-2/3} du = \frac{\Gamma(1/3)^2}{\Gamma(2/3)}, \quad \int_0^x u^{-2/3}(1-u)^{-2/3} du = 3x^{1/3} {}_2F_1(1/3, 2/3; 4/3; x).$$

The second identity follows by integrating the binomial series on $[0, x]$, and $\Gamma(4/3) = \Gamma(1/3)/3$. Substitution into (9.8) yields exactly (1.1). Together with Proposition 9.6, this proves Theorem 1.1 for the stated full-plane annealed law and the closed-cell convention.

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